

Dissertation

Micromachined Viscosity Sensors

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*To my parents
Doris and Herbert Riesch.*

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Nomenclature

0.1 List of Abbreviations

Symbol	Description
ac	alternating current
AFM	atomic force microscope
Al	aluminium
APC	automatic power control
Au	gold
CD	compact disc
Cr	chromium
dc	direct current
DI-water	deionized water
DRIE	deep reactive ion etching
DVD	digital versatile disc
FEM	finite element method
FFT	fast Fourier transform
FPC	flexible printed circuit board
KOH	potassium hydroxide
LIA	lock-in amplifier
LPCVD	low pressure chemical vapour deposition
MEMS	microelectromechanical system
n-Si	n-doped silicon
PCB	printed circuit board
PDMS	polydimethylsiloxane

Symbol	Description
PECVD	plasma enhanced chemical vapour deposition
PSD	position sensitive detector
p-Si	p-doped silicon
PZT	lead zirconate titanate
QCM	quartz crystal microbalance
RIE	reactive ion etching
RTD	resistance temperature detector
SAW	surface acoustic wave
SEM	scanning electron microscope
SHO	simple harmonic oscillator
SiO ₂	silicon oxide
SOI	silicon on insulator
TSM	thickness shear mode
Ti	titanium

0.2 List of Symbols

Symbol	Description	Unit
A	surface area	m ²
$a_{\alpha\beta}$	elements of the transformation matrix	
B	magnetic flux density	T
B	susceptance	S
b	damping coefficients	kg/s
b_e	intrinsic damping	kg/s
b_i	induced damping due to liquid loading	kg/s
C	capacitance	F
$c_{1,2,3,4}$	fit parameters	

Symbol	Description	Unit
D	damping factor	
D	electric displacement field	As/m ²
d	dipping depth, distance, diameter	m
d_1, d_2, d_3	direction cosines	
$\mathbf{E}$	electric field vector	V/m
E	electric field	V/m
E	Young's modulus	Pa
E_α	elements of the electric field vector	V/m
e	Euler's number, $e = 2.7182\dots$	
F	force	N
FE	focus error signal	V
f	frequency	Hz
f_0	resonance frequency	Hz
f_e	excitation frequency	Hz
f_r	readout frequency	Hz
G	conductance	S
G	rigidity modulus	Pa
$\bar{G}$	complex shear modulus, $\bar{G} = G' + jG''$	Pa
G'	storage modulus	Pa
G''	loss modulus	Pa
g	gravitational acceleration, $g = 9.81 \text{ m/s}^2$	m/s ²
h	height, thickness	m
K	constant	
K	spring constant	N/m
K_0, K_1	modified Bessel functions of the second kind	
K_G	gauge factor	
k	index variable	
I	area moment of inertia	m ⁴

Symbol	Description	Unit
I, i	electric current	A
i_e	excitation current	A
$\mathbf{J}$	current density vector	A/m ²
J	objective function	
J_x, J_y	current density	A/m ²
J_α	elements of the current density vector	A/m ²
j	imaginary unit, $j = \sqrt{-1}$	
L	inductance	H
l	length	m
l_1, l_2, l_3	direction cosines	
M	torque	Nm
M_p	actuating moment	Nm
m	mass	kg
m'	mass per unit length	kg/m
m_1, m_2, m_3	direction cosines	
m_e	mass of the resonator	kg
m_i	induced mass due to the liquid loading	kg
N	dopant concentration	cm ⁻³
n_1, n_2, n_3	direction cosines	
P	power	W
P'	power per unit length	W/m
p	pressure	Pa
Q	quality factor	
q	flow rate	m ³ /s
R	electrical resistance	Ω
ΔR	change of resistance	Ω
Re	Reynolds number	
RF	radio frequency signal	V

Symbol	Description	Unit
r	circle radius in admittance locus plot	S
r	radius	m
T	temperature	°C
T	time constant	s
t	time	s
U, u	voltage	V
u_d	differential output voltage	V
u_d	driving voltage	V
u_e	excitation voltage	V
u_r	readout voltage	V
u_s	sensing voltage	V
v	velocity	m/s
W	work, energy	J
w	width	m
x, y, z	Cartesian coordinate axes	
x_1, x_2, x_3	Cartesian coordinate axes	
Y	admittance	S
α	angle	°
α	temperature coefficient of the electrical resistance	K ⁻¹
α, β	mode-shape factors	
α_k, β_k	coefficients of viscoelastic models	
$\alpha, \beta, \gamma, \delta$	indices denominating elements of tensors and matrices	
Γ	hydrodynamic function	
γ	strain	
$\dot{\gamma}$	shear rate, $\dot{\gamma} = \partial\gamma/\partial t$	
δ	depth of penetration	m
ε	permittivity	As/Vm

Symbol	Description	Unit
η	dynamic viscosity	Pa·s
θ	Heaviside step function	
ν	kinematic viscosity	m ² /s
ν	Poisson's ratio	
π	piezoresistance matrix	Pa ⁻¹
π	$\pi = 3.1415\dots$	
π_1	longitudinal piezoresistance coefficient	Pa ⁻¹
$\pi_{\alpha\beta}$	elements of the piezoresistance matrix	Pa ⁻¹
ρ	electrical resistivity vector	Ωm
ρ	mass density of liquids	kg/m ³
ρ_0	electrical resistivity of the mechanically unloaded semiconductor	Ωm
ρ_b, ρ_d, ρ_s	mass density of the beam/device/sphere	kg/m ³
ρ_{el}	electrical resistivity	Ωm
ρ_α	elements of the electrical resistivity vector	Ωm
$\rho_{\alpha\beta}$	elements of the electrical resistivity tensor	Ωm
σ	mechanical stress vector	Pa
σ	mechanical stress	Pa
σ_1	longitudinal mechanical stress in the piezoresistor	Pa
σ_α	elements of the stress vector	Pa
$\sigma_{\alpha\beta}$	elements of the stress tensor	Pa
τ	time constant	s
Φ	particle concentration	
ϕ	phase shift	°
φ	mode shape function	
ψ	deflection	m
Ω	correction function	
ω	angular frequency, angular speed	s ⁻¹

Symbol	Description	Unit
ω_0	angular resonance frequency	s^{-1}
$\omega_{0,\text{air}}$	angular resonance frequency in air	s^{-1}

Chapter 1

Introduction

The ability to measure the viscosity of liquids *in situ* enables a multitude of possibilities like the monitoring of processes and fluid conditions.

The monitoring of the deterioration of oil used as insulation fluid, lubricant, or hydraulic oil requires the online measurement of several parameters like water content, concentration of dissolved gases, or soot content [1, 2]. In principle, many of these parameters can be determined by the application of chemical sensors. These devices typically comprise a chemically sensitive layer. This layer selectively absorbs certain ions or molecules, increasing its mass density. The change in mass can then be determined by a suitable physical transducer, e.g., a quartz crystal microbalance (QCM). Unfortunately, most chemical sensors suffer from limited durability, selectivity, and long-term stability and are prone to poisoning by chemical substances that can be present in the sample fluid. As an alternative, *physical chemosensors* aim at determining the chemical state of a sample by monitoring several physical parameters and extracting the desired chemical information from these results [3]. The measurement of several physical quantities like permittivity, thermal heat conductance, and viscosity was suggested for the monitoring of engine oil and transformer oil [2]. Processes like polymerization or crystallization have a high impact on the viscosity of the respective fluid. It is therefore beneficial to apply a continuous measurement of this physical quantity to get an insight into the progress of the process [4, 5]. Furthermore, such measurements allow closed-loop control of these processes.

Conventional laboratory equipment is often not applicable for the online measurement of viscosity due to its cost, space requirements, and other pre-conditions, e.g., vibration-free mounting. Furthermore, sample taking for such devices often involves manual labour, tending to be time consuming and error-prone.

Microacoustic sensors like quartz thickness shear mode (TSM) resonators [6] and surface acoustic wave (SAW) devices, e.g., [7], have proved themselves particularly useful alternatives to traditional viscometers [8]. However,

these devices measure viscosity at relatively high frequencies and small vibration amplitudes. For non-Newtonian liquids, the results are therefore not directly comparable to those obtained from conventional viscometers. Furthermore, the penetration depth of the shear wave excited by these sensors is rather small and consequently only a thin film of liquid is probed. For complex liquids such as emulsions it has also been shown, that microacoustic devices may not be sufficient to detect rheological effects which are present only on the macroscopic scale [9]. Chapter 2 of this thesis further elucidates the consequences arising from the use of such sensors for rheologically complex liquids.

Micromachined vibrating structures usually feature lower resonance frequencies and higher vibration amplitudes, making them more suitable for measuring the steady shear viscosity of non-Newtonian and complex liquids [10]. Microcantilevers commonly used in atomic force microscopy [11–13] have been successfully used as liquid property sensors. They allow for simultaneous measurement of the liquid's viscosity and mass density, requiring sample volumes of less than 1 nL [13]. In other works, micromachined cantilevers and doubly clamped beams driven by Lorentz forces [10, 14, 15] or by the piezoelectric effect [16, 17] have been utilized as liquid property sensors, and the feasibility of these sensors has been demonstrated for viscosities in the range up to several Pa·s.

The objective of this thesis is the design, the modelling, and the investigation of novel miniaturized sensors for the viscosity of liquids. In particular, three different sensor designs will be presented in the course of this work. The first sensor, described in Chapter 3, is based on a piezoelectric trimorph beam. To the free end of the beam, tips of different geometrical dimensions have been attached. Although the beam is rather large in size (5 cm in length, the tip geometries are in the mm range), the setup enables the investigation of general principles and is used to devise a generalized model based on the commonly used oscillating sphere model. This model describes the relation between resonance frequency and damping of the cantilever and the viscosity and density of the sample liquid. After obtaining the model parameters in a calibration procedure, the model equations can be used to determine the viscosity and density of unknown liquids.

Micromachining technology allows the fabrication of miniaturized devices that comprise resonating parts. The resonators can be designed to operate in a low-frequency regime. Consequently, sensors can be built that measure a viscosity parameter that is comparable to the steady-shear viscosity. Chapter 4 is intended to give an overview on MEMS (microelectromechan-

ical systems) technology and viscosity sensors that are fabricated by micro-machining and operate at frequencies in the kHz range.

The second sensor design presented in this thesis is a micromachined beam in a clamped–clamped configuration (Chapter 5). The beam vibrations are excited by Lorentz forces. An optical readout based on the laser pickup head of a DVD player is presented. It is shown that the damping of the beam resonance is a measure for the viscosity of the liquid in which the sensor was immersed. Furthermore, this relation is not only valid for Newtonian liquids, but also holds for a set of viscoelastic sample liquids. The experimental results indicate, that the obtained viscosity parameter is comparable to the steady-shear viscosity probed by most laboratory viscometers. Finally, a model derived for atomic force microscope (AFM) cantilevers is applied to the clamped–clamped beam sensor, and its suitability for the particular device is discussed.

Immersed in a sample liquid, most cantilever-based sensors and also the clamped–clamped beam resonator are highly damped. The quality factor of the resonance is low and the mechanical vibration amplitude is small. Extremely sensitive readout principles like the optical detectors of atomic force microscopes are required to obtain the deflection amplitude of the resonating part. The novel micromachined viscosity sensor presented in Chapter 6 addresses this issue. The basic concept of this device is a thin plate vibrating in the in-plane direction. The plate excites mainly shear-waves in the liquid. Consequently, the dissipative losses associated with the liquid surrounding the sensor are kept low, whereas the mass of the plate and hence the energy stored in the resonant system is high. The quality factor of the sensor's resonance is increased and higher vibration amplitudes can be achieved compared to other micromachined viscosity sensors. The integrated piezoresistive readout eliminates the need for highly complex optical readout methods. A simplified model for the device is presented and the feasibility of the sensor is demonstrated by experiments. Finally, the generalized model devised in Chapter 3 is applied to the measurement results. Although the flow field around the vibrating plate is rather complex, this simple model enables the interpretation of the results, i.e., resonance frequency and quality factor, within a well-defined range of viscosity and density, enabling the simultaneous measurement of both these parameters.

Chapter 2

Viscosity of Liquids

This chapter is intended to give a short review of viscosity, rheology, measurement methods, and the consequences arising for miniaturized viscosity sensors.

2.1 Newtonian Liquids and Hookean Solids

Figure 2.1 shows a liquid between two parallel planes. The planes are separated by a distance d . The upper plane moves with a velocity v_0 in x -direction. A constant force F in the same direction is required to produce the constant motion of the upper plane. In his *Philosophiæ Naturalis Principia Mathematica* (1687) Isaac Newton published a hypothesis associated with the setup in Figure 2.1: “The resistance which arises from the lack of slipperiness of the parts of the liquids, other things being equal, is proportional to the velocity with which the parts of the liquids are separated from one another” (from [18]). Consequently, the force per unit area F/A is proportional to the velocity gradient v_0/d (the shear rate). The constant of proportionality, describing the “lack of slipperiness”, is called the *dynamic viscosity* η ,

$$\frac{F}{A} = \eta \frac{v_0}{d}. \quad (2.1)$$

We notice an important property of Newtonian fluids: The flow of the liquid persists as long as the force is applied to the upper plate. If the force is removed the shearing stops and the liquid stays in the deformed state. Using the shear stress $\sigma = F/A$ and the shear rate $\dot{\gamma} = v_0/d = \partial\gamma/\partial t$, (2.1) becomes

$$\sigma = \eta \dot{\gamma}. \quad (2.2)$$

In 1678 the behaviour of solids was described by Robert Hooke in his *True Theory of Elasticity*: “the power of any spring is in the same proportion with the tension thereof” (from [18]). The higher the applied mechanical stress is,

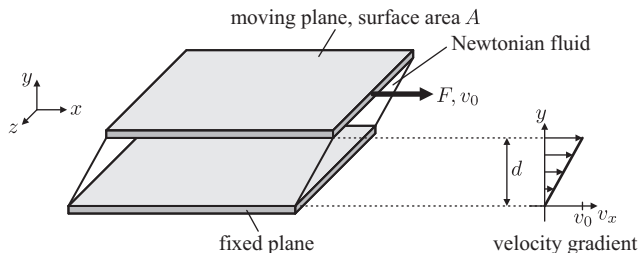


Figure 2.1: Dynamic viscosity of a fluid. The diagram shows two parallel planes at $y = 0$ and $y = d$. The upper plane moves with a velocity v_0 in x -direction. The space between the planes is filled with a liquid. In such a setup, a velocity gradient $v_x(y)$ is imposed on the liquid. A continuous force per unit area, F/A is required to maintain this velocity gradient in the liquid layer.

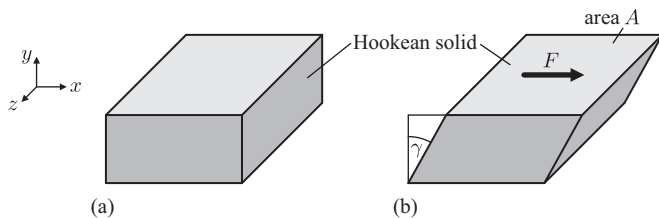


Figure 2.2: Deformation of a Hookean solid (simplified illustration). The application of a shear stress $\sigma = F/A$ results in a deformation by the angle γ .

the higher is the strain. This Hookean behaviour is illustrated in Figure 2.2. In the example a shear stress $\sigma = F/A$ is applied to a block of material. This stress results in an instantaneous deformation (Figure 2.2b). The angle in the diagram is the strain γ . According to Hooke the relation between the stress and the strain is

$$\sigma = G\gamma, \quad (2.3)$$

where G is the *rigidity modulus*. Once the deformed state is reached no further movement is observed. When the force F is removed the block returns instantaneously to its undeformed state (Figure 2.2a).

Up to the 19th century Hooke's law was used for solids and Newton's law described the behaviour of liquids. However, it turned out in experiments that there are liquids that are not perfectly Newtonian. For example, some liquids show a *shear-thinning* behaviour. Their viscosity decreases at higher shear

rates. Few liquids are *shear-thickening*, they have a higher viscosity when they are sheared. Hence, the viscosity of the liquid depends on the applied shear rate and (2.2) becomes

$$\sigma = \eta(\dot{\gamma})\dot{\gamma}. \quad (2.4)$$

Another effect is called *Thixotropy* and describes the behaviour of a liquid whose viscosity depends on the duration of the shearing.

Everyday examples of such non-Newtonian liquids are toothpaste or mayonnaise. Toothpaste is shear-thinning. It is easily squeezed from the toothpaste tube but then sits unmoving on the toothbrush [18]. Like the toothpaste, mayonnaise behaves like a solid when put on a bread. However, it can easily be spread on a bread with a knife. Under the shear force exerted by the knife the viscosity decreases [19].

Obviously, there are liquids that behave like solids under certain circumstances and solids which behave like liquids depending on the present conditions. It becomes clear that two different methods of viscosity measurement, therefore, do not necessarily reveal the same viscosity parameter.

2.2 Measurement of Viscosity

A measurement device that determines the viscosity of a fluid is called a *viscometer*. A multitude of measurement principles exists for viscosity [20]. The most important types are sketched in Figure 2.3.

Figure 2.3a depicts a capillary viscometer. If a Newtonian liquid flows through a circular tube of radius r at a flow rate q , the pressure gradient dp/dl along the length of the tube is

$$\frac{dp}{dl} = \frac{8q\eta}{\pi r^4}, \quad (2.5)$$

where η is the dynamic viscosity of the liquid [18]. If the tube is arranged vertically the pressure difference equals the hydrostatic pressure. In most capillary viscometers the flow rate is measured in terms of a time Δt required for a specified volume of the liquid to flow between two graduation marks. Therefore, from measurement of this time, the *kinematic viscosity* ν can be obtained as

$$\nu = K\Delta t, \quad (2.6)$$

where K is a constant for the viscometer [20]. ν is defined as the quotient of the dynamic viscosity η and the mass density ρ of the liquid, i.e., $\nu = \eta/\rho$.

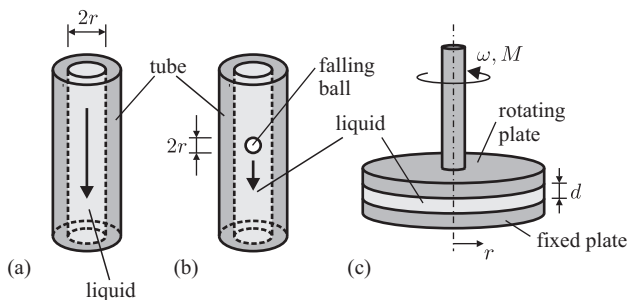


Figure 2.3: Common measurement principles utilized by conventional laboratory viscometers. (a) Capillary viscometer. (b) Falling ball viscometer. (c) Rotational viscometer.

For the measurement of η , the knowledge of the mass density of the liquid is required. However, for non-Newtonian liquids the situation is more complex since the shear rate varies from zero in the centre of the tube to a maximum near the wall [18].

The falling ball viscometer, depicted in Figure 2.3b, measures the viscosity based on the forces acting on a sphere that is allowed to fall under gravity through a liquid. After a period of acceleration the ball reaches a constant velocity when the gravitational force equals the viscous resistance of the fluid. From this terminal velocity v the viscosity of the liquid η is calculated using Stokes' law,

$$\eta = \frac{2gr^2(\rho_s - \rho)}{9\pi v}, \quad (2.7)$$

where ρ and ρ_s are the mass density of the liquid and the sphere, respectively, r is the sphere radius, and g is the gravitational acceleration [20].

Rotational viscometers are a third kind of instruments for the measurement of viscosity. Their operating principle is sketched in Figure 2.3c. It relies on a simple shear flow (Figure 2.1) generated by a rotational motion. Figure 2.3c shows a liquid between a fixed and a rotating plate. The upper plate rotates at a constant angular speed ω . Due to the viscosity of the liquid (the “lack of slipperiness” noted by Newton) a certain torque M is required to drive the plate. From measuring this torque the viscosity of the liquid is then calculated. There are different implementations of this principle available: parallel plate viscometers like shown in Figure 2.3c, concentric-cylinder viscometers featuring a cylinder rotating in a cup, rotating cylinders in a large

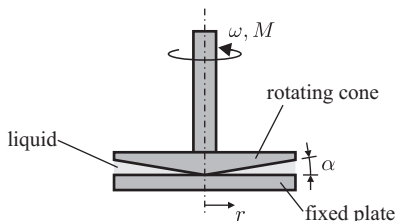


Figure 2.4: Cone-plate viscometer.

volume of liquids, and cone–plate viscometers [18, 20].

The shear rate $\dot{\gamma}$ produced by a plate–plate rheometer (Figure 2.3c) is

$$\dot{\gamma} = \frac{r\omega}{d}, \quad (2.8)$$

where ω is the angular velocity, r is the distance from the axis, and d is the distance between the two plates. Hence, the shear rate is zero at the centre of the plate and has a maximum at the plate edge. In the experiments presented in this work, a cone–plate viscometer was used for reference measurements. In contrast to the plate–plate setup, this instrument shears the liquid between an obtuse-angled cone and a plate (Figure 2.4). If the angle α between the cone and the plate is small the resulting shear rate is

$$\dot{\gamma} \approx \frac{r\omega}{r\alpha} = \frac{\omega}{\alpha}. \quad (2.9)$$

Consequently, the shear-rate is uniform throughout the liquid sample. The Brookfield LV-DV+CP cone–plate rheometer with the CP-40 cone used in the experiments featured an angle of 0.8° , $\alpha = 0.014$ rad, and a rotational speed in the range from 0.01 to 200 rpm ($\omega = 0.001 \dots 21$ s $^{-1}$). Hence, shear rates from 10^{-4} to 1500 s $^{-1}$ can theoretically be achieved. Actually, the measurement range of the instrument's torque sensor must be taken into account, limiting the applicable shear rates depending on the viscosity of the particular liquid.

The viscosity of a liquid is highly dependent on the temperature. Therefore, great care must be taken to control the temperature of the sample liquid. It should be noted that also the shearing of the liquid itself generates heat and may thus influence the viscosity of the sample. Most of the available viscometers, therefore, have facilities to control the temperature of the sample liquid,

e.g., by a circulating cooling/heating fluid. In the experiments presented in this work, Peltier devices were used for this purpose.

The term *rheometer* is often used in connexion with the measurement of the flow properties. In contrast to a viscometer, which aims at determining the viscosity η of a fluid, the rheometer is designed to detect the full rheological behaviour of a liquid, e.g., its viscoelastic behaviour.

2.3 Linear Viscoelasticity

Miniaturized viscosity sensors typically utilize oscillatory principles. Below the response of liquids to oscillatory motion will be discussed within the limits of *linear viscoelasticity*. In (2.2) the relation between the desired shear rate $\dot{\gamma}$ and the required stress σ is given by the viscosity η :

$$\sigma = \eta \frac{d\gamma}{dt}. \quad \equiv (2.2)$$

In linear viscoelasticity this equation is replaced by a general linear differential equation,

$$\begin{aligned} & \left(1 + \alpha_1 \frac{\partial}{\partial t} + \alpha_2 \frac{\partial^2}{\partial t^2} + \dots + \alpha_n \frac{\partial^n}{\partial t^n} \right) \sigma = \\ & = \left(\beta_0 + \beta_1 \frac{\partial}{\partial t} + \beta_2 \frac{\partial^2}{\partial t^2} + \dots + \beta_m \frac{\partial^m}{\partial t^m} \right) \gamma, \end{aligned} \quad (2.10)$$

that allows to take a more complex behaviour into account [18].

Some important special cases of this equation will be considered here. If β_0 is the only non-zero parameter, the result

$$\sigma = \beta_0 \gamma \quad (2.11)$$

is similar to (2.3) with $\beta_0 = G$. A material described by this equation is a perfect Hookean solid.

If β_1 is the only non-zero coefficient, we have

$$\sigma = \beta_1 \frac{\partial \gamma}{\partial t} = \beta_1 \dot{\gamma}. \quad (2.12)$$

This equation corresponds to (2.2) with $\beta_1 = \eta$. Thus the equation represents the shear flow of a perfect Newtonian liquid (Figure 2.1).

In the limits of linear viscoelasticity applying an oscillatory shear strain

$$\gamma(t) = \hat{\gamma} e^{j\omega t} \quad (2.13)$$

results in a stress response $\sigma(t) = \hat{\sigma} e^{j(\omega t + \phi)}$, where $\omega = 2\pi f$ is the angular frequency of strain and stress and ϕ is the phase shift between excitation and response. With a complex stress amplitude $\bar{\sigma} = \hat{\sigma} e^{j\phi}$ we have

$$\sigma(t) = \bar{\sigma} e^{j\omega t}. \quad (2.14)$$

With these definitions, (2.11) and (2.12) result in

$$\bar{\sigma} = G\bar{\gamma} \quad \text{and} \quad \bar{\sigma} = j\omega\eta\bar{\gamma}, \quad (2.15)$$

respectively, where $\bar{\gamma} = \hat{\gamma}$ was written for consistency. In linear viscoelasticity it is customary to use a *complex shear modulus* $\bar{G}$. Hence, the equations above can be written as

$$\bar{\sigma} = \bar{G}\bar{\gamma}, \quad (2.16)$$

where the complex shear modulus $\bar{G} = G$ for Hookean solids and

$$\bar{G} = j\omega\eta \quad (2.17)$$

for Newtonian liquids. In literature, $\bar{G}$ is often written as $\bar{G} = G' + jG''$ and G' and G'' are referred to as the *storage modulus* and the *loss modulus*, respectively.

If both β_0 and β_1 are non-zero and the other coefficients of (2.10) are zero, the result is

$$\sigma = \beta_0\gamma + \beta_1\dot{\gamma} = G\gamma + \eta\dot{\gamma}. \quad (2.18)$$

This model is called the *Kelvin model* or *Voigt model*. If a constant stress σ_0 is applied at $t = 0$, i.e., $\sigma(t) = \sigma_0\theta(t)$, where $\theta(t)$ is the Heaviside step function, the strain response of the Kelvin model is

$$\gamma(t) = \frac{\sigma_0}{G} \left(1 - e^{-t/\tau_k}\right), \quad (2.19)$$

where $\tau_k = \eta/G$. This behaviour is depicted in Figure 2.5. After application of the stress, the material deforms until the equilibrium strain σ_0/G is reached. The rate of growth of the strain is controlled by the ratio η/G , which has the dimension of time. In contrast, a Hookean solid, described by (2.11), would respond to such a step with an instantaneous deformation, i.e., a step in strain, $\gamma = \sigma_0/G\theta(t)$. The strain of a material that follows the Kelvin model

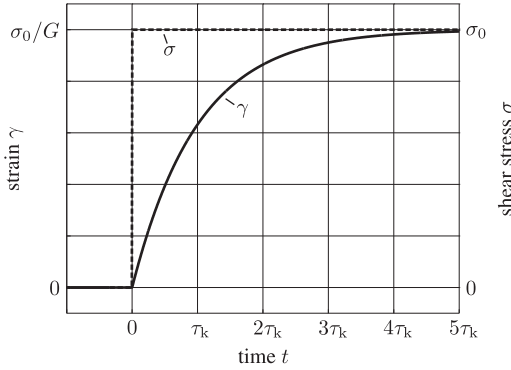


Figure 2.5: Response of the Kelvin model to a constant stress σ_0 applied at $t = 0$.

is retarded. The time constant $\tau_k = \eta/G$ is accordingly called the *retardation time*.

Another important model in linear viscoelasticity is the *Maxwell model*. It is obtained from (2.10) by making α_1 and β_1 the only non-zero parameters as

$$\sigma + \alpha_1 \dot{\sigma} = \beta_1 \dot{\gamma}. \quad (2.20)$$

Here, applying a constant strain $\gamma = \gamma_0 \theta(t)$ yields a stress

$$\sigma(t) = \gamma_0 \frac{\eta}{\tau_m} e^{-t/\tau_m} \theta(t), \quad (2.21)$$

where $\eta = \beta_1$ and $\tau_m = \alpha_1$. Both $\gamma(t)$ and $\sigma(t)$ are sketched in Figure 2.6a. Hence, at $t = 0+$ the material responds with an instantaneous stress to the constant strain. This behaviour is similar to that of a Hookean solid. However, the stress then starts to decrease exponentially. It relaxes with a time constant τ_m which is, therefore, called the *relaxation time*. After $5\tau_m$, the behaviour of the material reaches nearly that of a Newtonian liquid, that shows zero stress at a constant strain.

Using (2.16) a complex shear modulus is determined for both the Kelvin and the Maxwell model. A Fourier transform of the Kelvin model, (2.18), leads to

$$\bar{G}(j\omega) = G + j\omega\eta. \quad (2.22)$$

For the Maxwell model, (2.20), we have

$$\bar{G}(j\omega) = \frac{j\omega\eta}{1 + j\omega\tau_m}. \quad (2.23)$$

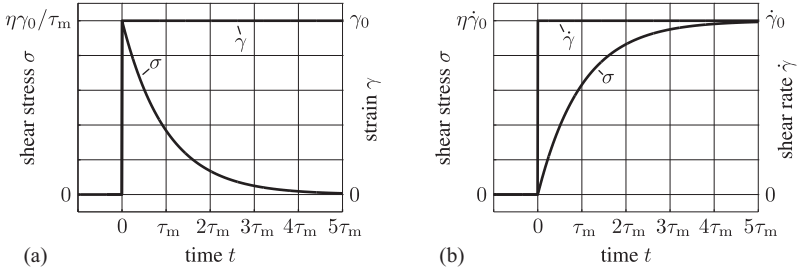


Figure 2.6: (a) Response of the Maxwell model to a constant strain γ_0 applied at $t = 0$. (b) A constant shear rate $\dot{\gamma}_0 = \partial\gamma/\partial t$ is applied to the Maxwellian fluid.

The complex amplitude of the stress response of the liquids to an oscillatory shear strain with the angular frequency ω is again

$$\bar{\sigma}(j\omega) = \bar{G}(j\omega)\bar{\gamma}(j\omega). \quad (2.24)$$

Below we will again consider a fluid that follows the behaviour of the Maxwell model. The measurement results of two types of viscometers will be discussed. First, we apply a constant shear rate $\dot{\gamma}_0 = \partial\gamma/\partial t = \text{const}$ at $t = 0$ to the fluid. This constant shear rate excitation is typical for the cone-plate viscometer (Figure 2.4). From solving (2.20) we have

$$\sigma(t) = \eta\dot{\gamma}_0 \left(1 - e^{-t/\tau_m}\right). \quad (2.25)$$

A plot of this solution is depicted in Figure 2.6b. After introduction of the constant shear rate, i.e., starting the measurement, the stress growth is retarded. The time constant determining the rate of growth is τ_m . After sufficient time, the terminal stress $\eta\dot{\gamma}_0$ is reached. The apparent viscosity measured by the viscometer is the same as that of a Newtonian liquid with the viscosity η . However, a different situation emerges from using an oscillatory viscometer, a principle, that is typically used by miniaturized viscosity sensors. For comparison, the complex shear moduli $\bar{G}$ of a Maxwellian and a Newtonian fluid, equations (2.17) and (2.23) are plotted in Figure 2.7a. For this illustration, the model parameters were $\eta = 1 \text{ Pa}\cdot\text{s}$ and $\tau_m = 1 \text{ s}$. It becomes clear, that at low frequencies the Maxwellian fluid shows the same behaviour as the Newtonian liquid. Consequently, an oscillatory viscometer operating at frequencies well below the relaxation frequency $1/\tau_m$ will show

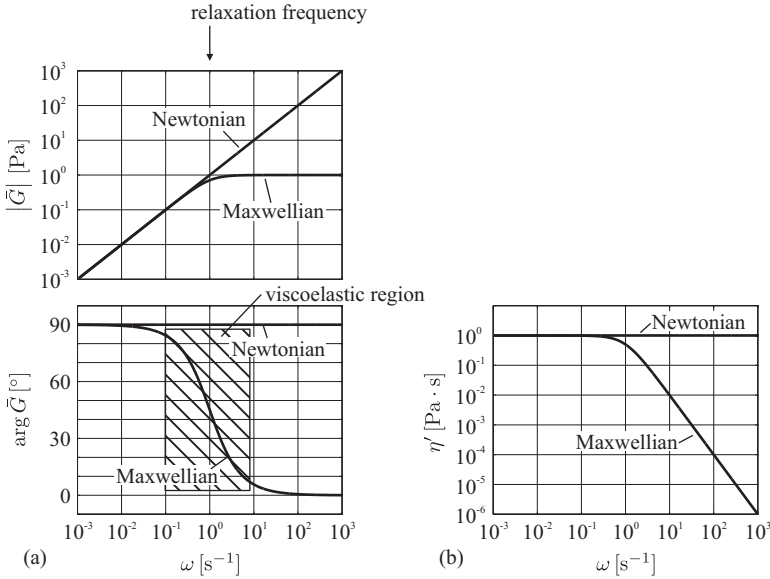


Figure 2.7: Relaxation spectrum of Newtonian and Maxwellian fluids.

a reading similar to that of the cone–plate instrument. However, at frequencies above $1/\tau_m$, the Maxwell model exhibits the complex shear modulus of an elastic material, $\bar{G}(j\omega) = \eta/\tau_m$. This situation is illustrated in Figure 2.7b, where the apparent viscosity $\eta' = G''/\omega$ is plotted versus the frequency [18]. At higher frequencies, the viscosity obtained by the oscillatory viscometer is far below the result of a cone–plate setup. The two decades of frequency around the relaxation frequency are called the viscoelastic region of the fluid.

The complex shear modulus $\bar{G}$ of many liquids can be described by a superposition of Maxwell models [18]. In [19], the behaviour of a polydimethylsiloxane (PDMS) sample is given in terms of a relaxation modulus in the time domain,

$$G(t) = \sum_{k=1}^N G_k e^{-t/\tau_k}. \quad (2.26)$$

The values of G_k and τ_k are listed in Table 2.1 and $N = 5$. The stress

Table 2.1: Material parameters for polydimethylsiloxane (PDMS) at 25°C (from [19]).

k	τ_k [s]	G_k [Pa]
1	0.01	$2 \cdot 10^5$
2	0.1	10^5
3	1	10^4
4	10	10^2
5	100	10^1

response $\sigma(t)$ to a strain $\gamma(t)$ is

$$\sigma(t) = \int_{-\infty}^t G(t-t') \dot{\gamma}(t') dt'. \quad (2.27)$$

From a Fourier transform of (2.26) and (2.27) we have

$$\bar{\sigma}(j\omega) = \sum_{k=1}^N \frac{j\omega G_k \tau_k}{1 + j\omega \tau_k} \bar{\gamma}(j\omega). \quad (2.28)$$

The behaviour of PDMS under oscillatory shear can therefore be described by a complex shear modulus that is a superposition of five Maxwell models,

$$\bar{G}(j\omega) = \sum_{k=1}^N \frac{j\omega \eta_k}{1 + j\omega \tau_k}, \quad (2.29)$$

where $\eta_k = G_k \tau_k$, see (2.23). $G(t)$, equation (2.26), and equation $\bar{G}(j\omega)$, (2.29), are plotted in Figure 2.8. The diagrams indicate that the liquid exhibits the behaviour of a Newtonian liquid when it is excited by an oscillatory strain at low frequencies. At higher frequencies PDMS becomes an elastic material.

2.4 Measurement of Emulsions and Suspensions

A common miniaturized viscosity sensor is the thickness shear-mode (TSM) quartz resonator. A detailed description of the sensor and the evaluation of the measurement results is given in Section 5.5.2. The sensor surface performs shear vibrations and excites shear waves in the liquid. The dissipative

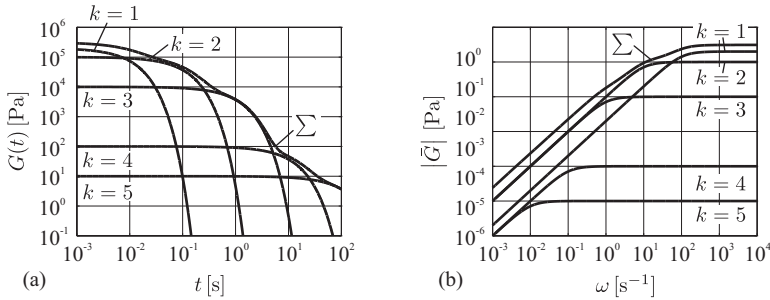


Figure 2.8: $G(t)$ and the complex shear modulus $\bar{G}(j\omega)$ of PDMS. The diagrams represent plots of the equations (2.26) and (2.29). The individual Maxwell models that represent the behaviour of the fluid are referred to as $k = 1 \dots 5$, corresponding to Table 2.1. The sums of these models, i.e., $G(t)$ and $\bar{G}$, are labelled Σ .

power associated with these waves leads to a damping of the sensor vibrations. At the same time the liquid that is moved with the quartz surfaces increases the resonator mass and, therefore, decreases the resonance frequency. Measurement of either the resonance frequency or the quality factor allows the determination of the viscosity–density product of the liquid [21].

The characteristic penetration depth δ of the shear waves in the liquid is

$$\delta = \sqrt{\frac{2\eta}{\omega\rho}}, \quad (2.30)$$

where η and ρ are the dynamic viscosity and the mass density of the liquid, respectively, and $\omega = 2\pi f$ is the angular frequency of the resonator vibration [22]. The resonance frequency f of typical TSM resonators is in the range of several MHz. For our calculation we assume $f = 6$ MHz. In a liquid with a mass density of $\rho = 1000 \text{ kg/m}^3$ and viscosities in the range from 1 to 100 mPa·s, the penetration depth is between 230 nm to 2.3 μm . Therefore, a viscosity sensor operating under such conditions probes the viscosity of only a thin film at the resonator surface.

The high strain frequency potentially excites the sample fluid in its viscoelastic or elastic region (see Figure 2.7). Therefore, the measurement results obtained from a TSM resonator can deviate from the steady-shear viscosity, e.g., probed by a cone–plate viscometer. Besides, the small penetration depth of the shear wave must be considered, too. Figure 2.9 depicts the measurement of oil-in-water emulsions with a TSM resonator. The two emul-

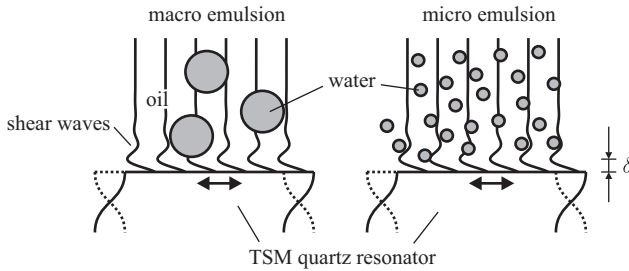


Figure 2.9: Measurement of micro and macro emulsions by a TSM quartz resonator (according to [23]).

sions differ in the size of the water droplets suspended in the oil. In case of the macroemulsion (left diagram in Figure 2.9) the diameter of the droplets is in the same order of magnitude as the penetration depth δ of shear wave. For the microemulsions (right diagram in Figure 2.9) the droplets are far smaller than δ . Measurement results presented by Jakoby *et al.* in [23] showed that the viscosity parameter probed by the TSM resonator correlated to that of a cone-plate viscometer for the microemulsions. For the macroemulsions, however, the penetration depth of the shear waves was too small. The sensor was not able to detect the macroscopic structure of the liquid. The quartz resonator did not detect the increase of viscosity due to the interaction of the droplets but measured the viscosity of the continuous phase (the oil) instead.

2.5 Conclusion

Due to the simplicity in fabrication, most miniaturized sensors for the viscosity of liquids utilize oscillatory measurement principles. The quartz resonators and surface acoustic wave devices which are commonly used exhibit high resonance frequencies and low vibration amplitudes. The high excitation frequencies tend to excite the viscoelastic or elastic behaviour of a fluid instead of its viscous properties. In case of complex structured liquids like emulsions and suspensions, the penetration depth associated with these devices is often too small to probe the macroscopic behaviour of the sample liquid.

The aim of this thesis is the investigation of miniaturized viscosity sensors that overcome these limitations. Effort is put in decreasing the operating fre-

quency of the devices. Hence, the viscous region of the liquid's spectrum is probed. As shown before, the viscosity parameter measured under such conditions is comparable to the steady-shear viscosity probed by conventional viscometers. Furthermore, decreasing the frequency ω increases the penetration depth δ . The measurement results of [23] indicate that a sensor operating at lower frequencies is better suited for probing complex liquids with larger sized structures. Finally, a sensor that could operate at several frequencies or could be swept over a wide bandwidth would potentially reveal the frequency characteristic $\bar{G}(j\omega)$ of the liquid's shear modulus [24]. Hence, not only a single viscosity parameter, but the entire viscoelastic behaviour could be obtained.

Chapter 3

Piezoelectric Trimorph Beam Sensor

In this chapter, resonating cantilevers are characterised for the measurement of mass density and viscosity of liquids. Their operating frequency is around 100 Hz. Therefore, they are expected to measure viscosity in a rheological domain which is comparable to that probed by conventional laboratory instruments (see the explanation given before in Chapter 2).

Only the tip of the cantilever is immersed in the liquid. Hence, the induced damping of the cantilever vibration is kept low. The sensors exhibit high quality factors ranging from 20 to 60 even for highly viscous liquids. Therefore, the detection of the resonances can be accomplished by a simple readout circuit. Furthermore, the measurement range is greatly extended in comparison to micromachined viscosity sensors which usually suffer from low Q-factors (Chapter 4). When the cantilever tip is immersed in a liquid, the resonance frequency and the damping of the cantilever are influenced by the viscosity and density of the liquid. The relation between damping and resonance frequency of the cantilevers and the liquid parameters is complex. Recently, several models have been devised to give a proper description of the interaction of a vibrating cantilever and the surrounding liquid, e.g., [25, 26], but most of these models assume fully immersed cantilevers, which are long and thin, i.e., their width w is much smaller than their length l . For the designs considered in this work, only the cantilever tip is immersed in the liquid. For modelling the sensor–fluid interaction, the cantilever length l must therefore be replaced by the dipping depth d , which is in the same range as w . A solution for cantilevers featuring $w \approx l$, and accordingly $w \approx d$ is given in [27]. In [16] and other works the influence of the liquid loading on the cantilever’s frequency characteristics has been successfully modelled by approximating the forces acting on the cantilever tip by those acting on a sphere oscillating in a fluid. The results indicate that the cantilever tip is subject to an additional

mass loading and an additional damping caused by the surrounding liquid.

Below a vibrating cantilever sensor and a setup for the measurement of viscosity and density is presented. The interaction of the sensor and the liquid in which the tip is immersed is modelled by an oscillating sphere. Possible simplifications of the model are discussed, and a general model is devised. This model allows for simple calibration of the sensor in a set of liquids with well-known properties. Therefore, the knowledge of mechanical and electrical properties of the cantilever is not required, the model parameters are instead obtained from the calibration procedure.

Since only the cantilever tip is immersed in the liquid, the sensor principle allows attaching different tips of well-defined geometries to the cantilevers. The cantilevers feature piezoelectric excitation as well as piezoelectric read-out. The vibrating part is about 55 mm long. This sensor is, therefore, not a miniaturized sensor, but could be used as an experimental platform for studying the interaction of miniaturized geometries, which are attached as a tip to the cantilever, with liquids. Furthermore, the model devised in this chapter was successfully applied to the MEMS resonator device described in Chapter 6, see in particular Section 6.9.

3.1 Sensor Fabrication

The cantilever sensors used in this work are based on commercially available PZT (lead zirconate titanate) trimorph bending actuators (Argillon GmbH, Redwitz, Germany). They feature a length of 49.95 mm, a width of 7.2 mm, and a total thickness of 0.8 mm [28]. The cantilevers consist of two piezoelectric PZT layers on both sides of a carbon fibre substrate (Figure 3.1a). The PZT layers are polarized in thickness direction. Electrodes (1, 2, 3) allow for excitation of the actuator. Applying a voltage between the top electrode (electrode 1) and the centre electrode (2 in Figure 3.1a), which is used as ground electrode (Figure 3.1b) causes a contraction or elongation in the upper PZT layer but not in the substrate, and therefore deforms the cantilever. In our setup (Figure 3.2) a sinusoidal voltage is applied, which leads to bending vibrations of the beam. A maximum voltage of 200 V can be applied to the cantilever. The actual beam deflection is determined by measuring the voltage at the sensing electrode.

The bending actuator is clamped at one end, whereas different tips of well-defined cross-sections are attached to the free end. These tips are immersed in the sample liquid. The tip geometries and materials are given in

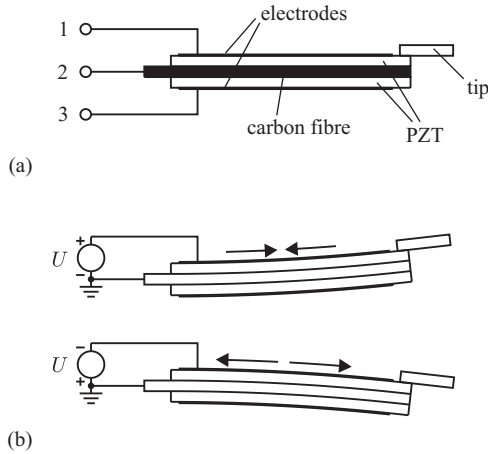


Figure 3.1: (a) PZT bending actuator with attached tip. The bending actuator consists of a carbon fibre substrate and two piezoelectric layers. The electrodes (1, 2, 3) allow for excitation and readout of the sensor. (b) Applying a voltage to the upper layer leads to contraction or elongation of the upper layer as indicated by the arrows.

Table 3.1. The clamping fixture is mounted on a rigid frame allowing for vertical (x -direction) positioning of the sensor and preventing vibrations of the entire setup. A lock-in amplifier (LIA) measures the sensor voltage u_s , resulting in the cantilever's frequency response. As voltage source u_d the internal oscillator of the lock-in amplifier is used.

The cantilever exhibits several resonant vibration modes. As an example, Figure 3.3a shows the deflection (z -direction) for a cantilever with tip A (Table 3.1) vibrating in air. The corresponding sensing electrode voltage u_s is shown in the diagram below (Figure 3.3b). The resonance frequencies are 100 Hz and 851 Hz for the first and second mode, respectively. Further resonances were found at 2456 Hz (3rd mode), and 4870 Hz (4th mode). The mode shapes for a uniform cantilevered beam without a load at the tip are given in Figure 3.4 [29].

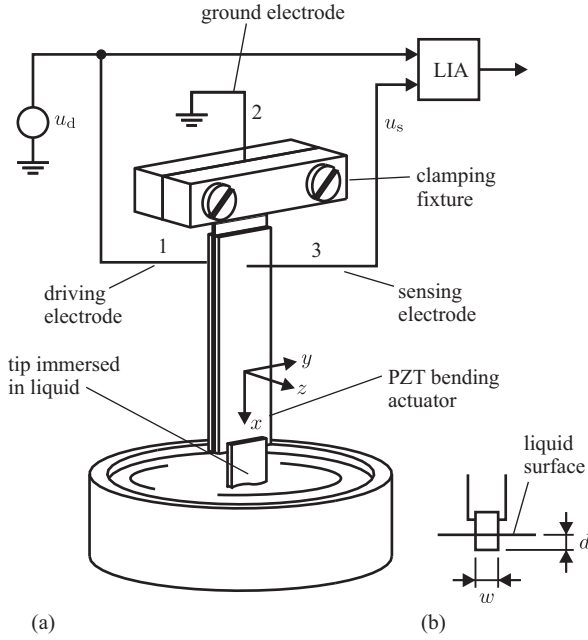


Figure 3.2: Measurement setup. The PZT bending actuator is rigidly clamped at one end. To the free end of the cantilever, a tip of well-defined geometry (width w) is attached and immersed in the sample liquid (dipping depth d). The interaction of the cantilever tip and the liquid changes the frequency characteristics, which is measured by the lock-in amplifier (LIA).

Table 3.1: Tip geometries (rectangular cross-section) and tip materials

Tip	Width w	Thickness	Material
Tip A	7 mm	0.5 mm	silicon
Tip B	5 mm	0.5 mm	silicon
Tip C	4 mm	0.3 mm	brass
Tip D	2 mm	0.5 mm	silicon

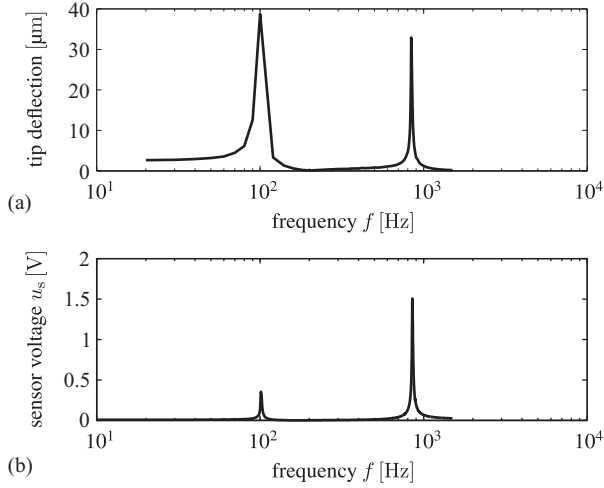


Figure 3.3: Frequency characteristics of a PZT cantilever with tip A (Table 3.1), $V_d = 400 \text{ mV}_{\text{rms}}$ driving voltage, vibrating in air. (a) The tip deflection (peak-peak) was measured using a Polytec OFV-5000/OFV-505 laser vibrometer. (b) The diagram below shows the corresponding sensing electrode rms voltage.

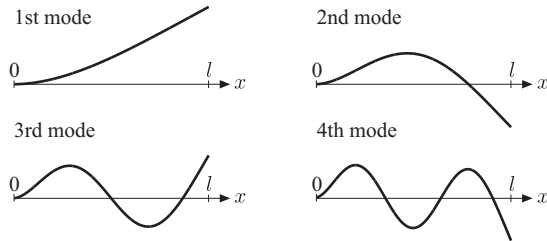


Figure 3.4: Mode shapes of a uniform cantilevered beam. The mode shapes have been obtained from solutions of the Euler-Bernoulli beam equation and are roughly also valid for the piezoelectric trimorph.

3.2 Theoretical Model

The vibration behaviour of the piezoelectric bending actuator is described by the Euler-Bernoulli beam equation [29],

$$EI \frac{\partial^4 \psi(x, t)}{\partial x^4} + m' \frac{\partial^2 \psi(x, t)}{\partial t^2} = \frac{\partial^2 M_p(x, t)}{\partial x^2}, \quad (3.1)$$

where EI and m' are the effective bending stiffness and the effective mass per unit length of the composite beam, $\psi(x, t)$ is the beam deflection in z -direction (Figure 3.2), and M_p is the actuating moment due to the piezoelectric effect. Since M_p is considered to be a constant moment along the entire beam, $\partial^2 M_p / \partial x^2 = 0$. Consequently, the boundary conditions for the clamped-free beam are

$$\begin{aligned} \psi(0, t) &= 0, & \frac{\partial \psi(0, t)}{\partial x} &= 0, \\ \frac{\partial^2 \psi(l, t)}{\partial x^2} &= \frac{1}{EI} M_p, & \text{and} & \quad \frac{\partial^3 \psi(l, t)}{\partial x^3} = -\frac{1}{EI} F, \end{aligned} \quad (3.2)$$

where l is the length of the beam, and F is the force acting on the tip due to the interaction with the surrounding liquid. The actuating moment M_p is given by [30]

$$M_p = \int_{A_p} Y_p d_{31} E_z z_{\text{d}} dA = w_b Y_p d_{31} u_{\text{d}} z_{\text{m}} \quad (3.3)$$

where A_p , Y_p , and d_{31} are the cross sectional area, the Young's modulus, and the piezoelectric modulus of the actuating layer, respectively, and E_z is the electric field in this layer in z -direction. w_b is the width of the bending actuator, z_{m} the mean distance of the actuating layer from the beam centre, and u_{d} is the excitation voltage (Figure 3.2).

The measured voltage u_s is calculated by applying

$$D_z = d_{31} \sigma_{xx} + \varepsilon E_z, \quad \text{and} \quad (3.4)$$

$$\sigma_{xx} = \frac{1}{K_{11}} (\gamma_{xx} - d_{31} E_z) \quad (3.5)$$

to the sensing PZT layer, where D_z , σ_{xx} , and γ_{xx} are electric displacement, stress, and strain. ε and K_{11} are the permittivity and the compliance of the PZT layer. Since u_s is measured by means of a voltage amplifier with a

high input impedance in our setup, the current i_s from the sensing electrode vanishes, and therefore the charge

$$\int_{A_e} D_z dA = 0, \quad (3.6)$$

where A_e is the surface area of the sensing electrode. From (3.4), (3.5), (3.6), and $\gamma_{xx} = -z\partial^2\psi(x, t)/\partial x^2$

$$u_s(t) = K_v \frac{\partial\psi(l, t)}{\partial x} \quad (3.7)$$

is obtained, where K_v is a constant depending on material and geometry parameters [30]. The voltage at the sensing electrode is given by the slope of the beam deflection $\psi(x, t)$ at the free end, $x = l$, of the cantilever. This fact is greatly confirmed by the measurements given by Figure 3.3. Despite the lower tip deflection amplitude of the second mode the resulting output voltage is higher than for the first mode of vibration.

The interaction with the liquid surrounding the cantilever tip can be modelled by approximating the vibrating cantilever as an oscillating sphere with a radius r immersed in a liquid [16]. The force F acting on such a sphere is given by [22]

$$F = 6\pi\eta r \left(1 + \frac{r}{\delta}\right) v + 3\pi r^2 \sqrt{\frac{2\eta\rho}{\omega}} \left(1 + \frac{2r}{9\delta}\right) \frac{dv}{dt}, \quad (3.8)$$

where v is the velocity of the sphere, r is the sphere radius, ω the angular oscillation frequency, and δ the depth of penetration of the acoustic wave, which is given by

$$\delta = \sqrt{\frac{2\eta}{\omega\rho}}. \quad (3.9)$$

The tips are attached to the free ends of the cantilevers, therefore the velocity is $v(t) = \partial\psi(l, t)/\partial t$.

In principle, solving the equations given above yields the frequency characteristics of the vibrating cantilever with a tip immersed in liquid for all modes of vibration. As the cantilever represents a composite structure involving layers featuring different material properties, effective parameters have to be used in the beam equation, e.g., for the Young's modulus. As also the material parameters of the layers are not, or only partly, available in the required accuracy, and since the solution of the beam equation would require

the solution of a higher order system in order to determine the coefficients by a suitable expansion (e.g., eigenmode expansion), we have approximated the resonance behaviour of the sensor in the vicinity of the first mode resonance frequency as a second order system, given by

$$(m_e + m_i) \frac{d^2\psi}{dt^2} + (b_e + b_i) \frac{d\psi}{dt} + K\psi = F_0 e^{j\omega t}, \quad (3.10)$$

where ψ is the deflection of the cantilever tip in z -direction (Figure 3.2), m_e and b_e are the effective mass and the intrinsic damping of the cantilever and the tip, K is the spring constant, and F_0 and ω are the driving force's amplitude and angular frequency. m_i and b_i are the induced mass and damping due to the liquid loading, given by (3.8),

$$m_i = 3\pi r^2 \sqrt{\frac{2\eta\rho}{\omega}} \left(1 + \frac{2r}{9\delta}\right), \quad \text{and} \quad (3.11)$$

$$b_i = 6\pi\eta r \left(1 + \frac{r}{\delta}\right). \quad (3.12)$$

For sample liquids exhibiting high viscosities, and low vibration frequencies, r is negligible compared to δ , leading to a simplification of (3.11) and (3.12) [16]. For the considered cantilever tips, and dipping depths (Table 3.1) the effective sphere radius r is in the range of a few millimetres. The expected penetration depths δ for the sample liquids used in this work were calculated for a vibration frequency of 100 Hz, and are given in Table 3.2. The results show, that δ is in the same range as r , and the prerequisites for said simplification are not fulfilled.

Consequently, the consideration of the characteristic penetration depth δ leads to a frequency dependence of the liquid mass loading m_i , equation (3.11), and the liquid damping b_i , equation (3.12). In the following both m_i and b_i are considered constant in the vicinity of the resonance frequency, which is justified by the high quality factors, and the accompanied narrow bandwidths of the resonances. Therefore, solving the differential equation (3.10) using the Laplace transform yields

$$\psi(s) = \frac{1}{1 + \frac{2D_n}{\omega_n} s + \frac{1}{\omega_n^2} s^2} F(s), \quad (3.13)$$

where $\psi(s)$ and $F(s)$ are the Laplace transforms of the tip deflection $\psi(t)$ and the driving force $F(t)$,

$$\frac{1}{\omega_n^2} = \frac{m_e + m_i}{K}, \quad (3.14)$$

and

$$\frac{2D_n}{\omega_n} = \frac{b_e + b_i}{K}, \quad (3.15)$$

where ω_n is the resonance frequency and D_n the damping factor of the respective vibration mode n . From (3.8), (3.14), and (3.15), one obtains

$$\omega_n^2 = \omega_{n,\text{air}}^2 \frac{1}{1 + \frac{2\pi r^3}{3m_e} \rho + \frac{6\pi r^2}{\sqrt{2}m_e} \frac{1}{\sqrt{\omega_n}} \sqrt{\eta\rho}}, \quad (3.16)$$

and

$$\frac{2D_n}{\omega_n} = \frac{2D_{n,\text{air}}}{\omega_{n,\text{air}}} \left(1 + \frac{6\pi r}{b_e} \eta + \frac{6\pi r^2}{\sqrt{2}b_e} \sqrt{\omega_n} \sqrt{\eta\rho} \right), \quad (3.17)$$

where $\omega_{n,\text{air}}$ and $D_{n,\text{air}}$ are the resonance frequency and damping factor of the cantilever without liquid loading. The oscillating sphere model shows that the cantilever's resonance frequency is affected by a term related to the liquid density, and a second term containing the viscosity–density product, whereas the time constant $T_n = 2D_n/\omega_n$, representing a damping coefficient that has the dimension of time, depends on the viscosity and the viscosity–density product.

For the interpretation of the measurement results involving rectangular cross-sections having a more generalized model is advantageous. Based on (3.16) and (3.17) and by introducing four independent coefficients c_1 , c_2 , c_3 , and c_4 we have

$$\omega_n^2 = \omega_{n,\text{air}}^2 \frac{1}{1 + c_1 \rho + c_2 \frac{1}{\sqrt{\omega_n}} \sqrt{\eta\rho}}, \quad (3.18)$$

and

$$T_n = T_{n,\text{air}} (1 + c_3 \eta + c_4 \sqrt{\omega_n} \sqrt{\eta\rho}), \quad (3.19)$$

where $T_{n,\text{air}}$ is the in-air time constant $2D_{n,\text{air}}/\omega_{n,\text{air}}$. The values of the parameters c_k are determined by the tip size and geometry, the effective cantilever mass and damping, the dipping depth d (Figure 3.2b), and the respective mode of vibration n .

Figure 3.5 elucidates the model given by (3.18) and (3.19). We consider four liquids L_A , L_B , L_C , and L_D of which L_A and L_B are of the same density ρ_1 , whereas L_C and L_D are of the same viscosity η_3 . The expected values of ω_n and T_n are given by markers in the figure. The curves in the figure represent the results expected for liquids of the same viscosity and density, respectively.

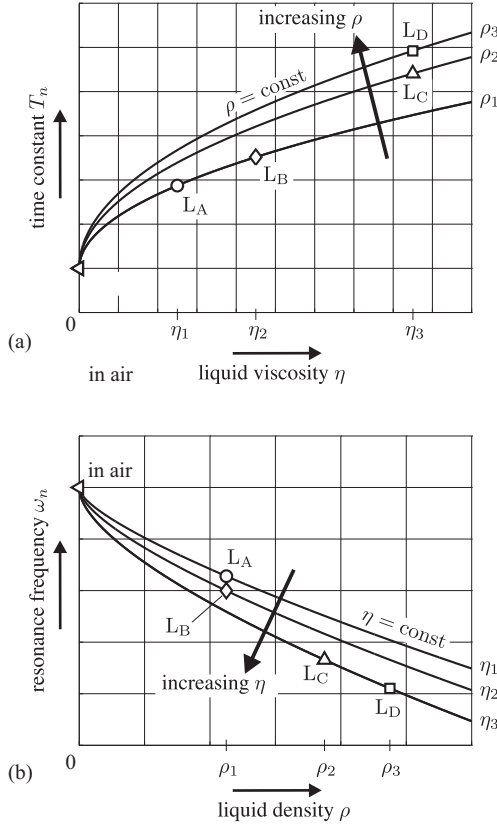


Figure 3.5: Time constant T_n and resonance frequency ω_n of a resonant cantilever dipping in 4 different sample liquids L_A , L_B , L_C , and L_D as given by the generalized model (3.18) and (3.19).

For the determination of viscosity and density of an unknown liquid (3.18) and (3.19) have to be solved for η and ρ . The system of equations can be written as

$$A = B\rho + C\sqrt{\eta\rho} \quad (3.20)$$

$$D = E\sqrt{\eta\rho} + F\eta, \quad (3.21)$$

where $A = \omega_{n,\text{air}}^2/\omega_n^2 - 1$, $B = c_1$, $C = c_2/\sqrt{\omega_n}$, $D = T_n/T_{n,\text{air}} - 1$, $E = c_4\sqrt{\omega_n}$, and $F = c_3$. Solving the system of equations yields

$$\begin{aligned} \rho_{1,2} = & -\frac{1}{2} \frac{ACE - 2ABF - C^2D}{B(BF - CE)} \\ & \pm \sqrt{\frac{1}{4} \frac{(ACE - 2ABF - C^2D)^2}{B^2(BF - CE)^2} - \frac{A^2F}{B(BF - CE)}}, \end{aligned} \quad (3.22)$$

and

$$\begin{aligned} \eta_{1,2} = & -\frac{1}{2} \frac{CDE - 2BDF - AE^2}{F(BF - CE)} \\ & \pm \sqrt{\frac{1}{4} \frac{(CDE - 2BDF - AE^2)^2}{F^2(BF - CE)^2} - \frac{BD^2}{F(BF - CE)}}. \end{aligned} \quad (3.23)$$

For the cantilevers used in this work it turns out that $BF - CE = c_1c_3 - c_2c_4 < 0$. Furthermore, all the coefficients of (3.20) and (3.21) are positive, i.e. $A, B, C, D, E, F > 0$. Therefore, in (3.22) and (3.23) only that $\rho_{1,2}$ and $\eta_{1,2}$, respectively, are positive which exhibit a plus sign before the root, i.e.

$$\begin{aligned} \rho = & -\frac{1}{2} \frac{(c_2c_4 - 2c_1c_3) \left(\frac{\omega_{n,\text{air}}^2}{\omega_n^2} - 1 \right) - \frac{c_2^2}{\omega_n} \left(\frac{T_n}{T_{n,\text{air}}} - 1 \right)}{c_1(c_1c_3 - c_2c_4)} + \\ & + \left\{ \frac{1}{4} \frac{\left[(c_2c_4 - 2c_1c_3) \left(\frac{\omega_{n,\text{air}}^2}{\omega_n^2} - 1 \right) - \frac{c_2^2}{\omega_n} \left(\frac{T_n}{T_{n,\text{air}}} - 1 \right) \right]^2}{c_1^2(c_1c_3 - c_2c_4)^2} - \right. \\ & \left. - \frac{c_3 \left(\frac{\omega_{n,\text{air}}^2}{\omega_n^2} - 1 \right)^2}{c_1(c_1c_3 - c_2c_4)} \right\}^{\frac{1}{2}}, \end{aligned} \quad (3.24)$$

and

$$\eta = -\frac{1}{2} \frac{(c_2 c_4 - 2c_1 c_3) \left(\frac{T_n}{T_{n,\text{air}}} - 1 \right) - c_4^2 \omega_n \left(\frac{\omega_{n,\text{air}}^2}{\omega_n^2} - 1 \right)}{c_3 (c_1 c_3 - c_2 c_4)} +$$

$$+ \left\{ \frac{1}{4} \frac{\left[(c_2 c_4 - 2c_1 c_3) \left(\frac{T_n}{T_{n,\text{air}}} - 1 \right) - c_4^2 \omega_n \left(\frac{\omega_{n,\text{air}}^2}{\omega_n^2} - 1 \right) \right]^2}{c_3^2 (c_1 c_3 - c_2 c_4)^2} - \right.$$

$$\left. - \frac{c_1 \left(\frac{T_n}{T_{n,\text{air}}} - 1 \right)^2}{c_3 (c_1 c_3 - c_2 c_4)} \right\}^{\frac{1}{2}}. \quad (3.25)$$

3.3 Measurements

With the setup depicted in Figure 3.2, the frequency response of the cantilevers is examined. The lock-in amplifier (Stanford Research SR830) is used to drive the cantilever and to measure the sensor voltage u_s and the phase shift $\phi(j\omega)$ between the driving voltage, i.e., the driving force, and the sensor voltage, i.e., the actual cantilever deflection. The measurements are carried out with a sinusoidal driving voltage of 200 mV rms and within a frequency range from 70 to 110 Hz. At the resonance frequency of the first mode a maximum tip deflection of 19 μm (peak–peak) was measured in air by means of a Polytec OFV-5000/OFV-505 laser vibrometer.

A variety of oils are used as test liquids: AK150, AK350 (from Wacker Chemie), and SIL300 are silicone oils, and Alcatel 120 (A120) oil. They exhibit liquid densities in the range from 881 to 1078 kg/m^3 and viscosities from 145 to 440 mPa·s. These nominal liquid parameters are obtained from data sheets and measurements by means of a Brookfield LVDV+II-CP cone/plate rheometer (Table 3.2). The parameters of the liquids vary with temperature. As especially the viscosity is highly temperature dependent, temperature control of the liquid container has been established by means of a Peltier heater/cooler system. All results presented here have been obtained at 23 °C.

The dipping depth (Figure 3.2b) was adjusted by lowering the sensor until the tip touches the liquid surface, and then adding the desired $d = 2 \text{ mm}$ using a micrometer screw. For liquids exhibiting low surface tensions, like those considered in this work, a concave meniscus is formed at the liquid–tip

Table 3.2: Reference values for dynamic viscosity η and mass density ρ of the sample liquids used in the measurements and the expected depth of penetration δ at an angular frequency of $\omega = 2\pi \cdot 100$ Hz, see (3.9). The dynamic viscosity was measured using a Brookfield LVDV+II-CP cone/plate rheometer at an ambient temperature of 23 °C.

Sample Liquid	Viscosity η [mPa·s]	Density ρ [kg/m ³]	δ (100 Hz) [mm]
AK150 silicone oil	130	965	0.65
AK350 silicone oil	440	979	1.2
SIL300 silicone oil	211	1078	0.79
Alcatel A120 oil	270	881	0.99

interface. The resulting liquid surface shape is expected to increase the effective dipping depth, and therefore to influence the sensor's behaviour. Consequently, the sensor principle would be limited to liquids with similar surface tensions, which can be assumed for the considered oils.

Figure 3.6 summarizes the measurement results at mode 1 for tip A in air and immersed in the sample liquids. As expected from (3.18) and (3.19), the liquid loading decreases the resonance frequency ω_1 and increases the damping factor D_1 .

From the cantilever's frequency response we extract the first mode resonance frequency $\omega_1 = 2\pi f_1$ and the damping factor D_1 by fitting a second order transfer function

$$\phi(j\omega) = \arg \left\{ \frac{1}{1 + j\omega \frac{2D_1}{\omega_1} - \frac{\omega^2}{\omega_1^2}} \right\} \quad (3.26)$$

to the phase shift measurements (Figure 3.6) with respect to the parameters ω_1 and D_1 .

Figure 3.7 and Figure 3.8 depict the results for the cantilevers with tips A, B, C, and D (Table 3.1) and the sample liquids (Table 3.2). The left columns of Figure 3.7 and Figure 3.8 show the measured time constant $T_1 = 2D_1/\omega_1$ versus the dynamic viscosity η of the sample liquids. The results for each cantilever appear to be nearly lying on a single trend curve, indicating that T_1 is dominantly influenced by the liquid's dynamic viscosity. According to model equation (3.19), the value of c_4 thus must be small compared to c_3 .

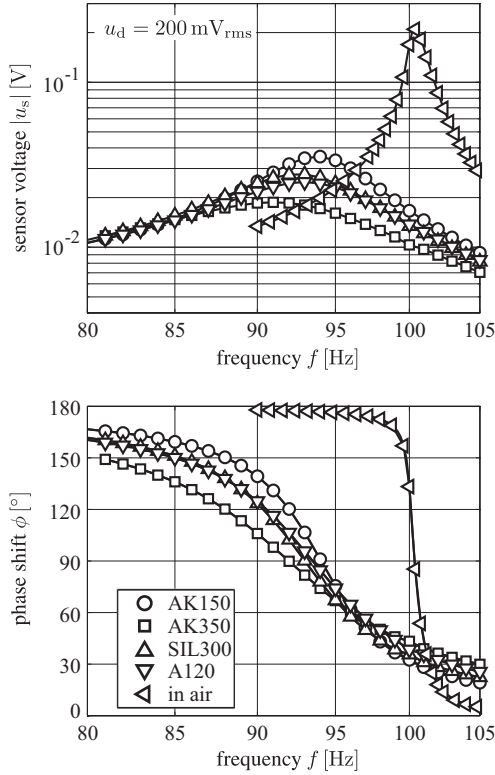


Figure 3.6: Magnitude and phase response of a cantilever with tip A vibrating in air and immersed in different liquids (Table 3.2). The dipping depth d is 2 mm, the driving voltage $V_d = 200 \text{ mV}_{\text{rms}}$.

The resonance frequency f_1 versus the liquid density can be seen in Figure 3.7 and Figure 3.8, right column. The results are widely spread in the ρ – f_1 plane. Obviously, the resonance frequency of the cantilever tip immersed in the liquid is strongly influenced by both the liquid's density and viscosity. In our model this relationship is described by (3.18).

Using the measurement results (Figure 3.7 and Figure 3.8), the model parameters c_1 , c_2 , c_3 , and c_4 can be extracted. Figure 3.9 illustrates the model equations (3.18) and (3.19) and the parameter fit procedure. The values of $\omega_{1,\text{air}}$ and $T_{1,\text{air}}$ are extracted from a single measurement of the vibrating

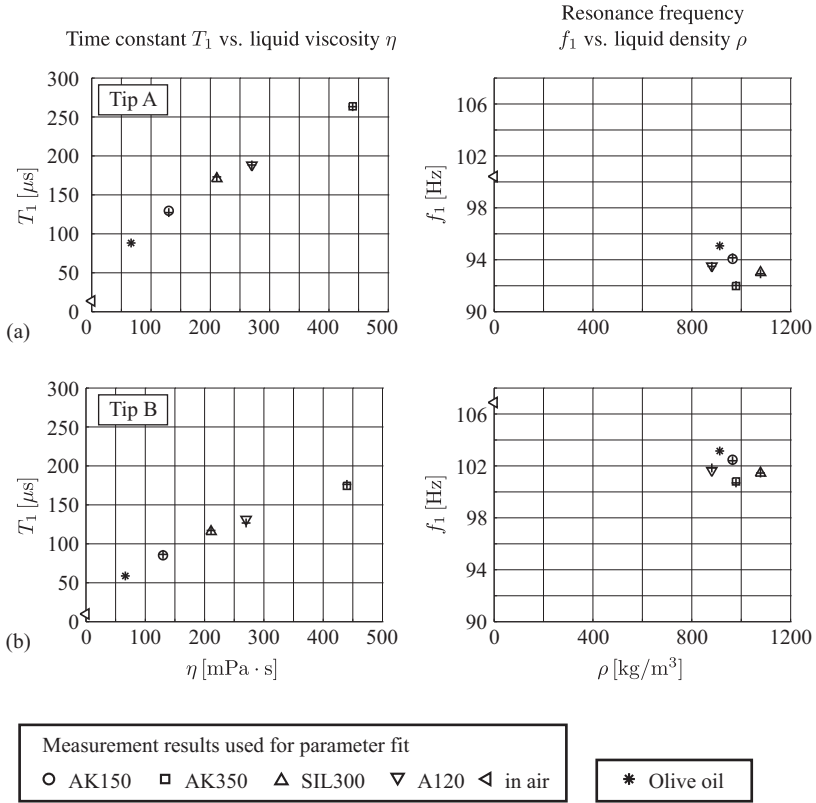


Figure 3.7: Resonance frequency ω_1 and time constant T_1 obtained for the tips A and B and a variety of sample liquids. The crosses (+) in the diagrams represent the calculated values obtained from the parameter fit and show good agreement between the experimental results and the model.

cantilever in air. Furthermore, each measurement in a sample liquid yields a pair of (ω_1, T_1) . The values of viscosity and density, η and ρ , of each sample liquid are also known (Table 3.2). Finally, a fit algorithm applied to the model equations results in the model parameters c_1 , c_2 , c_3 , and c_4 for a particular cantilever. The results of this parameter fit are depicted in Figure 3.10. The diagram shows the model parameters c_1 , c_2 , c_3 , c_4 for the cantilevers with different tips. It is important to note that the parameters do not only depend

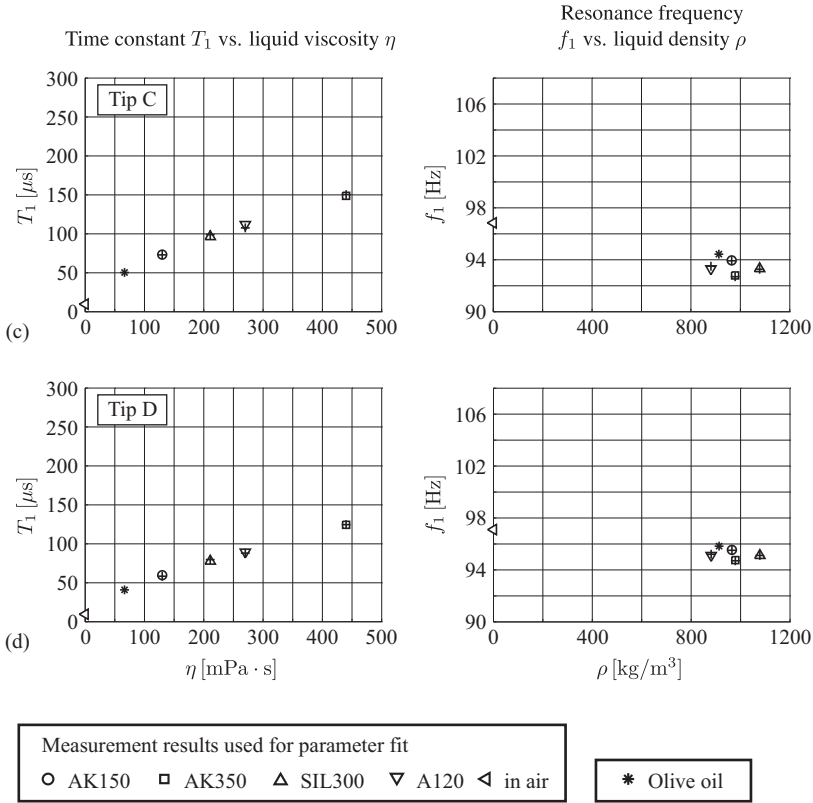


Figure 3.8: Resonance frequency ω_1 and time constant T_1 obtained for the tips C and D and a variety of sample liquids. The crosses (+) in the diagrams represent the calculated values obtained from the parameter fit and show good agreement between the experimental results and the model.

on the width of the attached tip, but also on the mass and the intrinsic damping of the entire cantilever. Therefore the parameter values do not necessarily decrease with the tip width, as shown by the location of the tip D marker in the c_3 – c_4 plane (Figure 3.10). The PZT bending actuator with tip D was not sealed by a protective coating, and thus features a lower intrinsic damping. However, the parameter fitting process described above yields parameter values considering such different behaviours of the cantilevers.

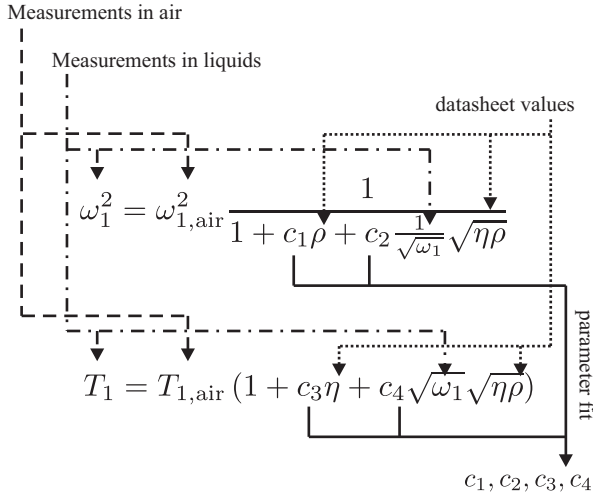


Figure 3.9: Obtaining the model parameters c_1 , c_2 , c_3 , and c_4 from measurements (Figure 3.7 and Figure 3.8) in air and in reference liquids with known viscosities and densities (Table 3.2).

For a validation of the model, the parameters determined above were used to calculate the values of ω_1 and T_1 for each tip-liquid combination. These calculated values are indicated by crosses (+) in Figure 3.7 and Figure 3.8, and are in nearly perfect agreement with the experimental data.

At last, we use the model parameters obtained above to determine the viscosity and density of an olive oil sample. The measurement results, i.e., the resonance frequencies ω_1 and the time constants T_1 of the cantilevers immersed in the sample liquid are given in Table 3.3. From these results the density and the viscosity of the liquid are calculated using (3.24) and (3.25), respectively.

3.4 Conclusion

The change of the dynamic behaviour of a vibrating cantilever allows to investigate the physical properties of liquids. Various types of small tips of different geometries are attached to the cantilever and immersed into the sample solutions. The liquid surrounding the cantilever tip changes both the reso-

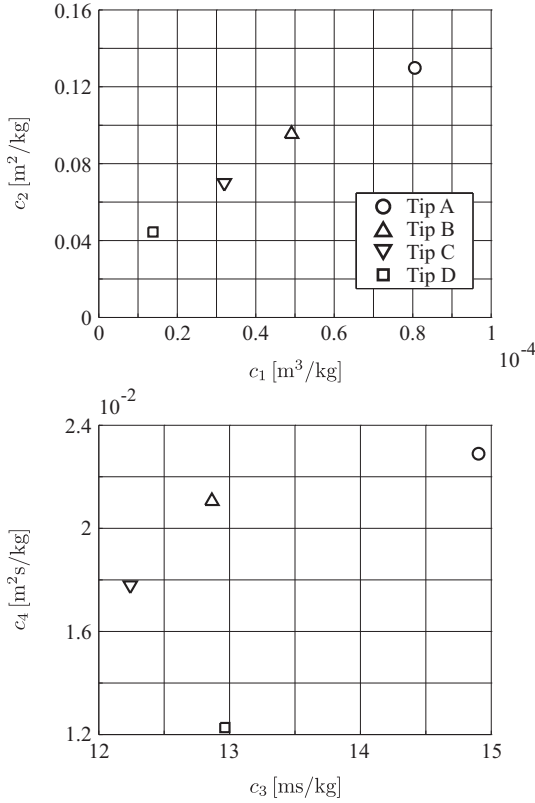


Figure 3.10: Fitted parameter values c_1 , c_2 , c_3 , and c_4 obtained by fitting the model equations to the measurement results.

nance frequency and the damping of the entire cantilever structure. To be able to conclude from the measured frequency response to the liquid's parameters an analytical model is needed. The developed model is based on the forces acting on an oscillating sphere in liquid, but generalized model parameters are used to consider the actual geometries of the applied cantilever tips. These parameters furthermore include the electrical and mechanical characteristics of the beam, which therefore need not be known. The model proved to be well-suited for the characterisation of various cantilevers and tip geometries by measuring in liquids with known density and viscosity. To extract

Table 3.3: Determination of viscosity and density of an olive oil sample by means of vibrating cantilevers. The density ρ and the viscosity η have been calculated from the measurement results $\omega_1 = 2\pi f_1$ and T_1 using (3.24) and (3.25). The density and viscosity obtained by means of weighting and a cone-plate rotational viscometer, respectively, are $\eta_{\text{ref}} = 66.4 \text{ mPa}\cdot\text{s}$ and $\rho_{\text{ref}} = 913 \text{ kg/m}^3$.

Cantilever	f_1 [Hz]	T_1 [μs]	Viscosity η [mPa $\cdot$ s]	Density ρ [kg/m 3]
Tip A	95.07	88.24	66.4	919.7
Tip B	103.15	58.80	62.9	921.4
Tip C	94.44	50.43	64.6	920.1
Tip D	95.83	40.88	67.6	908.6

the model parameters from the measured data a curve fitting procedure was performed. The obtained parameters are specific for each cantilever tip and allow the subsequent simultaneous determination of density and viscosity of unknown liquids.

Chapter 4

Micromachined Viscosity Sensors

It is the intention of this chapter to give an overview on viscosity sensors that are based on microelectromechanical systems (MEMS). Such micromachined sensors are an interesting alternative to other sensor systems and conventional laboratory equipment. The review section focuses on sensors that operate at lower frequencies than common thickness shear mode (TSM) resonators and surface acoustic wave (SAW) devices. Therefore, the sensors can potentially be used for the characterisation of non-Newtonian liquids, see the discussion in Chapter 2.

4.1 Motivation

The term *micromachining* can be defined as the use of “tools and techniques developed for the integrated circuit industry, such as microlithography, etching, etc., or some of the newer techniques developed specifically by and for the micromachining community” [31] to create micromechanical devices comprising small moving parts or fixed carrier structures. As base material, mostly silicon wafers or wafers that consist of layers of silicon, silicon oxide, and silicon nitride are used. A detailed introduction in the techniques, principles, and fabrication of MEMS is given in [31, 32].

Besides the properties that allow single crystal silicon to be used as material for semiconductor devices, it also features advantageous mechanical characteristics. Silicon is a nearly perfect Hookean solid (see Chapter 2) and shows a Young’s modulus, hardness, and tensile yield stress that are comparable to those of stainless steel [31]. The deformation of silicon is accompanied by low intrinsic losses. Therefore, resonators with high Q-factors can be fabricated.

For resonant structures, as needed for building oscillatory viscometers, a drive mechanism is required as well as a means for detecting the resulting vibration. Micromachining technology provides the possibility to realize different actuating principles like piezoelectric drives, electrostatic comb drives, Lorentz force actuation, and electrothermal drives. On the sensor side, materials can be deposited and structured by micromachining that readout a deformation by, e.g., the piezoelectric and piezoresistive effects, capacitive coupling, or induction. Furthermore, the presence of a semiconductor material allows the integration of the sensor and the readout circuitry on a single chip.

As the word *micro* in microelectromechanical system indicates, the smallest feature sizes of MEMS are typically several micrometres. In contrast the rather large oscillatory viscometer presented before in Chapter 3, miniaturized devices can be fabricated. This allows the integration of viscosity sensors in microfluidic platforms. Depending on the design and the dimensions of the device, MEMS resonators featuring resonance frequencies from below 1 kHz up to several hundreds of MHz can be fabricated. This fact makes micromachined sensors in particular interesting for the application as miniaturized viscosity sensors, for which the operating frequency must be carefully chosen (see Chapter 2).

A high number of devices are usually fabricated on a single wafer. Micromachining thus enables the production of sensors in parallel in large batches. However, the effort for packaging and testing, which often must be done for individual sensors, must be considered as well.

In summary, these considerations indicate that micromachining or MEMS technology allows the design and fabrication of miniaturized integrated viscosity sensors. The decision in favour of using micromachining technology was further supported by the availability of the required processes at the Institute of Sensor and Actuator Systems of the Vienna University of Technology (VUT), the Centre for Micro- and Nanostructures (ZMNS) of the VUT, and the Centre for Microtechnologies of the University of Applied Sciences Vorarlberg. In Chapter 5 and Chapter 6, two different types of MEMS viscosity sensors will be investigated.

4.2 Review of Viscosity Sensors

Below, an overview of micromachined devices for sensing the viscosity of liquids is given. In particular, devices and sensor principles are presented,

that allow for operation in the frequency range from about 1 to 500 kHz. As explained in Chapter 2, such sensors are of high interest as they measure a low-frequency viscosity parameter, that is potentially comparable to the steady-shear viscosity determined by laboratory viscometers. The sensors are therefore of high interest for examining non-Newtonian liquids.

Atomic force microscope cantilevers

Recently, a significant number of works were published on the measurement of viscosity with atomic force microscope (AFM) cantilevers. The dynamics of these probe beams surrounded by a fluid have been extensively studied. The obtained hydrodynamic models are of particular interest as they allow AFM imaging, e.g., of biological matter in its native medium, e.g., in liquid. Clearly, the results of these investigations can be employed to devise liquid properties from the mechanical behaviour of an AFM cantilever.

Oden *et al.* investigated the frequency spectra of AFM cantilevers immersed in liquids [11]. They used rectangular silicon bar cantilevers as well as two types of triangular cantilevers. The lengths of the cantilevers were between 85 and 225 μm . The vibrations of the beams were excited by thermal fluctuations and ambient noise. The small mechanical deflections were measured by an optical method commonly used in atomic force microscopes. Figure 4.1 depicts this readout principle. A laser beam is reflected off the tip of the cantilever. The cantilever deflection changes the angle of reflection of the laser beam which is directed to a position sensitive detector (PSD). The output signal of the PSD is then evaluated by a fast-Fourier transform (FFT) spectrum analyser. In air, the cantilevers exhibited resonance frequencies in the range from 70 to 100 kHz. Consequently, the frequency spectrum determined with the setup in Figure 4.1 showed amplitude peaks at these frequencies. When the AFM cantilevers were immersed in a liquid the resonance frequency as well as the peak amplitude decreased due to the additional mass loading and damping. As sample liquids, glycerol/water mixtures of varying concentrations were used. Their viscosities were between 1 and 150 mPa·s. The resulting resonances frequencies and quality factors were in the range from 3 to 50 kHz and from 0.2 to 8, respectively. For the interpretation of the results, an oscillating sphere modelling approach was employed [34]. The beam vibrating in a liquid environment was approximated by a sphere performing translational oscillations in the liquid. Whereas the modelling fitted the measurement results quite well in the viscosity range from 1 to 10 mPa·s, high deviations were noticed for more viscous liquids.

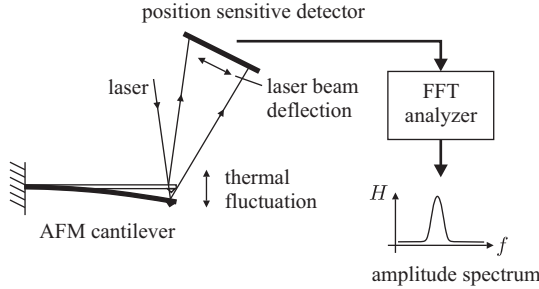


Figure 4.1: An AFM cantilever utilized for the measurement of viscosity and density. The cantilever is actuated by thermal fluctuation and ambient vibrations. The amplitude spectrum of the beam is then obtained from an optical readout. A laser beam is reflected at the back of the cantilever. Depending on the mechanical deflection, the reflection angle of the laser beam changes. This change is detected by a position sensitive detector, whose output signal is subsequently evaluated by an FFT spectrum analyser [11, 33]

A remarkable work is the hydrodynamic model of John E. Sader [25]. This model aims at describing the frequency spectrum of an AFM cantilever in a setup similar to Figure 4.1. The liquid surrounding the beam is considered by an additional mass loading and an additional damping of the cantilever. Both this mass and this damping can be calculated from a hydrodynamic function $\Gamma(\omega)$, where ω is the angular frequency of the vibration. The hydrodynamic function of an infinitely long beam that is circular in cross-section is well known [35], and is given by

$$\Gamma_{\text{circ}} = 1 + \frac{4jK_1(-j\sqrt{j\text{Re}})}{\sqrt{j\text{Re}}K_0(-j\sqrt{j\text{Re}})}, \quad (4.1)$$

where $\text{Re} = \rho\omega h^2/(4\eta)$, and K_0 and K_1 are modified Bessel functions of the second kind. η and ρ are the dynamic viscosity and the mass density of the liquid, respectively. For a beam with a rectangular cross-section, the exact solution of the hydrodynamic function is known, but its evaluation takes a significant amount of numerical calculation [25]. Therefore, Sader devised a correction function $\Omega(\omega)$ and expressed the hydrodynamic function of the AFM cantilevers using the solution for circular beams as

$$\Gamma_{\text{rect}} = \Omega(\omega)\Gamma_{\text{circ}}(\omega). \quad (4.2)$$

The fundamental assumptions of the model are: The beam has a uniform rectangular cross-section over its entire length, the length greatly exceeds the width of the beam, the thickness of the beam is negligible compared to its width, and the fluid is incompressible [12]. Using the mass loading and the additional damping due to the liquid surrounding cantilever, the amplitude spectrum $H(j\omega)$ (Figure 4.1) of the cantilever is calculated. In the literature, this model is often referred to as *Sader's model*.

In the limit of small dissipative effects a resonance of the AFM cantilever can be approximated by a simple harmonic oscillator (SHO). In [25], Sader also presents such an approximation. The characteristics of the SHO are defined by the resonance frequency f_0 and the quality factor Q , which are both calculated using the hydrodynamic function Γ_{rect} . The model based on this simplification is often called the *SHO model*.

Chon *et al.* presented an experimental validation of Sader's calculations [12]. They use two sets of AFM cantilevers. One set features ideal rectangular beam geometries, whereas the other cantilevers have some irregularities, e.g., imaging tips at their ends. Below some results of the ideal cantilevers will be given. The beams were 200 and 400 μm long, 20 μm wide, and 440 nm thick. The sample liquids acetone, carbon tetrachloride (CCl_4), water, and 1-butanol exhibit viscosities in the range from 0.308 to 2.47 mPa·s. Immersed in these liquids, the resulting resonance frequencies and quality factors were in the range from 6 to 40 kHz (first mode of vibration) and from 1.2 to 5.1, respectively. Chon *et al.* compare the experimental results with those of Sader's complete model and the SHO model. The deviation of the calculated resonance frequencies and quality factors from the experimental ones were below 10 % for both Sader's full model and the SHO model.

Another experimental verification of Sader's model was given by Bergaud and Nicu [36]. They used cantilevers which are comparable in size to Chon's cantilevers. These cantilevers were 40 μm wide and from 100 to 300 μm long (in 50 μm steps). The mechanical vibration was excited by shaking the sensor device. The fluidic cell containing the sample liquid and the cantilever was mounted on a piezoelectric disc actuator. An optical readout similar to the setup in Figure 4.1 was utilized to detect the beam vibrations. The resonance frequencies of the cantilevers immersed in water and ethanol were observed. The results were below 30 kHz and 160 kHz for the first and second mode of vibration, respectively. Good agreement is shown between the frequencies obtained from the experiment and the results of the SHO model. The error never exceeded 8 % for the longer cantilevers. However, for the 100 μm and the 150 μm long cantilevers, the error was up to nearly 20 %. This observation

agrees with [12], where also shorter cantilevers were investigated and high deviations were found for length-to-width ratios l/w less than 4.

Boskovic *et al.* demonstrate the simultaneous measurement of the viscosity and the mass density of a liquid with AFM cantilevers [13]. Like Bergaud and Nicu [36], they employ the SHO model. In a setup similar to Figure 4.1 they measured the frequency spectrum of an AFM cantilever. By curve-fitting the model equations to the spectrum the quality factor Q and the resonance frequency f_0 of the fundamental mode of vibration were obtained. Subsequently, these results were used to determine the viscosity and mass density of liquids and gases. Unknown model parameters were obtained from a calibration measurement in vacuum. The size of the cantilever was $397 \times 29 \times 2 \mu\text{m}^3$. As sample fluids, gases and the liquids acetone, carbon tetrachloride (CCl_4), water, and 1-butanol were used. These liquids have viscosities from 0.31 to 2.5 mPa·s and mass densities between 790 and 1590 kg/m³. The resulting resonance frequencies and quality factors were in the range from 4.51 to 6.55 kHz and from 1.2 to 2.9, respectively. Both the viscosity and the density of the sample liquids could be measured simultaneously with the sensor. Boskovic *et al.* report deviations of the results from the reference values below 14 %.

The evaluation of Sader's model equation [25] requires elaborate calculations. Maali *et al.* [37] devised simplified expressions which represent a good approximation of Sader's hydrodynamic function for the geometries and operation frequencies that are typical for AFM cantilevers. This model will be applied to a clamped-clamped micromachined beam in Section 5.6.

The main disadvantage of typical setups using AFM cantilevers is the optical readout. This readout must be highly sensitive to detect the small deflection amplitudes. Most groups, therefore, used the expensive equipment of an atomic force microscope. Hence, they could demonstrate the measurement principle and examine the validity of their models, but in order to obtain an integrated sensor system an other means of deflection sensing must be devised. In Chapter 5 such a low-cost setup based on the optical laser pickup head of DVD players is presented. In [38], Hennemeyer *et al.* developed a low-cost system based on the readout principle of Figure 4.1 and demonstrate its feasibility for the characterisation of sugar solutions.

Lorentz force actuated plate

Most sensor designs based on AFM cantilever rely on Brownian motion as source of the beam vibrations. The resulting deflection amplitudes are in

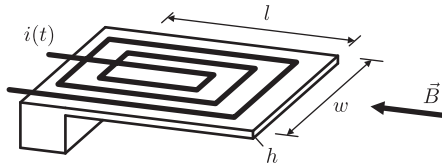


Figure 4.2: Lorentz force actuated plate sensor. A plate fabricated from monocrystalline silicon is deflected by the forces arising from the excitation current $i(t)$ and a magnetic field (flux density $\vec{B}$). The deflection amplitude is measured by piezoresistive elements (not shown in the sketch) [14].

the femtometre range [11]. A highly sensitive optical readout, in particular that of an atomic force microscope, is required to determine the frequency and the quality factor of the beam resonance. It is, therefore, advantageous to integrate an excitation mechanism on the beam. Hence, higher vibration amplitudes can be achieved. Furthermore, the sensor electronics gains control over the excitation force, which allows to utilize nonlinear effects [39] or to evaluate both the vibration amplitude and the phase shift between excitation force and beam deflection.

Goodwin *et al.* demonstrated the feasibility of a micromachined cantilever for the measurement of gases at high pressures [14]. The operating principle of their sensor is depicted in Figure 4.2. The vibrating part is a rectangular plate connected to a support along one edge. The plate is rather large compared to the AFM cantilevers described above. It features dimensions of $w = 2$ mm, $l = 1.5$ mm, and $h = 20$ μ m and was fabricated from monocrystalline silicon. Onto this plate, aluminium was deposited and structured to form a coil. For excitation of the structure, an ac current of 1 mA was driven through the coil and the device was placed in a magnetic field (flux density 100 mT). To detect the vibrations boron-doped polycrystalline silicon resistors were deposited on the plate surface. In vacuum, the resonator is characterised by a resonance frequency of 12 kHz and a Q-factor of 2800. The sensor was intended to be used in boreholes to investigate the physical properties of hydrocarbon reservoir fluids. Typically, the temperature of these fluids is up to 150 °C and the pressure ranges up to 200 MPa [14]. Goodwin *et al.*, therefore, present experimental results of the sensor in argon at temperatures between 323 and 423 K and pressures from 7 to 68 MPa. The viscosity of argon under these conditions varies between 26 and 67 μ Pa·s, its mass density is in the range from 79 to 767 kg/m³. Due to the low viscosity of the fluid, high quality

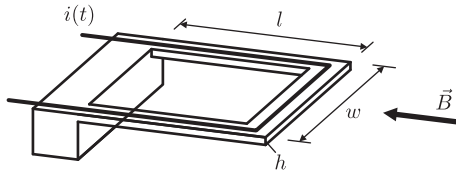


Figure 4.3: A U-shaped cantilever driven by Lorentz forces arising from the excitation current $i(t)$ and the magnetic field $\vec{B}$ [10]. An optical reflective sensor (not shown in the drawing) detects the deflection of the structure.

factors of the resonance from 74 to 84 are achieved. The resonance frequency is in the range from 3.5 to 8 kHz. Other experiments using the same sensor were presented in [40] with fluids showing viscosities in the range from 0.205 to 0.711 mPa·s and densities varying from 619 to 890 kg/m³. The resulting resonance frequency was between 3040 and 3580 Hz and quality factors from 22 to 36 were measured. Due to its geometries, the vibrating plate does not fulfil the preconditions for Sader's model. A suitable model for the sensor has been devised by Atkinson and Manrique de Lara [27].

A U-shaped cantilever with dimensions comparable to the rectangular plate described before was presented by Agoston *et al.* [10]. The sensor geometry is shown in Figure 4.3. The vibrations of the cantilever are excited by Lorentz forces arising from the field of a permanent magnet and a sinusoidal excitation current. The length of the cantilever is $l = 1500 \mu\text{m}$, the width is $w = 1100 \mu\text{m}$, and $h = 15 \mu\text{m}$. The beam deflection is detected by an optical reflective sensor. The resonance frequency of the device in air and immersed in liquid was in the range from 5 to 8 kHz. In their contribution, the authors compare the measurement results of a conventional laboratory viscometer (rotational type, see Section 2.2), a TSM quartz resonator (resonance frequency of several MHz), and the U-shaped cantilever. As sample liquids, different mixtures of diesel fuel and engine oil were used. One of the liquids contained a high molecular weight polymer additive. Hence, it exhibited a highly non-Newtonian rheological behaviour, whereas the other fluids behaved Newtonian in the measurements. In Chapter 2 it was indicated that due to the high operation frequency and the low penetration depth associated with the quartz sensor the results obtained for non-Newtonian liquids potentially deviate from those of conventional viscometry. In contrast, a sensor operating at far lower frequencies will obtain a viscosity parameter that is close to the steady shear viscosity measured by the rotational viscometer [10].

Measurement of the viscoelastic behaviour

When AFM cantilevers are utilized for the measurement of fluid properties, they are often excited by Brownian motion or ambient noise [11]. Consequently, both the magnitude and the phasing of the force driving the beam are unknown. The measurement of the beam deflection is limited to the spectrum of the deflection magnitude, from which the resonance frequency and the quality factor can be extracted. In contrast, employing a dedicated excitation mechanism allows obtaining the entire frequency characteristic of the device, i.e., deflection magnitude and phase shift between excitation force and deflection as a function of frequency. Belmiloud *et al.* presented measurements using cantilevers which are excited by Lorentz forces [24,41]. The cantilever deflection is detected by a setup similar to Figure 4.1. The cantilevers were between 18 and 70 μm thick, between 200 and 600 μm wide, and between 3 and 4 mm long. As the setup allowed the measurement of both the deflection magnitude and the phase shift between the excitation force and beam deflection the experiments yielded a complex transfer function $H(j\omega)$. This transfer function was then evaluated separately at each frequency using Sader's model [25] (with the simplifications of Maali *et al.* [37]). Hence, a frequency dependent viscosity $\eta(j\omega)$, corresponding to the loss modulus $G''(j\omega)$, and the storage modulus $G'(j\omega)$ were obtained from the measurements (Section 2.3). Therefore, the viscosity sensor yields not only a single viscosity parameter but the entire frequency characteristic $\bar{G}(j\omega) = G' + jG''$ of the liquid.

Clamped–clamped beam sensors

In Chapter 5 a viscosity sensor based on a clamped–clamped silicon nitride beam is presented. The device utilizes Lorentz force excitation and optical readout.

A related setup was recently published by Etchart *et al.* [15]. Figure 4.4 depicts the vibrating structure of the viscosity sensor. The vibrating beams were made from a silicon-on-insulator (SOI) wafer featuring a device layer thickness of 20 μm . Into a membrane formed by the device layer holes were etched to create a silicon beam that is rigidly clamped on both sides. The beam thickness is, therefore, 20 μm . On top of the beam, an electrical conductor carries the excitation current. Different beams which are 30 and 50 μm wide and 1.5, 2, 3, and 5 mm long were manufactured. The vibrating structures are therefore much longer and much thicker than those presented in

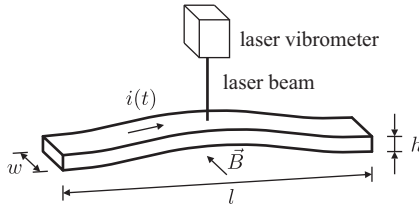


Figure 4.4: Clamped-clamped beam sensor principle used in [15]. The vibrations of the silicon beam are excited by Lorentz forces. An electrical conductor on top of the beam carries the excitation current $i(t)$. In a magnetic field, the forces on the conductor lead to a deflection of the beam. The vibration amplitude and phasing are detected by a commercially available laser vibrometer.

Chapter 5. These dimensions result in a high rigidity and low vibration amplitudes, but are associated with a high quality factor of the beam. However, a highly sensitive readout is required. In particular, a laser vibrometer was used in the experiments. For the 5 mm long beam surrounded by liquids with a viscosity from 0.2 to 100 mPa·s, Etchart *et al.* obtained quality factors between 2 and 11.

A semi-analytical model that describes the motion of a clamped–clamped cantilever in liquid was devised by Weiss *et al.* [26]. The Euler-Bernoulli beam equation is applied to the vibrating structure. The additional mass loading and the dissipative effects due to the liquid surrounding the sensor are calculated by assuming that the flow only occurs in a plane perpendicular to the beam. In this plane, the force acting on the beam is obtained by linearizing the Navier-Stokes equations and using a spectral domain/method of moments approach. In contrast to the work of Sader [25], the model also considers compressible liquids.

Membrane resonators

Brand *et al.* presented a membrane resonator for the monitoring of polymerization processes [4]. The square membranes are fabricated from silicon and silicon nitride and feature side lengths between 1 and 3 mm. Transverse vibrations of the resonator are excited by electrothermal excitation. A Wheatstone bridge of piezoresistors detects the actual deflection of the membrane. The resonance frequency in air was below 10 kHz. Immersed in samples of PDMS (polydimethylsiloxane), the resonance frequency decreased to approximately

1 kHz. The sensor showed high quality factors of up to 45 even in a liquid environment ($\eta \approx 1 \text{ mPa}\cdot\text{s}$).

Another membrane based viscosity sensor was reported by Martin *et al.* [42]. The flexural plate wave resonator shown in that work was excited by Lorentz forces. A meander-shaped transducers was utilized for the excitation. The plate vibrations were detected by measuring either the device impedance or the voltage induced at a second (output) transducer. The resonance frequency of the device was in the range of several hundreds of kHz. In [42] experimental results of the sensor in gaseous environments (viscosities below $0.22 \text{ mPa}\cdot\text{s}$ and mass densities below 1.8 kg/m^3) are shown.

In-Plane Mode Resonator Platform

In a recent article Seo and Brand present a disc resonator that operates in a rotational in-plane mode [43]. A schematic drawing of the resonator is depicted in Figure 4.5a. The design was intended to be used as a platform for biological and chemical sensing applications. For such purposes, resonators like SAW devices, TSM resonators, or cantilevers are often used and coated with a sensitive layer. By absorption of analytes in this layer or binding of biomolecules to them, the moving mass is increased. Hence, the resonance frequency of the sensor decreases. This change is then detected by a read-out circuit. In order to improve the sensitivity, a high frequency stability and, therefore, a high Q-factor are desired. In a liquid environment TSM and SAW devices are known to excite mainly shear waves. As the penetration depth of these waves is small, the damping is kept low. In contrast, the resonance of cantilever sensors are highly damped, deteriorating the quality factor. The device of Seo and Brand is based on a disc operating in a rotational mode of vibration. Like the TSM resonator, such a sensor excites mainly shear waves at its surfaces and thus features a high Q-factor. For reasons of suspension, excitation, and readout, the resonating part of the sensor actually is not a disk, but two semicircular disks supported by two anchor beams. The anchors carry heating resistors for thermal excitation and piezoresistive elements for detection of the disc deflection. The heating resistors are placed as such that mainly rotational modes of vibration are excited (Figure 4.5b and 4.5c). Although it was not explicitly shown by the authors, such the device could clearly be used as a sensor for the viscosity of liquids, similarly to the TSM resonator. Immersed in water ($\eta \approx 1 \text{ mPa}\cdot\text{s}$, $\rho \approx 1000 \text{ kg/m}^3$) the presented device featured a resonance frequency of approximately 580 kHz and a high Q-factor of 94.

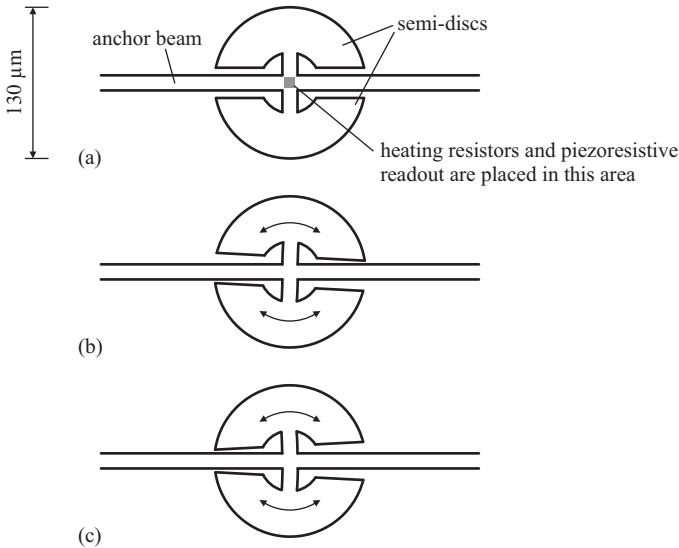


Figure 4.5: Semi-disk resonating in a rotational mode of vibration [43]. The sensor vibrations are excited by heater resistors at the joint between the semi-discs and the anchor beam. For readout of the deflection, the sensor is equipped with a Wheatstone circuit of piezoresistive elements. (a) Sensor principle. (b) First mode of in-plane vibration. (c) Second mode of in-plane vibration.

4.3 Conclusion

Micromachining is a suitable technology for the fabrication of miniaturized viscosity sensors. In the literature several designs have been presented. These sensors apply oscillatory motion. They are based on vibrating cantilevers, beams, plates, membranes, and disks.

For the excitation of the vibration, some sensors, in particular AFM cantilevers, rely on thermal fluctuations [11–13]. However, most devices that were specifically designed for use as viscosity sensors were equipped with actuators based on, e.g., Lorentz forces or thermal expansion [14, 15].

However, for the detection of the actual deflection, most sensor readouts use optical methods. Some devices, e.g., the AFM cantilevers presented in [11–13], are excited merely by thermal fluctuation. Their vibration amplitudes are in the sub-nanometer range, and a highly sensitive readout is

required. The optical systems of atomic force microscopes were utilized, yielding good results and a proof of principles of the sensor, but clearly can not be used in a mass fabricated sensor system. Driving the resonators by a designated actuation mechanism results in higher mechanical amplitudes and enables the use of low-cost optical systems. Such a system based on the laser pickup head of a DVD player is presented in Chapter 5. Still, this pickup head is an external component, and the sensor system is not a fully integrated one. Therefore, other means of measuring the MEMS deflection must be developed and integrated on the sensor chip [14, 43].

Another aspect is the quality factor associated with the operation of the sensors in liquid. A high quality factor increases the vibration amplitude at resonance frequency. Consequently, the principle utilized to determine the deflection of the vibrating structure may be less sensitive. Also simple read-out circuits can be used, e.g., oscillators using the sensor as the frequency determining element. In the viscosity range from 1 to 150 mPa·s AFM cantilevers were shown to exhibit a Q-factor between 0.2 and 8, which is rather low compared to their quality factor in air being in the range of several hundreds [11–13]. To increase the quality factor of the mechanical resonance of MEMS devices, one must ensure that mainly shear waves are excited in the liquid surrounding the sensor. In Chapter 6, a plate resonator is presented, that is specifically designed as a viscosity sensor and also aims at exciting mainly shear waves. Furthermore, piezoresistive elements are integrated on the sensor chip. Hence, no external optical readout is required.

Chapter 5

Micromachined Clamped–Clamped Beam Sensor

In Chapter 4 of this thesis, several reasons for the use of microelectromechanical (MEMS) devices for the measurement of the viscosity and mass density of liquids are presented. Below, these considerations are verified by examining a MEMS sensor based on a thin long clamped–clamped silicon nitride beam. Figure 5.1 depicts a schematic of this device. To circumvent the low vibration amplitudes of AFM (atomic force microscope) cantilevers (Chapter 4), this device is operated at forced vibration by Lorentz forces. This force $\vec{F}(t)$ arises from a sinusoidal current $i(t)$ through a conductor on the beam and the magnetic field $\vec{B}$ of an external permanent magnet. A low-cost optical readout using the laser pickup head of a DVD player eliminates the need for an AFM setup or laser vibrometers.

Similar to the principle of other oscillatory viscometers, the liquid surrounding the vibrating beam influences its resonance frequency and its damping factor. In principle, the evaluation of these parameters allows the determination of both the viscosity and the density of the liquid [13].

The sensor operates at frequencies of several tens of kHz. According to Chapter 2 the sensor yields, therefore, a viscosity parameter that is better comparable to the steady shear viscosity determined by conventional laboratory viscometers than the results of high frequency resonators, which is confirmed below for a set of SiO₂-in-water suspensions.

The sensor principle, the measurement setup, and the experimental results presented in this chapter were partly published in [44–47].

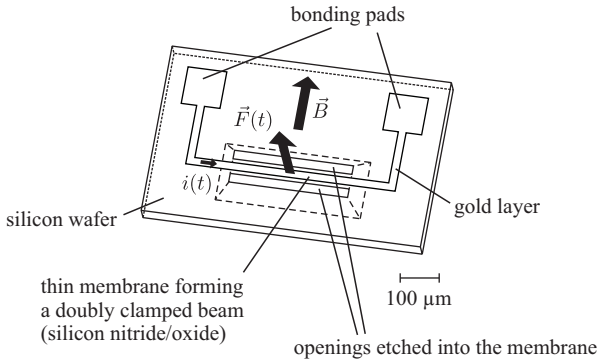


Figure 5.1: Clamped–clamped beam device featuring a micromachined silicon nitride (Si_3N_4 , SiN_x) beam and a conductive path (gold layer) allowing for Lorentz force excitation. The beam has a length of $350\text{ }\mu\text{m}$, a width of $40\text{ }\mu\text{m}$, and a thickness of $1.3\text{ }\mu\text{m}$.

5.1 Sensor Fabrication

The base material for the clamped–clamped beam sensor was a $350\text{ }\mu\text{m}$ thick, (100) oriented, 4 inch silicon wafer. The wafer was coated with a stack of 250 nm silicon oxide and 70 nm LPCVD silicon nitride on both sides (Figure 5.2a). A Ti–Au–Cr layer ($500\text{ }\text{\AA}$ titanium, $1000\text{ }\text{\AA}$ gold, and $500\text{ }\text{\AA}$ chromium) was deposited and structured to form the conductive path over the clamped–clamped beam and the bonding pads (Figure 5.2b). Then, a low stress silicon nitride protective film of 1000 nm was applied using a low temperature PECVD process (Figure 5.2c). After creating rectangular apertures in the wafer backside coating by means of reactive ion etching (Figure 5.2d), a thin membrane was manufactured by anisotropic KOH wet etching (Figure 5.2e) from the backside. In order to obtain clamped–clamped beams, the membrane was subsequently structured from the front side by reactive ion etching (Figure 5.2f). At the same time, the apertures for the bond pads were created. Finally, the chromium was removed from the bonding pads by wet-etching. The actual fabrication process also included the deposition of additional chromium and germanium layers needed for thermal sensors, that were fabricated on the same wafer. A detailed description including the process parameters can be found in [2]. Finally, the sensor devices were die-bonded to a printed circuit board (PCB) which provides the electrical connections. The

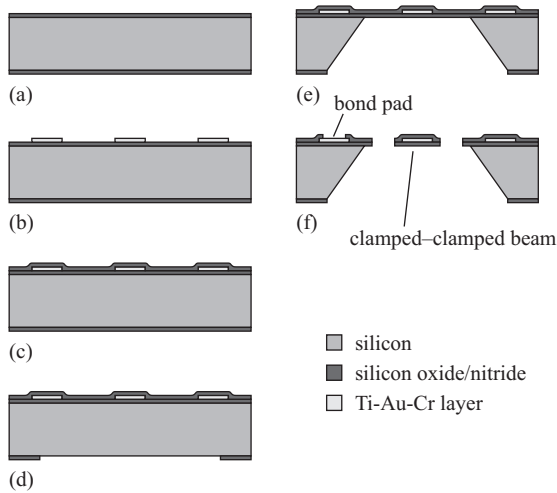


Figure 5.2: Simplified schematic of the fabrication process of the clamped-clamped beam device.

chip and the PCB were electrically connected by wire-bonding. The bond wires were protected by an epoxy compound. The PCB with the sensor chip was then mounted on a device holder to prevent vibrations of the entire device. Figure 5.3 is a photo of the sensor and this device holder. Figure 5.4 shows the dimensions of the fabricated device. Beams with different lengths l varying from 240 to 720 μm were fabricated. The thickness of the beam was $h = 1.3 \mu\text{m}$.

5.2 Optical Readout

To obtain the resonance frequency and the damping of a micromachined structure it is required to determine the beam deflection. This measurement could be achieved by providing, e.g., piezoelectric or piezoresistive layers on the beam, or designing structures for capacitive readout. For the beam excited by Lorentz forces also an inductive readout could be utilized due to the presence of a permanent magnetic field. The beam represents a conductive layer moving in a magnetic field with the flux density $\vec{B}$. The voltage thereby induced leads to a change of the effective impedance of the beam

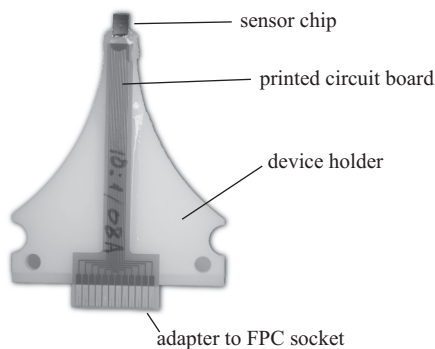


Figure 5.3: The clamped–clamped beam device die-bonded to the PCB that provides the electrical connections. The device holder ensures that the sensor chip is rigidly clamped and does not move during the measurements using the optical readout. The PCB is designed to fit into a FPC (flexible printed circuit board) socket.

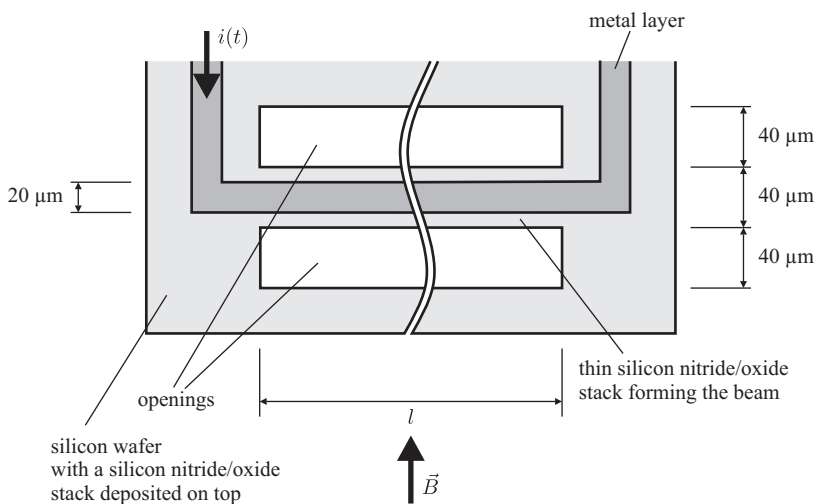


Figure 5.4: Dimensions of the fabricated clamped–clamped beam devices. Devices with lengths l in the range from 240 to 720 μm were fabricated.

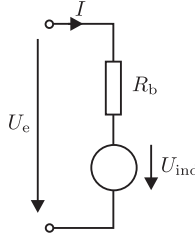


Figure 5.5: Lumped element equivalent circuit for an inductive readout of the vibrating beam. R_b is the resistance of the beam conductor and is approximately 75Ω . By moving the beam in the magnetic field, a voltage U_{ind} is induced in the conductor. In theory, this induced voltage could be measured at the terminals, but as it is several orders of magnitude smaller than the excitation voltage U_e the readout principle is hardly feasible.

(Figure 5.5). For the considered design, the vibration amplitude in air at a resonance frequency of $f_0 = 70 \text{ kHz}$ is $\hat{\psi} = 400 \text{ nm}$. The length of the beam is $l = 400 \mu\text{m}$, the magnetic flux density is $B = 200 \text{ mT}$, and the excitation current is $i(t) = Ie^{j\omega t}$, $I = 3 \text{ mA}$. Therefore, the beam deflection in complex notation is

$$\psi = KIe^{j\phi}, \quad (5.1)$$

where $K = \hat{\psi}/I = 133 \cdot 10^{-6} \text{ m/A}$, and ϕ is the phase shift between excitation current and beam deflection. For the complex amplitude of the induced voltage in the beam we have

$$U_{\text{ind}} = j\omega B l K I e^{j\phi} = 14 \mu\text{V} e^{j\phi}. \quad (5.2)$$

Given a typical electrical resistance of the beam $R_b = 75 \Omega$ (Figure 5.5), the voltage drop at the beam resistance is 225 mV at 3 mA . The excitation voltage U_e is, therefore, several orders of magnitude higher than the induced voltage U_{ind} . The measurement of this voltage would require measurement equipment with a very high dynamic range [44]. When immersed in liquids, the vibration amplitude and thus the induced voltage decreases drastically. Consequently, the measurement results suffer from a low signal-to-noise ratio, and other sensing principles must be considered.

Optical readout methods have been successfully used for micromachined sensors in liquids. A reflective sensor and the evaluation of the signals by a synchronous demodulator [10] offer a cheap means for the measurement of

micromachined beam deflection. However, for the small deflection amplitudes of the clamped–clamped beam, a more sensitive readout is required. In several works, the optical readout of atomic force microscopes (AFM) has been used [11], but this approach is only feasible for laboratory experiments. The optical pickup heads of CD and DVD drives offer a highly integrated and low-cost alternative [48]. For our experiments, the Sanyo SF-HD68V pickup head of a GDR-8164B DVD-ROM drive from LG Electronics, Inc. (November 2005 series) was used. It features two laser diodes with wave lengths of 780 nm (infrared, for reading CDs) and 650 nm (red, normally used for reading DVDs). In the setup, the 650 nm laser was used, since the absorption of red light in liquid is much lower, and the manual adjustment of the laser to the centre of the vibrating beam is much easier using a wavelength in the visible range [48].

Figure 5.6 depicts a schematic of the optical readout system. For a detailed explanation of the operating principles of laser pickup heads the reader is referred to [49, 50]. The laser power is controlled by an automatic power control (APC) circuit. This APC uses a photodiode situated next to the laser diode to measure the intensity and adjusts the laser diode current such that the output power is constant. Using a polarization filter, a beam splitter, and a lens, the laser beam is focused on the moving surface of a MEMS device. The reflected beam is projected to a 4-quadrant photo diode array. The optical system causes a distortion of the beam shape. If the reflecting surface coincides with the focal plane of the pickup lens, i.e., $d = d_0$, where d is the distance between the lens and the surface, and d_0 is the distance between the lens and the focal plane, the beam shape appears as a circle on the 4-quadrant detector. If the position of the moving surface does not match the focal plane of the optical system, the beam shape becomes elliptic. The orientation and length of the ellipse's major axis depends on the d to d_0 ratio (Figure 5.6). From the photo detector's output voltages u_A , u_B , u_C , and u_D , the focus error

$$\text{FE} = (u_A + u_C) - (u_B + u_D) \quad (5.3)$$

signal is obtained using a circuit of operational amplifiers. Figure 5.7 illustrates the relation between FE and d according to the theory. In a CD/DVD drive, this focus error signal is used to keep the laser beam focused on the CD or DVD: A focus controller adjusts the vertical position of the lens using a voice coil motor, and aims at keeping the focus error FE zero. Hence, a circular beam shape and $d = d_0$ is maintained.

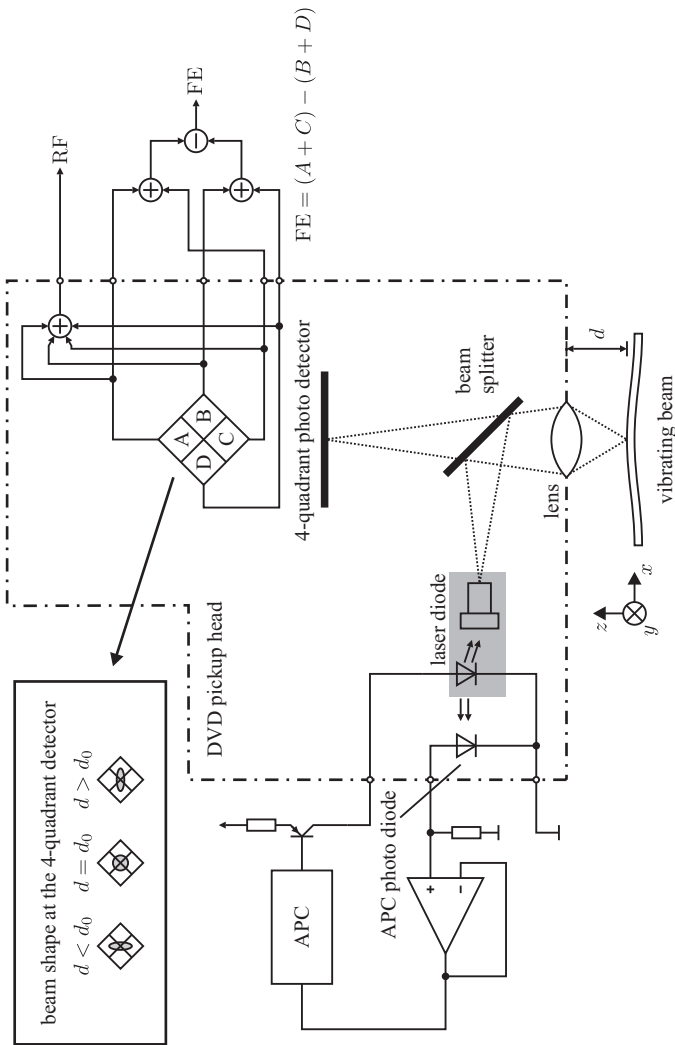


Figure 5.6: Simplified schematic of the optical readout utilizing a DVD pickup head.

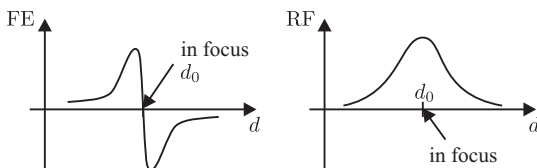


Figure 5.7: RF (radio frequency) and FE (focus error) signals of a CD/DVD pickup unit vs. distance d between the pickup's lens and the device under test. d_0 denominates the position of the focal plane.

Similarly, a second signal is generated by summing the photo detector outputs,

$$\text{RF} = u_A + u_B + u_C + u_D. \quad (5.4)$$

Usually, this signal is used to read the actual data from a CD or DVD. Due to the high bit rates and frequencies of this signal, it is labelled RF (radio frequency). RF also depends on the distance between the pickup lens and the opposite surface (Figure 5.7).

For the measurement of MEMS deflections, the lens of the pickup was rigidly clamped. The pickup head was laterally positioned over the centre of the vibrating silicon nitride beam. The laser beam was roughly focused, i.e., $d \approx d_0$. Deflections of the MEMS resonator from the rest position therefore lead to a change of d and can be detected by the resulting variations of FE and RF. As indicated in Figure 5.7, FE exhibits a much steeper slope in the vicinity of d_0 than RF and was therefore used to measure the beam deflections in the experiments.

The optical system was characterized by mounting the laser pickup head on a setup of three PLS-85 Precision Linear Stages (Micos GmbH, Eschbach, Germany). These stages feature a position repeatability of 100 nm and allow for precise positioning of the optical readout in x -, y -, and z -direction. The pickup's lens, which is normally moved by two voice coil motors for tracking and focusing, was firmly bonded to the housing of the pickup with an epoxy compound. Subsequently, the laser beam was adjusted to a flat opposite surface and the distance between this surface and the pickup's lens was varied in 50 nm steps by moving the PLS-85 in z -direction. The output voltages FE and RF of the laser pickup were obtained by digital voltmeters. Figure 5.8 depicts the result of these measurements. They show, that the actual behaviour of both output signals differs from the theory. However, the FE signal exhibits a steep falling slope of approximately $-0.5 \text{ V}/\mu\text{m}$ (at $100 \mu\text{m}$

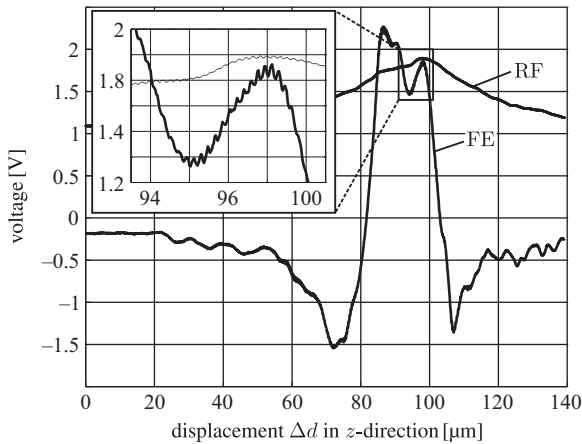


Figure 5.8: RF and FE output signals of the laser pickup head and the readout electronics versus the variation Δd of the distance between the lens and the opposite surface.

in Figure 5.8) and offers, therefore, great sensitivity for the measurement of the deflection of MEMS devices.

The inset in Figure 5.8 shows, that both FE and RF were superimposed by a small ripple. This ripple depended on the z -displacement, and its period length is in the range of half the wave length of the laser source (650 nm). Therefore, this ripple was attributed to an interference effect, but since only few information is available on the pickup head, the source of this effect could not be determined. The expected deflection of the beam is in the range of several ten nanometres and, therefore, far below the wave length of the laser light. Figure 5.9 summarizes the consequences of the interference effect. The left diagram in the figure resembles the inset of Figure 5.8. The vibrating clamped-clamped beam, on which the laser is focused, performs a sinusoidal deflection around a rest point z_0 . Due to thermal drift or vibrations of the measurement setup, the position of z_0 varies slowly. Depending on the gradient of FE at these new positions z_1 , z_2 , or z_3 , the resulting waveform is shifted by 180° (compare Figure 5.9a with Figure 5.9c), or shows significant harmonics (Figure 5.9b). These circumstances must be considered when processing the measurement data. In some cases, the gradient dFE/dz is increased by the interferences, and the sensitivity of the readout is improved, whereas in other

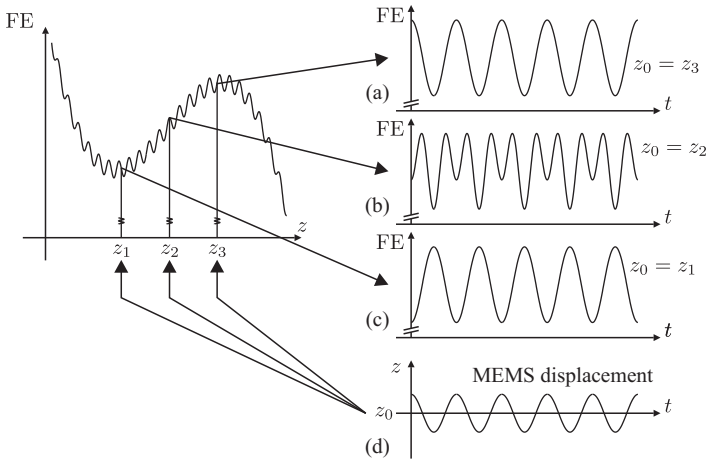


Figure 5.9: Consequences of the interference effect of the DVD pickup for MEMS readout. In (a), (b), and (c), FE is shown with normalized amplitude to point out the resulting waveforms, whereas (d) illustrates the deflection of the MEMS device.

cases it is decreased. The fluctuating gradient makes it impossible, at least for the presented setup, to specify an absolute value for the ratio of output voltage to beam deflection. Therefore, the measurement setup is incapable of obtaining the deflection amplitude of the beam, but yields the phase shift between excitation current and mechanical deflection.

5.3 Measurement Setup

Figure 5.10 depicts a schematic of the measurement setup. The beam vibrations are excited by applying a constant magnetic field (in y -direction) and a sinusoidal current (x -direction) to the device. The excitation current $i(t)$ is supplied by an Agilent 33220A function generator. A resistor of $R = 391 \, \Omega$ is connected in series to the beam sensor to limit this excitation current. The voltage at this resistor is used as the reference signal for the lock-in amplifier. The laser beam of the DVD pickup head is approximately focused on the centre of the vibrating structure. The readout circuit controls the laser power and obtains the distance dependent FE signal (see also Figure 5.6). Subsequently, a Signal Recovery SR830 lock-in amplifier (LIA) determines the phase shift

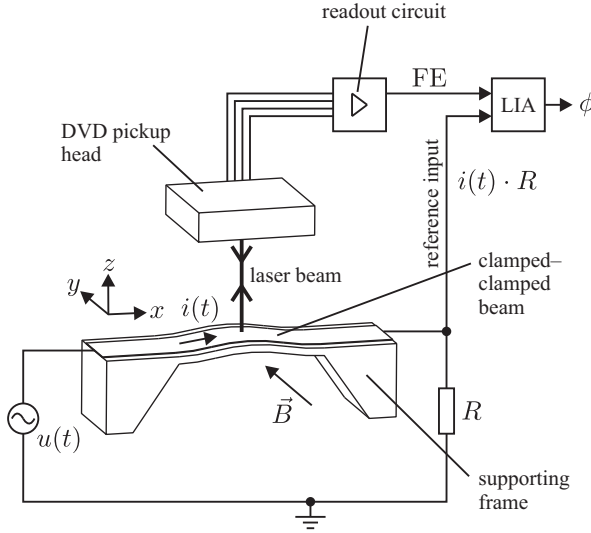


Figure 5.10: Schematic diagram of the sensor system. The beam vibrations are excited by Lorentz force caused by $i(t)$ and the magnetic field $\vec{B}$. From the FE signal the actual beam deflection in z -direction is obtained. The phase shift ϕ between the excitation current and FE (beam deflection) is measured by a lock-in amplifier (LIA).

ϕ between the beam current $i(t)$ and the beam deflection in z -direction given by FE. The result of a frequency sweep, $\phi(j\omega)$, is recorded by a computer program. Then, both the resonance frequency $f_0 = \omega_0/(2\pi)$ and the damping factor D are obtained by curve fitting

$$\phi(j\omega) = \arg \left\{ \frac{1}{1 + j\omega \frac{2D}{\omega_0} - \frac{\omega^2}{\omega_0^2}} \right\} \quad (5.5)$$

to these measurement results.

Figure 5.11a shows a photo of the prototype setup. For positioning of the laser beam during the experiments, the pickup is mounted on the translation stage described above. The vibrating clamped-clamped beam sensor is glued and wire-bonded to a printed circuit board (PCB). The device is mounted in the centre of a liquid container (Figure 5.11b). At the bottom side of the liquid container a permanent magnet has been attached to provide the required magnetic field ($B_y \approx 200$ mT). On the front side the container has a transparent

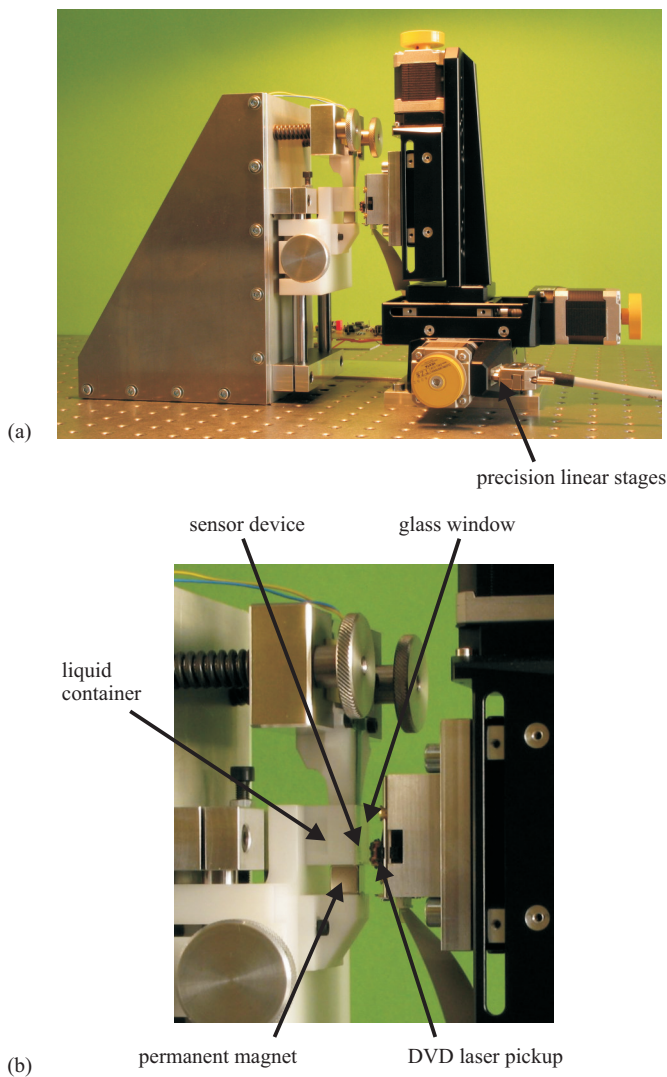


Figure 5.11: Prototype setup with the micromachined device immersed in the sample liquid. The static magnetic field is excited by the permanent magnet below.

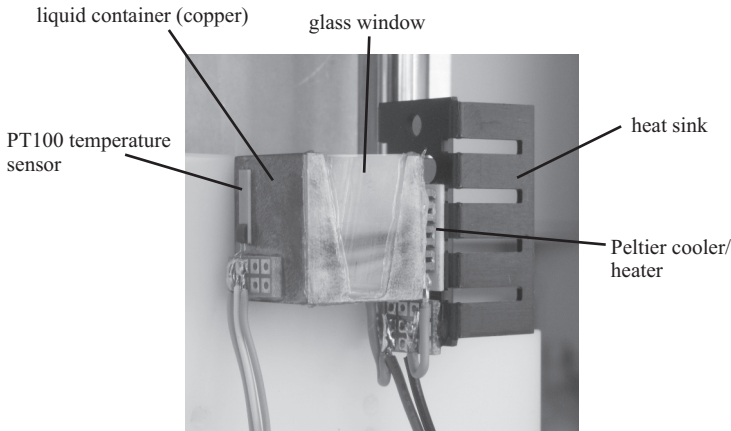


Figure 5.12: Liquid container allowing for temperature control. A PT100 temperature sensor (RTD) measures the temperature of the liquid container. The controller changes the supply current of the Peltier cooler/heater accordingly.

window, allowing for optical readout.

The viscosity of a fluid highly depends on the temperature [18]. Therefore, a means of controlling the temperature of the sample liquid must be employed. In the experiments presented below, a liquid container made of copper was used (Figure 5.12) and a Peltier heater/cooler and a resistance temperature detector (RTD) were attached to this container. A temperature controller maintained a temperature of 25 °C. The liquid container was filled with 1 ml of liquid.

5.4 Measurement of Newtonian Liquids

To demonstrate the feasibility of the sensor system, measurements were carried out with liquids that show a Newtonian behaviour. As sample liquids, a homologous series of alcohols and other pure chemicals have been used: ethanol, 1-propanol, 1-butanol, 1-pentanol, 1-hexanol, 1-heptanol, 1-octanol, cyclohexane, and toluene. They exhibit viscosities in the range of 0.565 to 7.368 mPa·s at 25 °C (Table 5.1).

The clamped–clamped beam sensor was immersed in the respective sample liquid and an excitation current of 3.1 mA rms was applied. A frequency

Table 5.1: Reference values for viscosity η and density ρ of the sample liquids. The values have been obtained from [20] (25°C temperature) and [51], respectively.

sample liquid	η [mPa · s]	ρ [kg/m ³]
toluene	0.57	867
cyclohexane	0.90	779
ethanol	1.06	789
1-propanol	1.90	803
1-butanol	2.52	810
1-pentanol	3.44	814
1-hexanol	4.49	814
1-heptanol	5.94	822
1-octanol	7.37	827
1-nonanol	9.10	827
1-decanol	10.97	830

sweep measurement yields the device's frequency characteristics $\phi(j\omega)$. Figure 5.13 shows the phase shift between the excitation current and the beam deflection near the first mode of resonance (dotted lines). To obtain the resonance frequency f_0 and the damping factor D , the second order transfer function, equation (5.5), was fitted to the measurement results (solid lines in Figure 5.13).

To first order, the damping factor D of the vibrating cantilever is dominantly influenced by the viscosity of the respective liquid, which is confirmed by the results given in Figure 5.14. Figure 5.15 shows the resonance frequency shift $(f_0 - f_{0,\text{air}})/f_{0,\text{air}}$, where $f_{0,\text{air}} = 96.7$ kHz is the beam's resonance frequency in air. The results indicate that the resonance frequency of the vibrating beam is not only influenced by the liquid's viscosity but also by the density. Compared to the results obtained for cyclohexane, the measurement of a beam immersed in toluene results in an additional frequency shift due to the higher density of the liquid (Table 5.1).

5.5 Measurement of SiO₂ Nano-Suspensions

As demonstrated above the clamped–clamped beam sensor is suitable for the measurement the viscosity of Newtonian liquids in the range from 0.57 to

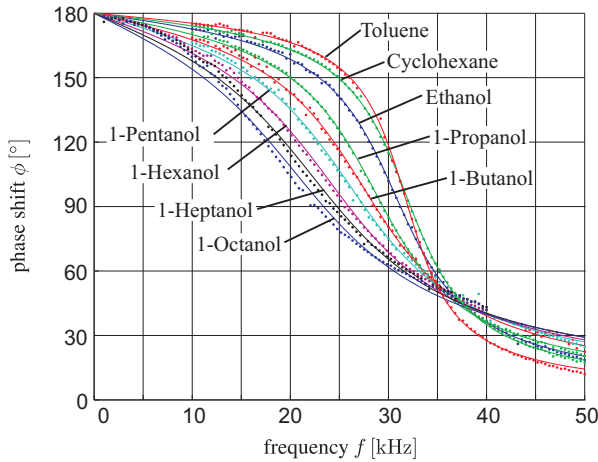


Figure 5.13: Phase shifts near resonance between the excitation current $i(t)$ and the actual deflection of the beam immersed in various sample liquids (dotted lines). The phase shift provides a better signal to noise ratio than the amplitude characteristics, and is thus used to extract the resonance frequency and the damping by curve fitting (solid lines). Due to the low resonance frequency in 1-heptanol and 1-octanol, the frequency sweeps were recorded only up to 40 kHz for these liquids.

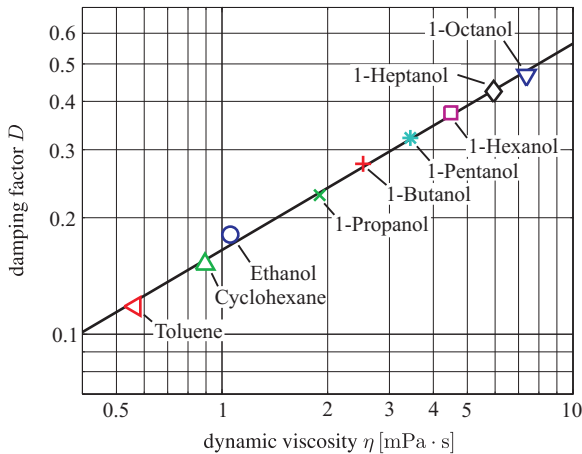


Figure 5.14: Measured damping D of the vibrating beam vs. dynamic viscosity η of the sample liquids.

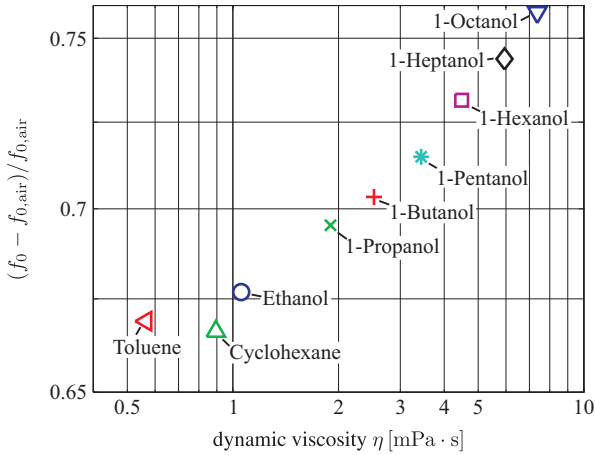


Figure 5.15: Shift of the resonance frequency of the vibrating beam vs. dynamic viscosity η of the sample liquids. The results indicate that the resonance frequency is not only influenced by the viscosity of the respective liquid, but also by its density.

7.32 mPa·s. A simple relation between the damping factor D and the viscosity η could be found. The resonance frequency of the beam was around several tens of kHz, much lower than the operating frequency of microacoustic sensors like the thickness shear mode (TSM) resonator. At the begin of this chapter it was supposed that the sensor, therefore, measures a viscosity parameter that is comparable to the steady shear viscosity determined by a conventional laboratory equipment. Below, this assumption will be confirmed for suspensions of silicon dioxide particles in water which exhibit a non-Newtonian behaviour [52]. The measurement results of the clamped–clamped beam sensor will be compared to those of a TSM quartz resonator.

5.5.1 Sample Liquid Preparation

A solution with a particle volume concentration of 22.4 % (silicon dioxide amorphous gel, 40 % weight concentration, ammonium stabilizing counter ion, average particle size 20–22 nm, from ABCR GmbH & Co. KG, Karlsruhe, Germany) was diluted with deionized (DI) water. The liquid quantities and concentrations of the mixtures are listed in Table 5.2. Figure 5.16 depicts the dynamic viscosity η of the suspensions at ambient temperature,

Table 5.2: SiO₂ in H₂O suspensions with different particle volume concentrations Φ . The viscosity was measured using a Brookfield LV-DV+CP cone-plate viscometer. The mass densities of the mixtures were calculated from the densities of the starting solution and water.

	Concentration Φ	Starting solution [ml]	DI-water [ml]	dynamic viscosity η [mPa·s]	mass density ρ [kg/m ³]
Sample 1	0.224	50	0	7.74	1290
Sample 2	0.2108	47	3	6.65	1273
Sample 3	0.2019	45	5	5.64	1261
Sample 4	0.1794	40	10	4.08	1232
Sample 5	0.1480	33	17	2.94	1191
Sample 6	0.1211	27	23	2.21	1156
Sample 7	0.493	11	39	1.23	1062
Sample 8	0	0	50	0.932	998

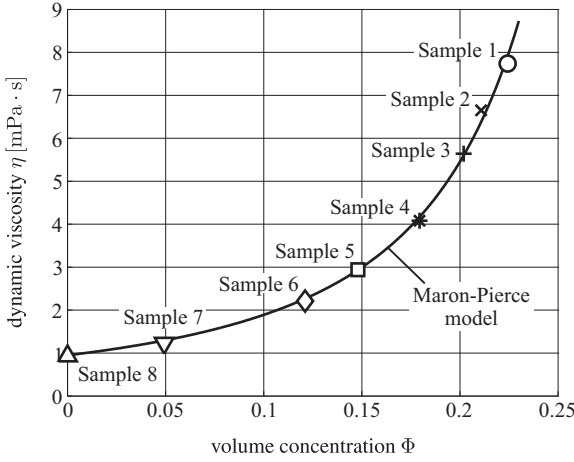


Figure 5.16: Dynamic viscosity η of the sample liquids (suspensions of SiO_2 in H_2O), measured using a cone-plate rheometer. The relationship between viscosity and particle volume concentration Φ can be described by the Maron-Pierce model (solid line), equation (5.6).

determined using a Brookfield LV-DV+CP cone-plate rheometer. The results reveal, that the relationship between the particle volume concentration Φ and η is properly modelled by the Maron-Pierce model [52, 53],

$$\eta = \eta_0 \left(1 - \frac{\Phi}{\Phi_{\max}} \right)^{-2}, \quad (5.6)$$

with $\eta_0 = 0.9475 \text{ mPa}\cdot\text{s}$, and $\Phi_{\max} = 0.3431$. In Figure 5.16, this model is represented by the solid line. The mass densities of the suspension samples have been calculated from the respective mixing ratios and the density values of the starting solution and of DI-water, 1290 kg/m^3 and 998 kg/m^3 , respectively (Table 5.2).

For comparison, the alcohols of Table 5.1 cover approximately the same viscosity range as the considered suspensions, but show Newtonian behaviour. They exhibit a constant dynamic viscosity, independent of the shear rates applied to them.

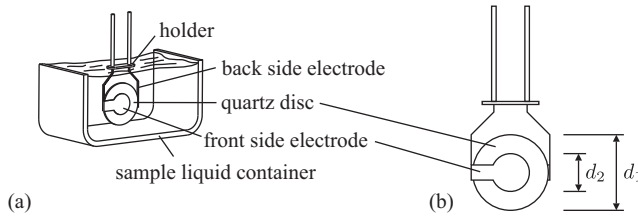


Figure 5.17: TSM quartz resonator for measuring the viscosity of liquid. The quartz resonator is immersed in the liquid. The sensor utilized in the measurements featured a quartz diameter of $d_1 = 8$ mm and an electrode diameter of $d_2 = 4$ mm. Its first mode resonance frequency is 6 MHz.

5.5.2 Quartz Resonator Measurements

For the quartz TSM resonator measurements, an AT-cut quartz was used, exhibiting a shear mode with a resonance frequency of 6 MHz (Figure 5.17a). The sensor features a quartz diameter of $d_1 = 8$ mm and an electrode diameter of $d_2 = 4$ mm (Figure 5.17b). When the resonator is immersed in a liquid, its behaviour in the vicinity of the first mode resonance frequency can be modelled by the lumped element equivalent circuit depicted in Figure 5.18a [21]. The motional arm R_1 – L_1 – C_1 represents the mechanical resonance and the internal losses, and C_0 accounts for the static capacitance of the device. Since both electrodes are in contact with the sample liquid, and the suspensions examined in this work are electrically conductive, the liquid's conductivity must be considered by the conductivity G_0 [54, 55]. The acoustic load, i.e., the liquid surrounding the TSM quartz resonator, is modelled by R_2 and L_2 . To first order, both R_2 and L_2 are proportional to the square root of the viscosity–density product $\sqrt{\eta\rho}$ [21].

The sensor was immersed in the respective liquid and connected to the two ports of a HP 8753E network analyser (Figure 5.18b), and the S-parameter S_{21} was measured. The sensor admittance Y was then calculated from

$$Y = \frac{S_{21}}{100 \, \Omega \cdot (1 - S_{21})}. \quad (5.7)$$

A frequency sweep yields $Y(j\omega_k)$, where ω_k are the angular frequencies which are covered. In principle, the expression for the quartz admittance

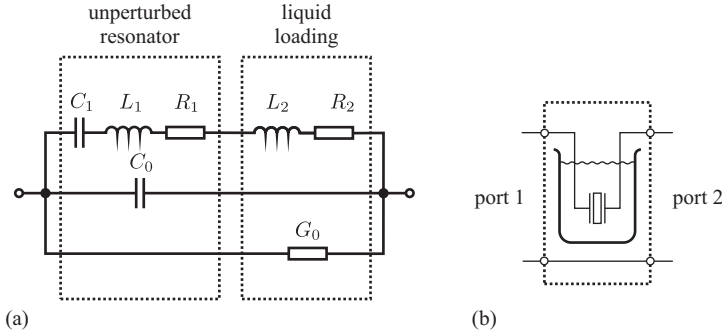


Figure 5.18: TSM quartz resonator measurements. (a) Lumped element equivalent circuit of the quartz resonator immersed in a conductive liquid. (b) Two-port element to the network analyser.

deduced from Figure 5.18a,

$$Y(j\omega) = \frac{1}{R_1 + R_2 + \frac{1}{j\omega C_1} + j\omega(L_1 + L_2)} + j\omega C_0 + G_0, \quad (5.8)$$

could be fitted to the $Y(j\omega_k)$ to determine the values of R_2 and L_2 , but due to the number of circuit elements involved this fit only converged for thoroughly chosen starting values. In the vicinity of the resonance frequency, $\omega_1 < \omega_k < \omega_2$, the admittance locus plot of the quartz can be approximated by a circular arc (Figure 5.19). The radius r of this arc is $r = 1/[2(R_1 + R_2)]$, and its centre is denoted by $G + jB$. Therefore, the value of $R_1 + R_2$ can be obtained by minimizing the objective function

$$J(G, B, r) = \sum_k (|Y(j\omega_k) - G - jB| - r)^2 \quad (5.9)$$

with respect to G , B , and r [56]. For the liquids considered in this work, R_1 is negligible compared to R_2 , and therefore $R_2 = 1/(2r)$.

Figure 5.20 compares the measurement results obtained for both sets of sample liquids described in Section 5.5.1. As expected, the equivalent circuit resistance R_2 of the quartz resonator immersed in the alcohols, which show Newtonian behaviour, is proportional to the square root of the viscosity–density product, $\sqrt{\eta\rho}$. However, the results obtained for the suspensions, do not follow this trend. The values of R_2 indicate, that the effective viscosities

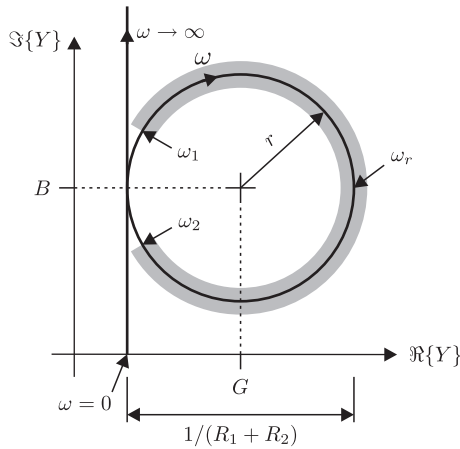


Figure 5.19: Sensor admittance locus plot $Y(j\omega)$: In the vicinity of the resonance frequency, $\omega_1 < \omega < \omega_2$, the quartz admittance can be approximated by a circular arc with the radius r , and the center $G + jB$.

probed by the quartz resonator are lower than those determined by conventional laboratory viscometers (see Chapter 2).

5.5.3 Clamped–Clamped Beam Sensor Measurements

The examination of the SiO₂ suspensions with the clamped–clamped beam sensor was carried out according to Section 5.3 and 5.4. From fitting (5.5) to the result of the frequency sweep, $\phi(j\omega)$, again the damping factor D and the resonance frequency f_0 were obtained.

The damping factor D obtained for the vibrating beam immersed in the sample liquids is plotted in Figure 5.21. Again, the damping factor D is dominantly influenced by the liquid's viscosity η . In contrast to the measurement results obtained by the TSM quartz resonator (Figure 5.20), the clamped–clamped beam device exhibits a similar behaviour for both sets of liquids. The measured damping factors as a function of the viscosity appear on the same trend line for both the Newtonian liquids and the suspensions (solid line in Figure 5.21). Small deviations can be attributed to the influence of the liquid's differing mass densities.

The evaluation of the measurement results also yields the resonance fre-

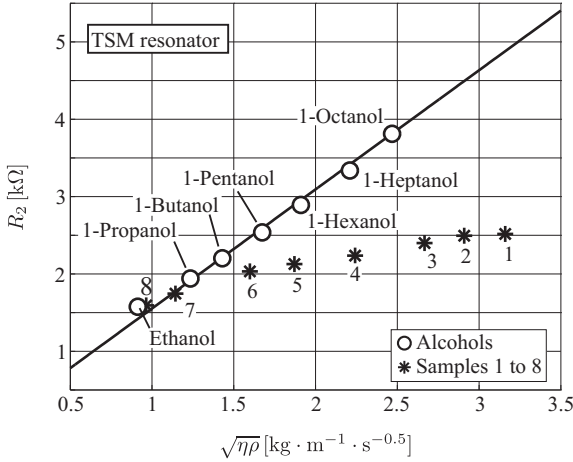


Figure 5.20: Equivalent circuit resistance R_2 (Figure 5.18a) of the quartz TSM resonator (6 MHz, 8 mm diameter AT-cut), immersed in the sample liquids, vs. square root of the viscosity–density product of the respective liquid. The results indicate that a TSM resonator is not suitable for the measurement of viscosity or concentration of the considered SiO_2 –in– H_2O suspensions (Sample 1 to 8), whereas the results for Newtonian liquids (alcohols) follow $R_2 \sim \sqrt{\eta\rho}$.

quency $f_0 = \omega_0/(2\pi)$. For the alcohol sample liquid set it has been shown, that f_0 is influenced by both the liquid’s dynamic viscosity and mass density [45]. In principle, using a suitable model for the vibrating beam, the mass density of the liquids could thus be determined simultaneously [13]. However, for the suspensions, the results were widely spread over a range of several kHz and no clear trend could be observed, requiring further investigation and special treatment of the measurement results for f_0 .

5.6 Device Modelling

Above, in Section 5.4, the measurement results demonstrated the feasibility of the clamped–clamped beam resonator. Figure 5.14 depicts the damping factor D versus the viscosity η of the liquid. The results indicate that this damping factor is dominated by the viscosity, and the relation is

$$D = c_1 \eta^{c_2}, \quad (5.10)$$

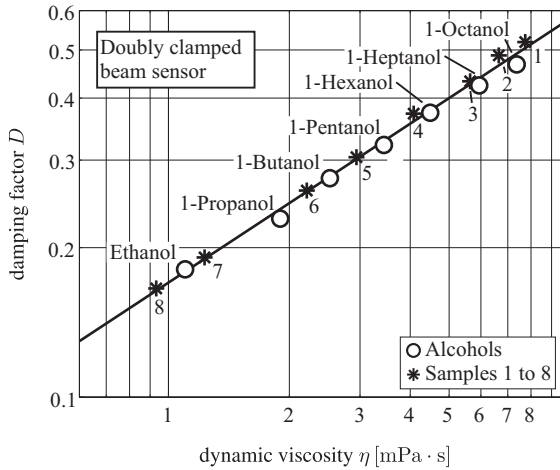


Figure 5.21: Measurement results demonstrating the feasibility of the clamped–clamped beam viscosity sensor for use in complex liquids. The figure shows good agreement between the damping factor D obtained from the sensor system and the results of conventional viscosity measurement, η . Small deviations from the curve fit can be attributed to the influence of the liquid’s differing mass densities.

where c_1 and c_2 are model parameters obtained from a calibration using liquids with well-known properties. However, the resonance frequency f_0 is influenced by both the viscosity η and the mass density ρ of the liquid (Figure 5.15). A suitable model is needed to properly describe the relation between f_0 , η , and ρ . Once such a model is found, the simultaneous measurement of density and viscosity with the clamped–clamped beam sensor is possible [13]. As already mentioned in Section 4.2, several models exist to describe the behaviour of vibrating beams in liquid. In the works of Sader [25] and Weiss *et al.* [26] the interaction of beams with rectangular cross-sections the surrounding liquids is modelled. Below the applicability of these models, in particular Sader’s model, to the sensor will be investigated. The measurements presented below were carried out with the setup described in Section 5.3, but using a device featuring a longer beam, $l = 480 \mu\text{m}$ (Figure 5.4). Due to fabrication tolerances, the beam was only $w = 35 \mu\text{m}$ wide. The clamped–clamped beam fulfils the requirements for the Sader model, as the length to width ratio is $l/w \approx 14$, and the width to thickness ratio is $w/h \approx 27$.

The behaviour of the vibrating beam in the vicinity of the resonance fre-

quency was modelled by a second order system,

$$(m_e + m_a) \frac{d^2\psi}{dt^2} + (b_e + b_a) \frac{d\psi}{dt} + K\psi = F_0 e^{j\omega t}, \quad (5.11)$$

where ψ is the beam deflection, m_e and b_e are the beam's effective mass and intrinsic damping, K is the beam's spring constant, m_a and b_a are the additional mass loading and damping due to the liquid surrounding the cantilever, and $F_0 e^{j\omega t}$ is the driving force. The phase shift between the excitation current and the beam deflection is, therefore,

$$\begin{aligned} \phi(j\omega) &= \arg \left\{ \frac{1}{K + j\omega(b_e + b_a) - \omega^2(m_e + m_a)} \right\} = \\ &= \arg \left\{ \frac{1}{1 + j\omega \frac{2D}{\omega_0} - \frac{\omega^2}{\omega_0^2}} \right\}, \end{aligned} \quad (5.12)$$

where $\omega_0 = 2\pi f_0$ is the resonance frequency, and the damping factor $D = 1/(2Q)$, where Q is the quality factor. In the first experiments, the phase shift $\phi(j\omega)$ was obtained from a frequency sweep and ω_0 and D were extracted by fitting (5.12) to the measurement results (see Section 5.3).

Figure 5.22 depicts the measurement results (obtained with the 480 μm clamped–clamped beam) and the corresponding fit results. The model properly matches the behaviour of the clamped–clamped beam immersed in the liquids ethanol, 1-propanol, and 1-butanol. In contrast, the phase shift detected in liquids exhibiting higher viscosities like 1-nonanol or 1-decanol cannot be modelled by (5.12). At frequencies distant from the resonance frequency the modelled curves strongly deviate from the measured ones, indicating, that a refined model is required to improve measurement accuracy.

Again, the damping factor D obtained from the curve fit is dominantly influenced by the dynamic viscosity η of the surrounding liquid (Figure 5.23), although a strong deviation from the trend line is noticed at higher viscosities. Figure 5.24 indicates that the resonance frequency is determined by both the mass density and the viscosity of the liquid.

A vibrating thin beam with rectangular cross-section immersed in a liquid is subject to an additional mass loading m_a and an additional damping b_a [25],

$$m_a = \rho \frac{\pi}{4} w^2 l \Gamma', \quad (5.13)$$

$$b_a = \rho \frac{\pi}{4} w^2 l \omega \Gamma'', \quad (5.14)$$

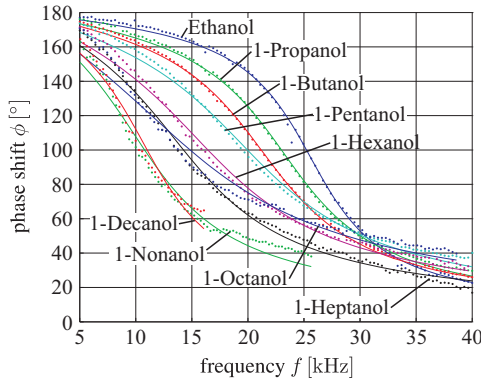


Figure 5.22: Phase shift between excitation current $i(t)$ and the actual deflection of the beam $\psi(t)$ immersed in various sample liquids (dotted lines). A second order system was fitted to the measurement results to obtain the resonance frequency ω_0 and the damping factor D of the vibrating beam. The fit results are depicted by solid lines.

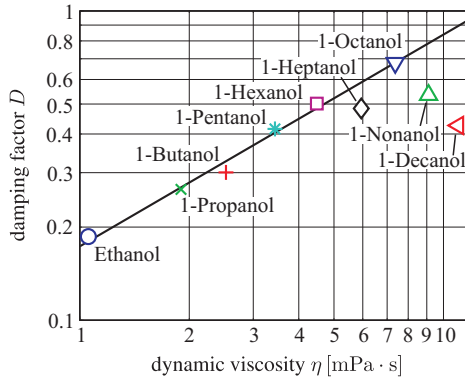


Figure 5.23: Damping factor D of the clamped-clamped beam sensor immersed in various sample liquids versus dynamic viscosity η of the respective liquids. The results have been obtained by fitting a second order transfer function, (5.12), to the frequency sweep measurement results. The diagram indicates that the damping factor is dominantly influenced by the liquid's viscosity.

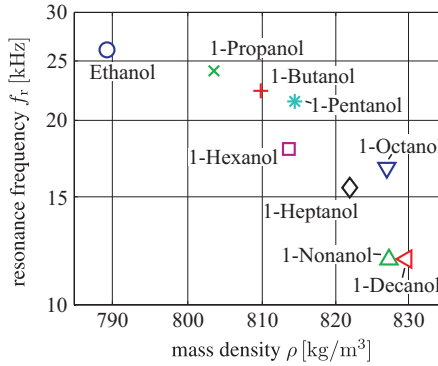


Figure 5.24: Measurement results showing the resonance frequency f_0 of the clamped–clamped beam immersed in a variety of alcohols versus the mass density of the liquids ρ .

where w and l are the width and the length of the beam, respectively, ω is the vibration frequency, ρ is the mass density of the liquid, and $\Gamma = \Gamma' + j\Gamma''$ is the hydrodynamic function reflecting the specific characteristic of the fluid–structure interaction, i.e., a rectangular beam vibrating in an infinitely extended medium. Using the simplifications of Maali *et al.* [37], the hydrodynamic function is

$$\Gamma' = a_1 + a_2 \frac{\delta}{w}, \quad (5.15)$$

$$\Gamma'' = b_1 \frac{\delta}{w} + b_2 \left(\frac{\delta}{w} \right)^2. \quad (5.16)$$

The parameters for the infinitely thin beam are $a_1 = 1.0553$, $a_2 = 3.7997$, $b_1 = 3.8018$, and $b_2 = 2.7364$. Again, δ is the characteristic penetration depth,

$$\delta = \sqrt{\frac{2\eta}{\rho\omega}}, \quad (5.17)$$

and w is the width of the beam.

Figure 5.25 gives both the dependence of m_a and b_a on the frequency for the beam immersed in 1-decanol according to (5.13) and (5.14) in the operating frequency range of the cantilever. The diagram elucidates that the loading of the sensor is highly frequency dependent. Therefore, the approximation of

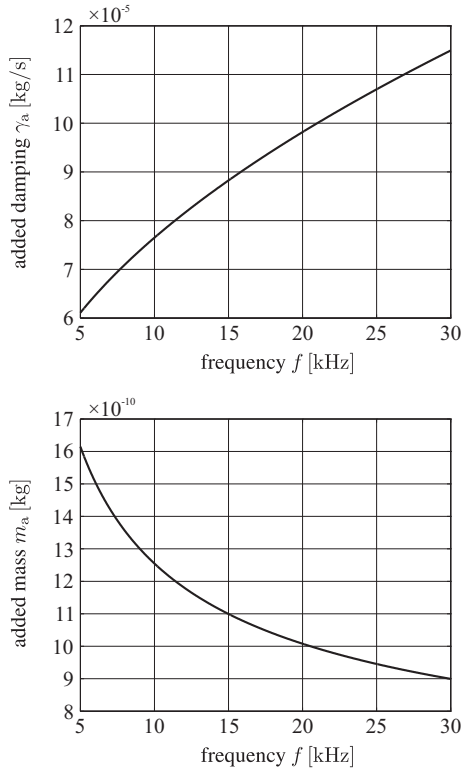


Figure 5.25: Added mass loading m_a and added damping b_a of a rectangular beam immersed in 1-decanol ($\eta = 11 \text{ mPa}\cdot\text{s}$, $\rho = 829.7 \text{ kg/m}^3$) as given by (5.13) and (5.14) [37]. The diagram demonstrates that both parameters exhibit a strong frequency dependence requiring an improved model for the evaluation of measurement results.

the cantilever's frequency response by a second order system, (5.12), that is a differential equation with constant coefficients m_a , m_e , b_a , b_e , and K , does not match the measurement results sufficiently. The deviation increases as the viscosity and the density of the liquid increase, see Figure 5.22.

To account for the frequency dependence of m_a and b_a an improved model is introduced. Again, equation (5.12) is fit to the measurement results, but this time m_a and b_a are not considered constant, but are replaced by (5.13) and (5.14). A calibration of the cantilever was done by using the val-

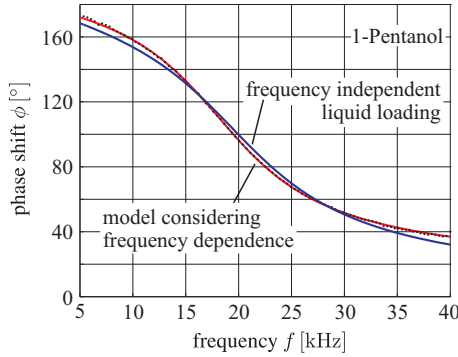


Figure 5.26: Phase shift ϕ between the excitation current $i(t)$ and the beam deflection $\psi(t)$ of a clamped–clamped beam immersed in 1-pentanol (dotted). The second order system model assuming frequency independent liquid loading of the cantilever exhibits deviations from the measurement results, whereas the improved model considering the hydrodynamic function, (5.13) and (5.14), fits the measurement results well.

ues of mass density and dynamic viscosity of the liquids as given in Table 5.1, and curve fitting (5.12), (5.13) and (5.14) with respect to the intrinsic mass m_e , damping b_e , and spring constant K of the vibrating beam. Figure 5.26 presents the results from a measurement using 1-pentanol as a calibration liquid. They indicate that the improved model considering said frequency dependence describes the beam’s frequency response properly. The curve fit yields numerical values for the mass of the beam $m_e = 1.01 \cdot 10^{-10}$ kg, the intrinsic damping $b_e = 3.02 \cdot 10^{-5}$ kg/s, and the spring constant $K = 14.3$ N/m. An estimation of the cantilever mass,

$$\begin{aligned}
 m_{e,\text{estimated}} &= \underbrace{480 \cdot 35 \cdot 1.3 \cdot 10^{-18} \text{ m}^3}_{\text{silicon nitride geometry}} \cdot 3440 \frac{\text{kg}}{\text{m}^3} \\
 &\quad + \underbrace{480 \cdot 20 \cdot 0.2 \cdot 10^{-18} \text{ m}^3}_{\text{gold layer}} \cdot 19.32 \cdot 10^3 \frac{\text{kg}}{\text{m}^3} = \\
 &1.12 \cdot 10^{-10} \text{ kg}, \quad (5.18)
 \end{aligned}$$

considering the beam geometries and the 200 nm gold layer on the beam, demonstrates that the mass obtained from the calibration procedure in 1-pentanol is trustworthy. Also the range of the spring constant seems to be

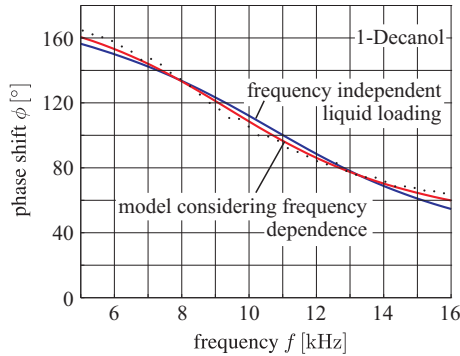


Figure 5.27: Phase shift measurement results (dotted) for immersion of the cantilever in 1-decanol. The improved model fits the measured ϕ better than the second order model assuming frequency independent coefficients, but the figure also indicates that more complex damping mechanisms are involved.

reasonable, as from laser Doppler vibrometer measurements and estimations a static beam deflection of several nanometres and a driving force in the range of several hundred nN is assumed. However, the intrinsic damping b_e obtained from the curve fit is too high compared to the damping due to the liquid loading, Figure 5.25. Also measurements with the beam immersed in 1-decanol, Figure 5.27, yield a similarly high intrinsic damping.

Reasons for the deviation in the damping behaviour can be found in the complex damping mechanisms involved. In the actual sensor layout the opening in the membrane is only $40\text{ }\mu\text{m}$ wide (Figure 5.4). The distance between the beam and the bulk silicon is, therefore, in the range of the characteristic penetration depth δ , (5.17), that is between 5 and $30\text{ }\mu\text{m}$ for the considered sample liquids and the frequency range in which the measurements were carried out (see Figure 5.28). The hydrodynamic function Γ as given by (5.15) and (5.16), however, is only valid for a thin beam in an infinitely extended volume of liquid and does not capture the interaction with the surrounding structure.

5.7 Conclusion

The resonant behaviour of the clamped–clamped micromachined vibrating beam is influenced by the surrounding liquid. In particular, the measurement

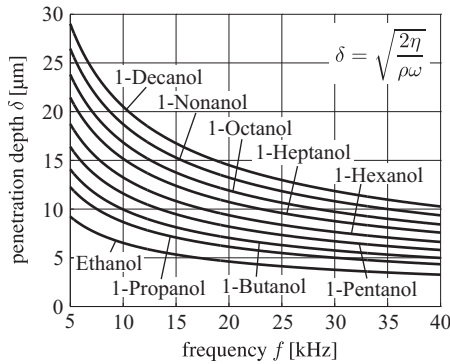


Figure 5.28: Characteristic penetration depth δ of the acoustic wave excited by the clamped–clamped beam sensors. These results were obtained by evaluating (5.17) for the liquid parameters of Table 5.1 and the frequency range in which the sensor is operated.

of the resonance frequency and the damping factor allows the determination of the viscosity and the density of the liquid. For this purpose, the actual beam deflection must be obtained. An optical system is suitable for the detection of the small deflections of the strongly damped beam. The presented prototype setup utilizes a low-cost optical readout based on a DVD laser pickup head and a customized electronics. The measurement results show that the obtained damping factor is mainly dominated by the liquid's viscosity, whereas the resonance frequency is also influenced by the density of the respective liquid. Therefore, the sensor system can directly be used to determine the viscosity of a liquid, which was successfully demonstrated for liquids exhibiting viscosities in the range of 0.565 to 7.368 mPa·s. However, to measure the liquid's density simultaneously, a suitable model for the sensor device must be applied.

In first results it was demonstrated that fitting a second order differential equation with constant coefficients results in a damping factor that can be used as a measure for the viscosity of a liquid. However, the added mass and the added damping due to the fluid surrounding the beam is a function of the Reynolds number, and the simple model used previously does not consider the resulting frequency dependence. Another model based the hydrodynamic function of the beam [25] fits the beam's frequency response more properly, but yields an intrinsic damping of the MEMS device in the same range as the

damping caused by the surrounding liquid, indicating the presence of further damping effects. The model could probably be further improved by including the surrounding structure in the calculation of the hydrodynamic function and account for additional loss mechanisms.

Liquids exhibiting complex rheological behaviour must be deformed significantly and with sufficiently low frequencies to reveal a viscosity parameter comparable to the steady-shear viscosity that is obtained from conventional laboratory viscometers. Suspensions of silicon dioxide in water were used as a model system for such complex liquids. A thickness shear mode quartz resonator operating at a frequency of several MHz and low vibration amplitudes probes these fluids in a different rheological regime compared to a laboratory cone-plate rheometer. The results of this microacoustic sensor do not follow the expected viscosity–density product dependence. In contrast, the damping factor obtained from a clamped–clamped beam sensor with a resonance frequency in the range of several ten kHz and vibration amplitudes in the nanometre range is comparable to the dynamic viscosity probed by the laboratory viscometer. Since the relationship between the viscosity and the particle volume concentration of the considered silicon–dioxide–in–water suspensions is properly described by the Maron–Pierce model, the clamped–clamped beam sensor can also be used to determine the concentration of the observed suspensions.

Chapter 6

Suspended Plate In-Plane Resonator

Based on the experiences gained from recent research with clamped–clamped beam resonators (Chapter 5) which perform out-of-plane vibrations, a new sensor design has been devised. The sensor avoids several drawbacks of other viscosity sensors presented in this thesis and in other literature. This chapter presents the design, modelling, fabrication, and experimental verification of this suspended plate viscosity sensor.

Figure 6.1a depicts a clamped–clamped beam cantilever. With respect to the larger two of its lateral surfaces (that are perpendicular to the x -direction), the beam vibrates out-of-plane. Such a mode of vibration leads to a velocity field in the fluid that is associated with high damping (see Section 5.4). This damping leads to small vibration amplitudes and, therefore, requires sophisticated readout techniques and circuitry. In contrast, sensors exciting mainly shear waves, like the TSM resonator, feature much higher quality factors [21].

The suspended plate sensor described in this chapter is aimed at improving the quality factor of micromachined viscosity sensors. The basic concept of this device is a thin plate vibrating in an in-plane mode. The plate is suspended by thin silicon beam springs, see Figure 6.1b. If the plate is thin enough, mainly shear waves would be excited in the surrounding liquid. Consequently, the damping of the vibrating part would be kept low and high Q-factors can be achieved. For sufficient mechanical stability, the fabricated MEMS devices actually feature rather thick plates and beams. In fact, the geometric parameters of the springs are comparable to those of the clamped–clamped beam in Chapter 5. As a result, the damping of the entire device is again dominated by the damping of the beams with rectangular cross-section, and the increase of the quality factor is moderate. However, the results given below indicate, that higher Q-factors can be achieved by varying the device dimensions, e.g., fabricating longer plates, see Figure 6.1c. Also, the devices

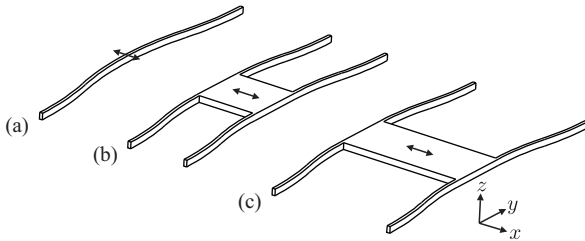


Figure 6.1: Principles of micromachined devices for the measurement of viscosity. (a) The clamped–clamped beam sensor (Chapter 5) is subject to a high damping since its mode of vibration is an out-of-plane mode with respect to its larger lateral surface. (b) Placing a plate at the centre of the beam leads only to a insignificantly higher damping, since mainly shear waves are excited at the plate surface. The increased mass, however, results in higher quality factors. (c) Varying the plate dimensions leads to improved Q-factors and allows the fabrication of devices with different resonance frequencies.

have proven sufficiently stable. Therefore, the thickness of the moving part could be further decreased.

The vibrating structure consists of p-doped silicon, which is intrinsically piezoresistive. This effect is utilized to read out the plate deflection. The applied design eliminates the need for additional fabrication steps to deposit piezoelectric or piezoresistive material or additional doping processes [57]. The sensor device including excitation and readout is fabricated with a three-mask process. The output voltage of the Wheatstone bridge formed by the piezoresistive elements is in the range of only some μV . However, the excitation frequency is well known and lock-in amplifiers and modulation techniques can be used to achieve a satisfying signal-to-noise ratio. The findings presented here were partly published in [58–60].

6.1 Sensor Design

Figure 6.2 depicts the schematics of the novel sensor device. The main element is a rectangular plate suspended by four beam springs. The springs are $5\,\mu\text{m}$ wide, and $20\,\mu\text{m}$ high. Their favoured direction of deflection is in the x – y plane, leading to in-plane vibrations of the suspended plate. The beams carry a conductive layer with the sinusoidal excitation current $i_e(t)$. In the

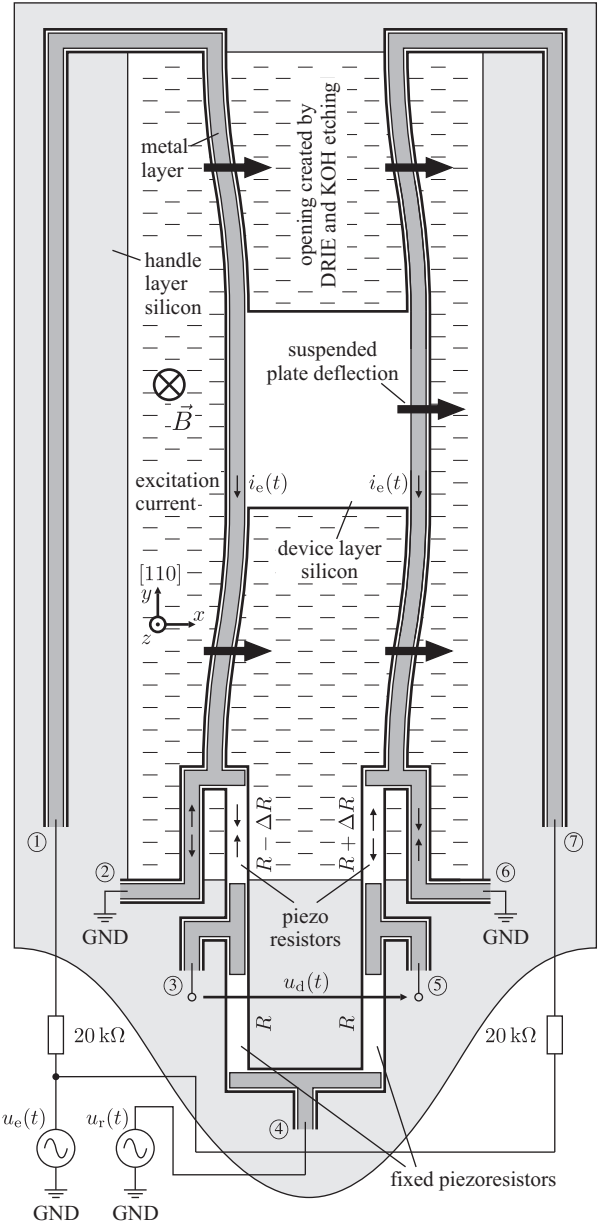


Figure 6.2: Schematic of the micromachined viscosity sensor. The in-plane movement of the plate is excited by the Lorentz forces arising from the excitation current $i_e(t)$ (sinusoidal, frequency f_e) and the magnetic flux density $\vec{B}$. The actual deflection is detected by a Wheatstone bridge of four piezoresistors (p-Si). Two of them are fixed, and two are mechanically stressed by the beam deflection. The Wheatstone bridge is fed with a sinusoidal voltage $u_e(t)$ at a frequency f_r . A lock-in amplifier locked to the difference frequency $f_r - f_e$ detects the beam deflection. This principle eliminates crosstalk from the excitation current and the baseline of the readout voltage.

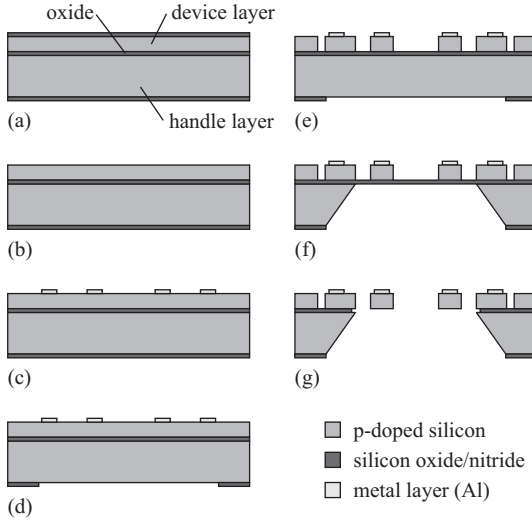


Figure 6.3: Simplified schematic of the fabrication process.

field of a permanent magnet (flux density $\vec{B} = -\vec{e}_z B_z$), Lorentz forces excite lateral vibrations of the plate and the springs. The ends of two beam springs (bottom of Figure 6.2) are forked. Two of the prongs carry the excitation current whereas the other two form piezoresistive elements. Depending on the deflection of the vibrating plate, those piezoresistors are subject to either compressive or tensile stress. Their electric resistances change accordingly. With two additional constant resistors, they form a Wheatstone bridge circuit. The sensor readout is driven by a voltage $u_r(t)$ and yields a differential output voltage $u_d(t)$.

6.2 Fabrication

The sensor was fabricated on a 4-inch (1 00) silicon-on-insulator (SOI) wafer. The thicknesses of the device layer, the buried oxide layer, and the handle layer were $20\text{ }\mu\text{m}$, $2\text{ }\mu\text{m}$, and $350\text{ }\mu\text{m}$, respectively. Both sides of the wafer were coated with a stack of silicon nitride and silicon oxide (Figure 6.3a). The device silicon layer was p-doped featuring a conductivity in the range from $0.11\text{ }\Omega\text{cm}$ to $0.21\text{ }\Omega\text{cm}$. First, the top side coating was removed by re-

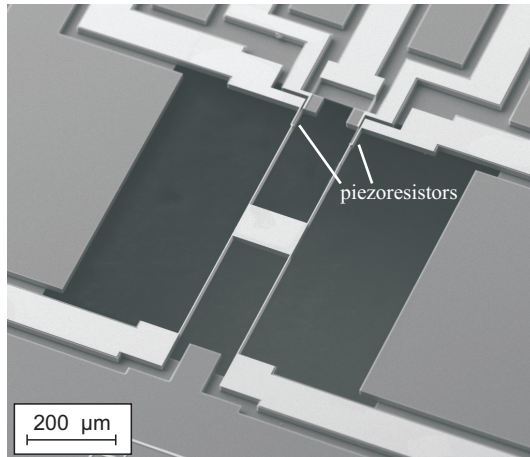


Figure 6.4: Scanning electron microscope (SEM) image of the suspended plate viscosity sensor. A magnified image of one of the piezoresistors is depicted in Figure 6.5.

active ion etching (RIE, Figure 6.3b). A 500 nm aluminium layer was vapour deposited and patterned using the lift-off technique (Figure 6.3c) to form the electrical connections. Subsequent annealing in vacuum was required to establish ohmic contacts between the metal layer and the silicon. Then openings were etched using RIE into the back-side nitride–oxide stack, which were required as mask for the KOH etching process (Figure 6.3d). The free-standing structure, the piezoresistors, and the conducting paths were formed by deep reactive ion etching (DRIE) on the wafer's front side (Figure 6.3e). Subsequently, the wafer was KOH etched from the back-side (Figure 6.3f), and the plate and the suspension springs were released by wet etching with buffered hydrofluoric acid (Figure 6.3g). Figures 6.4 and 6.5 are scanning electron microscope (SEM) images of the sensor device and of one of the piezoresistive elements. After wafer dicing, the sensor devices were die-bonded on small printed circuit boards (PCB), and the electrical connections from chip to PCB were established by gold wire-bonding. Finally, the wire bonds were protected by an epoxy compound.

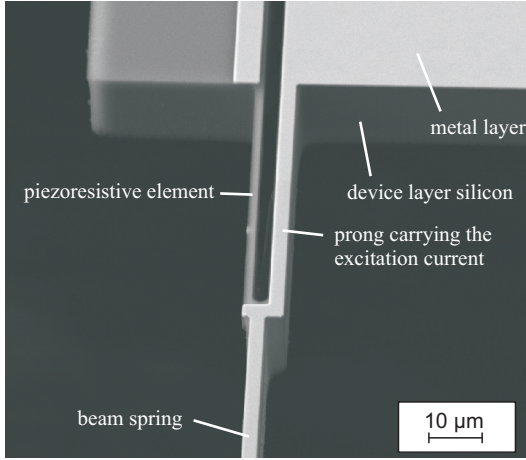


Figure 6.5: SEM image of the forked end of a beam spring, featuring a conductive layer for carrying the excitation current, and a piezoresistive element.

6.3 Modelling

For the sensor design, a simple model for the interaction of the vibrating parts with the surrounding fluid was used. Whereas the device operation in vacuum can be easily determined by finite element tools, it is preferable to describe its behaviour in a fluid by analytical or semi-analytical models to reduce the required computing power.

The interaction of the four springs (Figure 6.6) with the surrounding liquid was modelled by approximating the rectangular beam by a beam with circular cross-section [25, 35]. For the added mass due to the liquid loading per unit length we have

$$m'_{a,\text{spring}} = \rho \frac{\pi}{4} h^2 \Gamma', \quad (6.1)$$

and for the added damping coefficient per unit length

$$\gamma'_{a,\text{spring}} = \rho \frac{\pi}{4} h^2 \omega \Gamma'', \quad (6.2)$$

where ρ is the mass density of the liquid, and ω is the angular vibration frequency. $\Gamma = \Gamma' + j\Gamma''$ is the hydrodynamic function

$$\Gamma = 1 + \frac{4jK_1(-j\sqrt{j\text{Re}})}{\sqrt{j\text{Re}}K_0(-j\sqrt{j\text{Re}})}, \quad (6.3)$$

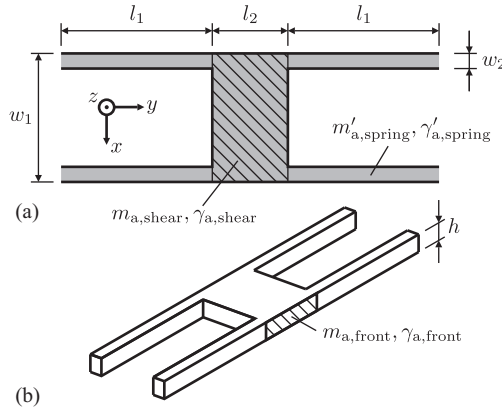


Figure 6.6: Model of the suspended plate device

where $\text{Re} = \rho\omega h^2/(4\eta)$, and K_0 and K_1 are modified Bessel functions of the second kind [25]. η is the dynamic viscosity of the liquid.

The in-plane movement of the rectangular plate (hatched area in Figure 6.6a) is expected to excite mainly shear waves in the liquid. Therefore, a one-dimensional model was employed that is commonly used for thickness shear mode resonators [61]. The mechanical stress acting on a plane at $z = 0$ that is oscillating harmonically in x -direction, and is in contact with a liquid, is given by [22]

$$\sigma_{zx} = \sqrt{\frac{\omega\eta\rho}{2}}(\text{j} - 1)v_x, \quad (6.4)$$

where v_x is the velocity of the plate, $v_x = \hat{v}_x \exp(-\text{j}\omega t)$. Therefore, the force acting on a vibrating rectangular plate (Figure 6.6) which is in contact with the liquid on both sides is approximated by

$$F_x = 2w_1 l_2 \sigma_{zx} = -2w_1 l_2 \sqrt{\frac{\omega\eta\rho}{2}} v_x + 2w_1 l_2 \sqrt{\frac{\eta\rho}{2\omega}} \text{j}\omega v_x. \quad (6.5)$$

This force F_x represents an additional mass loading of the sensor element,

$$m_{\text{a,shear}} = 2w_1 l_2 \sqrt{\frac{\eta\rho}{2\omega}}, \quad (6.6)$$

and an additional damping

$$\gamma_{a,\text{shear}} = 2w_1l_2\sqrt{\frac{\omega\eta\rho}{2}}. \quad (6.7)$$

Clearly, the front faces of the rectangular plate (hatched area in Figure 6.6b) with respect to the direction of motion must be considered, too. In this simplified model, we use equations (6.1) and (6.2) again and multiply by the length of the plate,

$$m_{a,\text{front}} = m'_{a,\text{spring}}l_2, \quad (6.8)$$

and

$$\gamma_{a,\text{front}} = \gamma'_{a,\text{spring}}l_2. \quad (6.9)$$

The quality factor Q of a resonator is given by

$$Q = 2\pi \frac{W_{\text{kin}}}{W_{\text{loss},T}}, \quad (6.10)$$

where W_{kin} represents the peak kinetic energy of the system, and $W_{\text{loss},T}$ is the dissipated energy per cycle (per time period $T = 2\pi/\omega$) [62].

We approximate the two pairs of springs as clamped–clamped beams, vibrating at their first mode of vibration. The kinetic energy contribution of such a beam immersed in liquid is

$$\begin{aligned} W_{\text{beam}} &= \frac{1}{2} \int_0^l m' \left[\omega \hat{\psi} \varphi(y) \right]^2 dy = \\ &= \frac{1}{2} (\rho_d w_2 h + m'_{a,\text{spring}}) \omega^2 \hat{\psi}^2 \int_0^l \varphi^2(y) dy, \end{aligned} \quad (6.11)$$

where $\hat{\psi}$ is the peak deflection of the resonator in x -direction, $l = 2l_1$ is the length of the beam, $\varphi(y)$ is the normalized mode shape of first mode of resonance of a clamped–clamped rectangular beam, and ρ_d is the mass density of the device. $\varphi(y)$ is calculated by applying the boundary conditions $\varphi(0) = 0$, $d\varphi(0)/dy = 0$, $\varphi(l) = 0$, $d\varphi(l)/dy = 0$, to the Euler-Bernoulli beam equation [63], yielding

$$\begin{aligned} \varphi(y) &= 0.6186 \sin\left(\frac{4.73}{l}y\right) - 0.6297 \cos\left(\frac{4.73}{l}y\right) \\ &\quad - 0.6186 \sinh\left(\frac{4.73}{l}y\right) + 0.6297 \cosh\left(\frac{4.73}{l}y\right), \end{aligned} \quad (6.12)$$

and

$$\int_0^l \varphi^2(y) dy = \alpha l, \quad (6.13)$$

with $\alpha = 0.39714$. It should be noted that $\varphi(y)$ represents only an approximation of the actual mode shape of the beam springs. The rectangular plate suspended by the springs acts as a concentrated force at $y = l/2$, and should therefore be considered by an improved model.

In a similar way, the peak kinetic energy in the plate suspended by the springs can be derived as

$$W_{\text{plate}} = \frac{1}{2} (l_2 w_1 h \rho_d + m_{\text{a,shear}} + m_{\text{a,front}}) \omega^2 \hat{\psi}^2. \quad (6.14)$$

The total kinetic energy is

$$W_{\text{kin}} = 2W_{\text{beam}} + W_{\text{plate}}. \quad (6.15)$$

Again, we consider the two pairs of springs as two rectangular clamped-clamped beams. The averaged dissipated power per unit length due to the liquid surrounding such a beam is

$$P' = \frac{1}{2} \gamma'_{\text{a,spring}} \omega^2 \hat{\psi}^2 \varphi^2(y), \quad (6.16)$$

where $\gamma'_{\text{a,spring}}$ is the damping coefficient according to (6.2). Integrating equation (6.16) over the time period T and the length of the beam yields

$$W_{\text{loss,beam}} = \int_0^l \int_0^T P' dt dy = \pi \omega \hat{\psi}^2 \gamma'_{\text{a,spring}} \alpha l. \quad (6.17)$$

Accordingly, the dissipated energy of the vibrating plate is

$$W_{\text{loss,plate}} = \pi \omega \hat{\psi}^2 (\gamma_{\text{a,shear}} + \gamma_{\text{a,front}}). \quad (6.18)$$

Hence, the total energy loss per cycle is

$$W_{\text{loss},T} = 2W_{\text{loss,beam}} + W_{\text{loss,plate}}. \quad (6.19)$$

The resonator device itself is considered free of intrinsic losses, which is justified by the high quality factor of the MEMS device vibrating in air.

Equation (6.10), therefore, results in

$$Q = \omega \frac{2(\rho_d w_2 h + m'_{\text{a,spring}}) \alpha l + \rho_d l_2 w_1 h + m_{\text{a,shear}} + m_{\text{a,front}}}{2\gamma'_{\text{a,spring}} \alpha l + \gamma_{\text{a,shear}} + \gamma_{\text{a,front}}}. \quad (6.20)$$

At resonance frequency, the peak kinematic energy of the resonator, W_{kin} , equals the peak strain energy of the spring beams [62]. This strain energy associated with the bending of the beams is

$$W_{\text{strain}} = EI\hat{\psi}^2 \int_0^l \left[\frac{d^2\varphi(y)}{dy^2} \right]^2 dy = \frac{EI\hat{\psi}^2\beta}{l^3}, \quad (6.21)$$

where E and $I = tw_2^3/12$ are the Young's modulus and the area moment of inertia, respectively [63], and the mode-shape factor $\beta = 198.782$. From $W_{\text{kin}}(\omega_0) = W_{\text{strain}}$, equations (6.15), and (6.21), the angular resonance frequency of the first mode of the suspended plate device is derived yielding

$$\omega_0^2 = \frac{2EI\beta}{l^3 \left[2(\rho_d w_2 h + m'_{\text{a,spring}})\alpha l + \rho_d l_2 w_1 h + m_{\text{a,shear}} + m_{\text{a,front}} \right]}. \quad (6.22)$$

It should be noted, that the additional masses due to the liquid loading depend on the angular frequency. Therefore, equation (6.22) represents an implicit equation for ω_0 .

6.4 Modelling Results

Based on the derived model, a variety of suspended plate sensors with different dimensions have been designed and fabricated. Table 6.1 summarizes the geometrical and mechanical parameters of the sensor that was used in the first experiments. Using equations (6.20) and (6.22), we can estimate the quality factor and the resonance frequency of the sensor. In air (viscosity $\eta = 1.8 \cdot 10^{-5}$ Pa·s, density $\rho = 1.2$ kg/m³), the calculated first mode resonance frequency is 17.2 kHz and the quality factor is 292. As sample liquids, a variety of alcohols were used in the measurements. Table 6.2 lists their dynamic viscosities and mass densities. Immersed in these liquids, the plate resonator's resonance frequency f_0 will be in the range of 12.7 kHz–9.4 kHz according to the model and the quality factor Q will drop to 3.1 for ethanol and 1 for decanol (Table 6.3). The calculated resonance frequencies are thus in a similar range as those of the clamped–clamped beam sensor (see Figure 5.24). The quality factors predicted by the models are fairly low, but it will be shown that the achieved deflection amplitudes are still high enough for good readout sensitivity.

Another interesting modelling result is given by the added masses and damping coefficients due to the liquid loading. Table 6.3 lists the contribu-

Table 6.1: Geometrical and mechanical parameters of the sensor device (Figure 6.6). The sensor is labelled AD05 N-14 on the wafer mask. The neutral axes of the beam springs coincide with the $[1\ 1\ 0]$ crystal direction of the device. In this direction, the Young's modulus of silicon is 169 GPa [64].

Parameter	Value	
w_1	100	μm
w_2	5	μm
L_1	600	μm
L_2	100	μm
h	20	μm
ρ_d	2330	kg/m^3
E	169	GPa

Table 6.2: Dynamic viscosity and mass density of the sample liquids used for model calculations and experiments [20, 51].

Liquid	η [mPa·s]	ρ [kg/m ³]
ethanol	1.06	789
1-propanol	1.90	804
1-butanol	2.52	810
1-pentanol	3.44	814
1-hexanol	4.49	814
1-heptanol	5.94	822
1-octanol	7.37	827
1-nonanol	9.10	827
1-decanol	10.97	830

Table 6.3: Added mass and damping coefficients at resonance frequency for the suspended plate resonator according to Table 6.1 and the liquid parameters of Table 6.2. For comparison, the effective mass and the damping coefficient of the spring are given as $m_{a,\text{spring}} = 2m'_{a,\text{spring}}\alpha l$ and $\gamma_{a,\text{spring}} = 2\gamma'_{a,\text{spring}}\alpha l$, see also equation (6.20).

Liquid	f_0 [kHz]	Q	$m_{a,\text{spring}}$ [ng]	$m_{a,\text{shear}}$ [ng]	$m_{a,\text{front}}$ [ng]	$\gamma_{a,\text{spring}}$ [10 ⁻⁶ kg/s]	$\gamma_{a,\text{shear}}$ [10 ⁻⁶ kg/s]	$\gamma_{a,\text{front}}$ [10 ⁻⁶ kg/s]
ethanol	12.66	3.14	514	22.9	53.9	27.7	1.82	2.91
1-propanol	12.06	2.34	629	31.7	66.0	39.2	2.40	4.11
1-butanol	11.68	2.04	701	37.3	73.6	46.5	2.74	4.88
1-pentanol	11.30	1.75	794	44.4	83.3	56.1	3.15	5.89
1-hexanol	10.92	1.54	888	51.6	93.1	66.0	3.54	6.92
1-heptanol	10.46	1.35	1015	60.9	106	78.8	4.01	8.26
1-octanol	10.12	1.22	1128	69.2	118	90.4	4.40	9.48
1-nonanol	9.78	1.11	1253	78.3	132	103	4.81	10.9
1-decanol	9.44	1.02	1386	87.6	145	117	5.20	12.2

tions of the beam springs ($m_{a,\text{spring}}$, $\gamma_{a,\text{spring}}$), the rectangular plate ($m_{a,\text{shear}}$, $\gamma_{a,\text{shear}}$), and the plate front face ($m_{a,\text{front}}$, $\gamma_{a,\text{front}}$) as defined in Section 6.3. The results indicate that the influence of the beam springs on the overall damping is much higher than that of the suspended plate itself. Increasing the width of the suspended plate w_1 would therefore have a low impact on the power loss per vibration cycle. However, the associated increase of the moving mass would clearly yield a higher quality factor, see equation (6.20). Given the same geometry of the beam springs, and therefore the same spring constant, the resonance frequency is decreased, making the sensor feasible for non-Newtonian liquids. As an example, a different sensor geometry is designed using the model equations: Shorter beam springs ($L_1 = 350\mu\text{m}$) and a larger rectangular plate ($L_2 = 150\mu\text{m}$, $w_1 = 500\mu\text{m}$) are considered. The mass of such a plate would be $3.5\mu\text{g}$. Surrounded by the same set of sample liquids (Table 6.2), the new sensor exhibits resonance frequencies in the range from 15.78 kHz to 14.34 kHz, and strongly increased quality factors in the range from 10.9 to 3.1.

6.5 Lorentz Force Excitation

For the excitation of MEMS-based viscosity sensors, the use of Lorentz forces has some advantages over other actuating principles. Piezoelectric drives require additional deposition processes to create the actuators and sometimes also subsequent polarisation steps. Electrostatic excitation would eliminate the need for such materials, but the common finger structures are accompanied by squeeze film damping, which decrease the quality factor. Electrothermal actuation could be utilized, but since the viscosity of liquids strongly changes with temperature, the influence of the heater resistors on the viscosity must be considered. Lorentz force excitation is easily accomplished, as only a conductive layer is required on the device to carry the excitation current. However, an external permanent magnet is needed to provide the magnetic field. With modern materials, flux densities of 200 mT and above are easily achieved. These flux densities allow to keep the excitation current small. From experiences with the clamped-clamped beam device of [45], a maximum excitation current in the range of several mA can be estimated.

To calculate the deflection amplitudes of the suspended plate a simple finite element (FEM) simulation was done. The geometry of the device (see Table 6.1 and Figure 6.2) was modelled in Comsol Multiphysics. In the *2D Conductive Media DC* application mode, the current distribution was simu-

lated. The resulting current densities $\vec{J} = (J_x, J_y)$ were then used to calculate the force per unit area acting on the resonator, $\vec{J} \times \vec{B}$. The mechanical deflection was obtained from Comsol Multiphysics' *Plain Strain* application mode and using the material parameters of single-crystal silicon. A dc excitation current of 1 mA on each spring was applied and a magnetic flux density of $B_z = 200$ mT was assumed. The simulation resulted in a static plate deflection of 36 nm.

6.6 Piezoresistive Readout

For the measurement of the resonance behaviour of the sensor, it is crucial to obtain the actual plate deflection. In Chapter 5 an optical readout based on a DVD player pickup head was presented. However, the readout is highly sensitive to ambient vibrations and can not be integrated on the sensor chip. As discussed in Section 4.1 several effects can be utilized by devices fabricated in micromachining technology. In Section 5.2 an inductive readout using the voltage induced in a conductor moving in a magnetic field was discussed, but the voltage magnitude was estimated to be very low and an insufficient signal-to-noise ratio was expected. The deflection of MEMS sensors like accelerometers is often detected by a capacitive readout. However, for operation in a liquid environment, the required finger structures will lead to a high squeeze film damping and reduce the quality factor of the resonance. In addition, the permittivity of the sample liquid would affect the output signal in some readout configurations. The use of piezoelectricity would be beneficial as it could be used for the excitation of the device vibration as well, but requires the deposition of piezoelectric layers and sometimes also subsequent polarization steps. The sensor design presented in this chapter is based on the piezoresistance effect of silicon. From the p-doped silicon device layer, piezoresistors were fabricated in a three-mask fabrication process.

6.6.1 The Piezoresistance Effect

The piezoresistance effect describes the change of an electrical resistance due to a mechanical compression or tension [65]. Under zero stress, the resistance of a rectangular conductor (Figure 6.7) is

$$R = \rho_0 \frac{l}{A}, \quad (6.23)$$

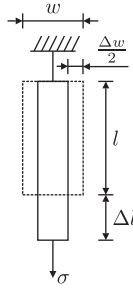


Figure 6.7: Deformation of a rectangular conductor due to a longitudinal stress σ . The dashed lines depict the undeformed bar.

where ρ_0 is the resistivity and l and $A = wh$ are the length and the cross-sectional area of the conductor, whereas w and h denominate its width and thickness, respectively. When the conductor is stretched, the length is increased, while A decreases. The relative change in resistance is

$$\frac{\Delta R}{R} = \frac{\Delta l}{l} - \frac{\Delta w}{w} - \frac{\Delta h}{h} \quad (6.24)$$

for small changes $\Delta l/l$, $\Delta w/w$, and $\Delta h/h$. The Young's modulus E relates the applied mechanical stress σ to $\Delta l/l$,

$$\frac{\Delta l}{l} = \frac{\sigma}{E}. \quad (6.25)$$

The contraction of the conductor normal to the stress is calculated using the Poisson's ratio ν [66],

$$\frac{\Delta w}{w} = \frac{\Delta h}{h} = -\nu \frac{\Delta l}{l}. \quad (6.26)$$

From (6.24), (6.25), and (6.26), the gauge factor K_G , i.e., the sensitivity of the conductor's resistance to the mechanical strain $\Delta l/l = \sigma/E$, is obtained as

$$\frac{\Delta R}{R} = K_G \frac{\sigma}{E} = (1 + 2\nu) \frac{\sigma}{E}. \quad (6.27)$$

For semiconductors gauges, however, it has been found that the change of resistance is larger than that expected from a pure dimensional change by a factor of 50 [65,66]. Mechanical stress causes a change of electron mobility in

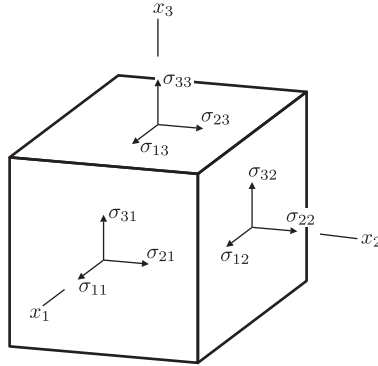


Figure 6.8: The stresses on the faces of a unit cube in a stressed body [67]. Note that the hidden faces are also subject to mechanical stress. For reasons of mechanical equilibrium the stresses acting on opposite sides of the cube are equal in magnitude and opposite in direction.

the conductor due to anisotropic effects. This effect, altering the resistivity in addition to the dimensional changes, is called the *piezoresistance effect* [65].

The stresses acting in an body are properly described by a tensor of the second rank with the components $\sigma_{\alpha\beta}$, $\alpha = 1, 2, 3$, $\beta = 1, 2, 3$. Figure 6.8 depicts these stress components, i.e., the forces acting on the faces of a unit cube in a stressed body [67]. x_1 , x_2 , and x_3 denote the three orthogonal axes and $\sigma_{\alpha\beta}$ is a stress acting in the x_α -direction on the face of the cube which is perpendicular to x_β . σ_{11} , σ_{22} , and σ_{33} are the normal stresses and σ_{12} , σ_{21} etc. are the shear stresses. From equilibrium conditions, we have $\sigma_{\alpha\beta} = \sigma_{\beta\alpha}$, i.e., $\sigma_{12} = \sigma_{21}$, $\sigma_{13} = \sigma_{31}$, and $\sigma_{23} = \sigma_{32}$. Hence, the stress on the infinitesimal small cube is properly described by six components,

$$[\sigma_{\alpha\beta}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{12} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{23} & \sigma_{33} \end{bmatrix} = \begin{bmatrix} \sigma_1 & \sigma_6 & \sigma_5 \\ \sigma_6 & \sigma_2 & \sigma_4 \\ \sigma_5 & \sigma_4 & \sigma_3 \end{bmatrix}. \quad (6.28)$$

These six components are conveniently written as a vector

$$\boldsymbol{\sigma} = (\sigma_1 \quad \sigma_2 \quad \sigma_3 \quad \sigma_4 \quad \sigma_5 \quad \sigma_6)^T. \quad (6.29)$$

Under zero stress, the resistivity of cubic crystals like silicon is isotropic [67]. The electric field vector $\mathbf{E} = (E_1 \ E_2 \ E_3)^T$ is parallel to the current

density $\mathbf{J} = (J_1 \ J_2 \ J_3)^T$. The magnitudes of the vectors are proportional and the proportionality factor, the electric resistivity ρ_0 , is a scalar,

$$\mathbf{E} = \rho_0 \mathbf{J}. \quad (6.30)$$

When a stress is applied to the material, the resistivity changes due to the piezoresistance effect. As the piezoresistance effect is anisotropic, the current density and the electric field are not necessarily parallel to each other and the scalar ρ_{el} is replaced by a tensor of the second rank,

$$[\rho_{\alpha\beta}] = \begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{13} \\ \rho_{21} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{32} & \rho_{33} \end{bmatrix}. \quad (6.31)$$

Instead of (6.30) we have

$$E_\alpha = \sum_{\beta=1}^3 \rho_{\alpha\beta} J_\beta \quad (6.32)$$

or, using the Einstein summation convention¹, that is leaving out the summation sign,

$$E_\alpha = \rho_{\alpha\beta} J_\beta, \quad (6.33)$$

where $\rho_{\alpha\beta}$ are the elements of the second rank tensor, (6.31). For thermodynamic reasons $\rho_{\alpha\beta} = \rho_{\beta\alpha}$ and the resistivity tensor has only 6 independent components [67],

$$\begin{bmatrix} \rho_{11} & \rho_{12} & \rho_{31} \\ \rho_{12} & \rho_{22} & \rho_{23} \\ \rho_{31} & \rho_{23} & \rho_{33} \end{bmatrix} \longrightarrow \begin{bmatrix} \rho_1 & \rho_6 & \rho_5 \\ \rho_6 & \rho_2 & \rho_4 \\ \rho_5 & \rho_4 & \rho_3 \end{bmatrix}. \quad (6.34)$$

Again, these six components are conveniently written as a vector,

$$\boldsymbol{\rho} = (\rho_1 \ \rho_2 \ \rho_3 \ \rho_4 \ \rho_5 \ \rho_6)^T. \quad (6.35)$$

In case of the unstressed silicon that is isotropic with respect to the electrical resistivity, the resistivity vector becomes

$$\boldsymbol{\rho}_0 = (\rho_0 \ \rho_0 \ \rho_0 \ 0 \ 0 \ 0)^T. \quad (6.36)$$

¹“When a letter suffix occurs twice in the same term, summation with respect to that suffix is to be automatically understood” (from [67]).

If the semiconductor is subject to a stress σ , the change of the components of the resistivity is²

$$\frac{\Delta\rho_\alpha}{\rho_{el}} = \pi_{\alpha\beta}\sigma_\beta, \quad (6.37)$$

where ρ_α and σ_β are the components of the resistivity vector and the stress vector, respectively. $\pi_{\alpha\beta}$ are the components of the piezoresistance matrix of silicon π [66],

$$\pi = \begin{pmatrix} \pi_{11} & \pi_{12} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{11} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{12} & \pi_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \pi_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \pi_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \pi_{44} \end{pmatrix}. \quad (6.38)$$

The resulting resistivity vector of a piezoresistive element is, therefore,

$$\rho = \rho_0 + \rho_0 \pi \sigma, \quad (6.39)$$

where ρ_0 , π , and σ are defined by (6.36), (6.38), and (6.29), respectively.

For a direction given by the direction cosines d_1 , d_2 , and d_3 with respect to the axes x_1 , x_2 , and x_3 , a scalar resistivity can be found [67]. To calculate this resistivity a current density with the magnitude J is applied in this direction and we have

$$\mathbf{J} = \begin{pmatrix} d_1 J & d_2 J & d_3 J \end{pmatrix}^T. \quad (6.40)$$

The resulting electric field is, with (6.33),

$$E_\alpha = \rho_{\alpha\beta} d_\beta J. \quad (6.41)$$

The component of $\mathbf{E}$ parallel to $\mathbf{J}$ is calculated using the scalar product,

$$E_{\parallel} = \frac{\mathbf{E} \cdot \mathbf{J}}{J} = \frac{E_\alpha J_\alpha}{J} = E_\alpha d_\alpha = \rho_{\alpha\beta} d_\alpha d_\beta J, \quad (6.42)$$

and the scalar resistivity in the direction $(d_1 \ d_2 \ d_3)^T$ is

$$\rho = \frac{E_{\parallel}}{J} = \rho_{\alpha\beta} d_\alpha d_\beta. \quad (6.43)$$

²Note that the Einstein summation convention is used here again. The suffix β appears twice in the right hand term, which implies summation of the term with respect to β .

In principle, the anisotropic nature of the piezoresistance effect allows numerous configurations of the sensor readout with different orientations of the current density, the electric field, and the mechanical stress [68]. In the following only designs will be considered featuring the electric field and the current density oriented in the same direction. Hence, the number of required electrical connections is minimized. This reduction is crucial, as the silicon structures which would carry these connections would bear a part of the mechanical load and, therefore, would reduce the mechanical stress in the piezoresistor.

When the rectangular plate of the sensor device is deflected, the piezoresistors are subject to either compressive or tensile stress (Figure 6.2). This stress is oriented in the same direction as the electric field and the current density. In this case, the piezoresistive effect can be described by the simple equation

$$E = \rho_0(1 + \pi_1\sigma_1)J, \quad (6.44)$$

where ρ_0 is the resistivity at zero stress, σ_1 is the applied mechanical stress, and π_1 is the longitudinal piezoresistance coefficient [69]. Now we must determine this coefficient π_1 . We choose a set of axes x'_1 , x'_2 , and x'_3 . The x'_1 -axis coincides with the orientation of the piezoresistive element, i.e., the direction of the current density, the electric field, and the mechanical stress. The current that flows through the piezoresistor is given by the current density

$$\mathbf{J}' = \begin{pmatrix} J & 0 & 0 \end{pmatrix}^T. \quad (6.45)$$

A uniaxial stress

$$[\sigma'_{\alpha\beta}] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (6.46)$$

acts on the piezoresistor. In (6.38) the piezoresistance matrix π was given with respect to the three unprimed axes x_1 , x_2 , and x_3 which coincide with the crystallographic directions $[1\ 0\ 0]$, $[0\ 1\ 0]$, and $[0\ 0\ 1]$ of the silicon crystal. To calculate the change of resistivity, the current density and the mechanical stress must be transformed to this set of axes. The transformation laws for the vector $\mathbf{J}'$ and the second rank tensor $[\sigma'_{\alpha\beta}]$ are [67]

$$J_\alpha = a_{\beta\alpha}J'_\beta \quad \text{and} \quad (6.47)$$

$$\sigma_{\alpha\beta} = a_{\gamma\alpha}a_{\delta\beta}\sigma'_{\gamma\delta}, \quad (6.48)$$

where $a_{\alpha\beta}$ are the elements of the transformation matrix [69]

$$(a_{\alpha\beta}) = \begin{pmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{pmatrix}. \quad (6.49)$$

In this transformation matrix, the elements of row α , these are l_α , m_α , and n_α , are the direction cosines of the x'_α axis with respect to the unprimed set of axes x_1 , x_2 , and x_3 [67]. For the transformed stress we have

$$[\sigma_{\alpha\beta}] = \begin{bmatrix} l_1^2\sigma_1 & l_1m_1\sigma_1 & l_1n_1\sigma_1 \\ l_1m_1\sigma_1 & m_1^2\sigma_1 & m_1n_1\sigma_1 \\ l_1n_1\sigma_1 & m_1n_1\sigma_1 & n_1^2\sigma_1 \end{bmatrix} \quad (6.50)$$

and for the current density

$$\mathbf{J} = \begin{pmatrix} l_1J & m_1J & n_1J \end{pmatrix}^T. \quad (6.51)$$

Applying the stress to (6.39) with the suffix substitutions of (6.28) yields electrical resistivity

$$\rho = \begin{pmatrix} \rho_0[1 + \sigma_1(l_1^2\pi_{11} + m_1^2\pi_{12} + n_1^2\pi_{12})] \\ \rho_0[1 + \sigma_1(l_1^2\pi_{12} + m_1^2\pi_{11} + n_1^2\pi_{12})] \\ \rho_0[1 + \sigma_1(l_1^2\pi_{12} + m_1^2\pi_{12} + n_1^2\pi_{11})] \\ \rho_0\sigma_1m_1n_1\pi_{44} \\ \rho_0\sigma_1l_1n_1\pi_{44} \\ \rho_0\sigma_1l_1m_1\pi_{44} \end{pmatrix} \quad (6.52)$$

with respect to the unprimed (crystallographic) axes. With (6.43), the electrical resistivity in the direction of current density and stress can be calculated by using the direction cosines $d_1 = l_1$, $d_2 = m_1$, and $d_3 = n_1$, yielding

$$\begin{aligned} \rho &= \rho_{\alpha\beta}a_{1\alpha}a_{1\beta} = \\ &= \rho_0l_1^2 + \rho_0\sigma_1(l_1^2\pi_{11} + m_1^2\pi_{12} + n_1^2\pi_{12})l_1^2 + \\ &\quad \rho_0m_1^2 + \rho_0\sigma_1(l_1^2\pi_{12} + m_1^2\pi_{11} + n_1^2\pi_{12})m_1^2 + \\ &\quad \rho_0n_1^2 + \rho_0\sigma_1(l_1^2\pi_{12} + m_1^2\pi_{12} + n_1^2\pi_{11})n_1^2 + \\ &\quad 2\rho_0\sigma_1\pi_{44}(l_1^2m_1^2 + m_1^2n_1^2 + n_1^2l_1^2). \end{aligned} \quad (6.53)$$

Since $l_1^2 + m_1^2 + n_1^2 = 1$ we obtain the longitudinal piezoresistance coefficient as [69]

$$\pi_1 = \pi_{11} - 2(\pi_{11} - \pi_{12} - \pi_{44})(l_1^2m_1^2 + m_1^2n_1^2 + n_1^2l_1^2). \quad (6.54)$$

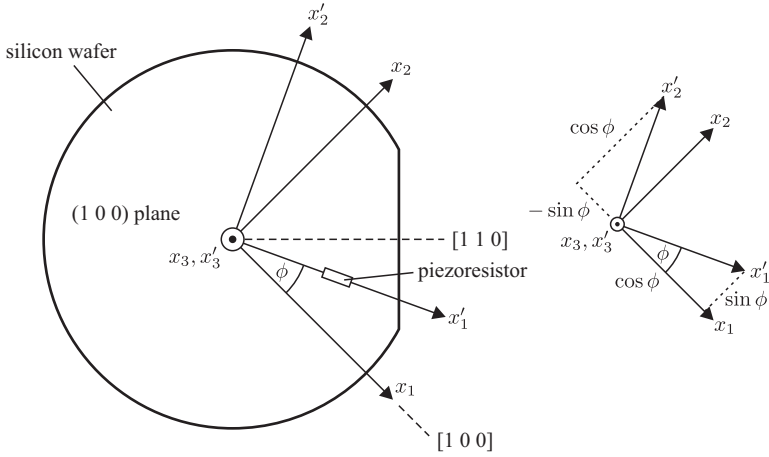


Figure 6.9: Orientation of a piezoresistive element on an (100) silicon wafer. x_1 , x_2 , and x_3 are the crystal axes of the silicon wafer [70]. The piezoresistor is oriented in the direction x'_1 specified by the angle ϕ . It is characterized by the longitudinal piezoresistance coefficient π_1 , describing the relation between a current density, an electric field, and a mechanical stress in the x'_1 direction. The figure on the right hand side depicts the determination of the transformation matrix $(a_{\alpha\beta})$ between the primed and the unprimed axes, (6.55).

6.6.2 Design of the Piezoresistors

The sensor device was fabricated on an SOI wafer with a (100) device layer. The orientation of the crystal axes x_1 , x_2 , and x_3 on such a wafer are depicted in Figure 6.9. The piezoresistor is oriented in the x'_1 -direction. To calculate the longitudinal piezoresistance coefficient π_1 , (6.54), the transformation matrix, (6.49), between the primed and the unprimed (crystallographic) axes is required. From Figure 6.9, this matrix is obtained as

$$(a_{\alpha\beta}) = \begin{pmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6.55)$$

Equation (6.54) results in

$$\pi_1 = \pi_{11} - 2(\pi_{11} - \pi_{12} - \pi_{44}) \cos^2 \phi \sin^2 \phi. \quad (6.56)$$

The values of π_{11} , π_{12} , and π_{44} are listed in Table 6.4. For the fabrication

Table 6.4: Piezoresistance coefficients of n-doped silicon (resistivity $11.7 \Omega\text{cm}$) and p-doped silicon (resistivity $7.8 \Omega\text{cm}$) [66] at room temperature.

Coefficient	n-Si [10^{-11} Pa^{-1}]	p-Si [10^{-11} Pa^{-1}]
π_{11}	-102.2	6.6
π_{12}	53.4	-1.1
π_{44}	-13.6	138.1

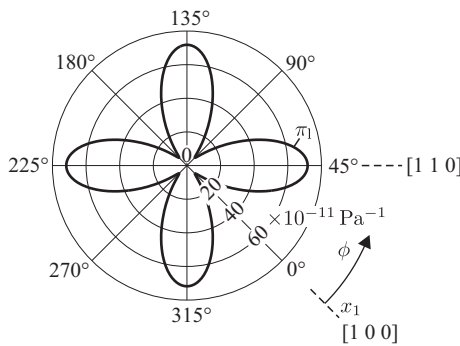


Figure 6.10: Longitudinal piezoresistance coefficient π_l on a p-doped (100) silicon wafer. The diagram shows the dependence of π_l from the orientation of the piezoresistor on the wafer specified by the angle ϕ (see Figure 6.9). The plot was obtained from the evaluation of (6.56) using the data from Table 6.4.

of the sensors, p-doped silicon was chosen, since it allows the formation of ohmic contacts between the semiconductor and the aluminium metallization [71]. Figure 6.10 shows the evaluation of (6.56) for angles ϕ from 0 to 360° . The diagram indicates that the maximum piezoresistance effect can be utilized by orienting the piezoresistors at $\phi = 45^\circ, 135^\circ, 225^\circ$, or 315° , that is in the $[1\ 1\ 0]$ and equivalent directions. In these directions, a coefficient of

$$\pi_l = 71.8 \cdot 10^{-11} \text{ Pa}^{-1} \quad (6.57)$$

is determined. The change of resistance of the piezoresistor is

$$\Delta R = \pi_l \sigma_1 R_0, \quad (6.58)$$

where R_0 is the resistance at zero stress.

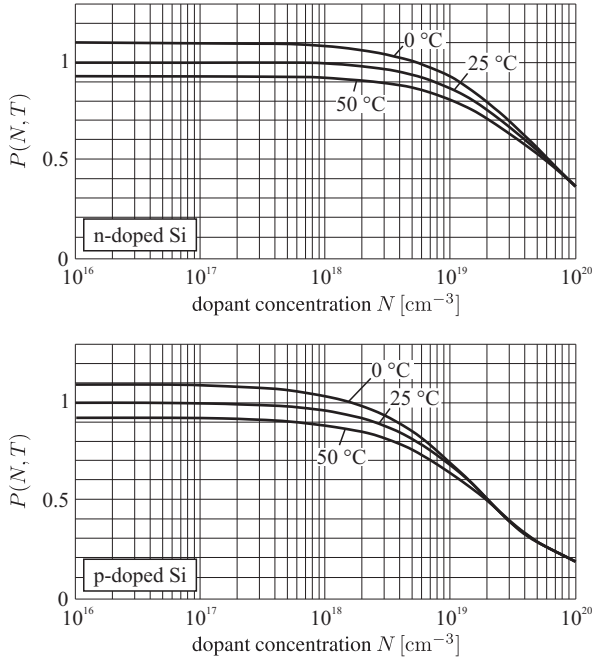


Figure 6.11: The function $P(N, T)$ illustrates the dependence of the piezoresistance coefficients of n- and p-doped silicon from the temperature T and the dopant concentration N (data from [72]).

The piezoresistance coefficients of silicon decreases, as either the dopant concentration N or the temperature T increases. In [66], this decrease is illustrated by a function $P(N, T)$,

$$\pi_{\alpha\beta} = P(N, T) \pi_{\alpha\beta, 300 \text{ K}}, \quad (6.59)$$

where $\pi_{\alpha\beta, 300 \text{ K}}$ are the coefficients from Table 6.4. Figure 6.11 indicates, that silicon exhibits maximum piezoresistive coefficients for dopant concentration of 10^{17} cm^{-3} or smaller (p-doped) and $5 \cdot 10^{17} \text{ cm}^{-3}$ or smaller (n-doped). In this range, $P(N, 300 \text{ K}) \approx 1$.

The formation of ohmic contacts between the semiconductor and the metal layer is crucial to reduce crosstalk from the excitation current (see the discussion in the next section). The dopant type and concentration have a high influence on the behaviour of the electric contact. In general, higher dopant

concentrations lead to ohmic behaviour, whereas for lower concentrations the contacts are of rectifying type. Very good results have been obtained from depositing aluminium layers on p-doped silicon. Applying subsequent thermal annealing leads to a highly p-doped region beneath the contact pads. Hence, the metal–semiconductor contacts show ohmic sheet resistances [71]. Therefore, p-doped silicon with a dopant concentration of 10^{17} cm^{-3} has been chosen. This concentration and dopant type leads to fairly high piezoresistive coefficients, and allows the formation of ohmic contacts to aluminium.

6.6.3 Wheatstone Bridge Circuit

Two of the beam springs suspending the plate are forked at their ends (Figure 6.2). When the plate is deflected, one of the prongs is compressed while the other is strained. Furthermore, one of the prongs carries a conductive layer and serves as ground connection for the excitation current, whereas the second one is not metallized, but is provided with electrical contacts to form a piezoresistive element [57].

Together with two additional fixed piezoresistors, a Wheatstone bridge circuit is formed and driven by the readout voltage u_r . Assuming that their resistances are equal, the resulting output voltage is

$$u_d(t) = \frac{1}{2} u_r \frac{\Delta R(t)}{R_0} = \frac{1}{2} u_r \pi_1 \sigma_1(t), \quad (6.60)$$

where π_1 is the longitudinal piezoresistance coefficient and σ_1 is the mechanical stress in the piezoresistor, see (6.58). Driving the Wheatstone bridge with a dc voltage u_r , exciting the resonator vibrations at a frequency f_e , and assuming linear operation of the sensor results in an output voltage

$$u_d = \hat{u}_d \cos(2\pi f_e + \phi), \quad (6.61)$$

which is proportional to the plate deflection. Here, ϕ represents the phase shift between the excitation current and the plate deflection. Since the excitation frequency is well-known, a lock-in amplifier can be used to separate the voltage amplitude of interest from a possible dc offset that stems from inequalities of the piezoresistors. However, the sensor readout will be disturbed by interference from the excitation current. Like in most electric systems, such interference can be caused by inductive or capacitive coupling. For the device layout given in Figure 6.2, another possible source of crosstalk must be considered: The excitation loop and the Wheatstone bridge use a common

ground connection with a resistance of approximately 3–4 Ω . In this ground connection the excitation current causes a voltage drop of several mV. Since the piezoresistors are not perfectly equal, this voltage drop will not be cancelled out completely in the Wheatstone bridge and will therefore interfere with the sensor readout.

It is, therefore, beneficial to drive the piezoresistors with a sinusoidal voltage at the readout frequency f_r , $u_r = \hat{u}_r \cos(2\pi f_r t)$. Then the output voltage of the Wheatstone bridge is

$$\begin{aligned} u_d &= \frac{1}{2} \pi_1 u_r(t) \sigma_1(t) = \frac{1}{2} \pi_1 \hat{u}_r \cos(2\pi f_r t) \hat{\sigma}_1 \cos(2\pi f_e t + \phi) = \\ &= \frac{1}{4} \pi_1 \hat{u}_r \hat{\sigma}_1 \{ \cos[2\pi(f_r - f_e)t - \phi] + \cos[2\pi(f_r + f_e)t + \phi] \}. \end{aligned} \quad (6.62)$$

Now, the plate deflection can be obtained from setting a lock-in amplifier to either $|f_r - f_e|$ or $|f_r + f_e|$ and the crosstalk from the excitation current to the sensor output is eliminated.

The dimensions of the piezoresistors are $50 \times 3 \times 20 \mu\text{m}^3$. The device layer of the SOI wafer, from which the piezoresistors are fabricated, is p-doped silicon with a dopant concentration of approximately 10^{17} cm^{-3} . This dopant concentration yields an optimum piezoresistive coefficient [66], and also allows the formation of ohmic contacts to the metal layer. The conductivity ρ_{el} of the device layer is in the range from 0.11 Ωcm to 0.21 Ωcm . Calculating the resistance yields

$$R_{\text{calc}} = \frac{\rho_{\text{el}} 50 \mu\text{m}}{3 \cdot 20 \mu\text{m}^2} = 917 \dots 1750 \Omega. \quad (6.63)$$

Since the quality factor of the resonance in liquid is expected to be very low, i.e., from 2 to 10, it is crucial for the piezoresistive readout to be sensitive enough to detect static deflections. The finite element simulation described in Section 6.5 resulted in a mean mechanical stress in the piezoresistors of $\sigma_1 = 74 \text{ kPa}$. Using a readout voltage amplitude $\hat{u}_r$ of 1 V, the resulting output voltage detected at the differential frequency $|f_e - f_r|$ is

$$\hat{u}_{d,|f_e-f_r|} = \frac{1}{4} \pi_1 \hat{u}_r \hat{\sigma}_1 = 13.3 \mu\text{V}. \quad (6.64)$$

6.6.4 Consequences of Nonlinear Contact Resistances

The first generation of devices was fabricated using a Cr–Au–Cr metallization instead of the aluminium contacts used later (see the fabrication process depicted in Figure 6.3). The resistances of the piezoresistive elements

were measured by recording their voltage–current characteristics. The measurements showed strong nonlinear characteristics, which were assumed to stem from the metal–semiconductor contacts and possibly from a remaining thin oxide–nitride layer between the metal and the silicon. Below, the consequences arising from these nonlinear contact resistances will be discussed.

The device³ used in a first experiment featured beam geometries of $l_1 = 300\text{ }\mu\text{m}$, $w_2 = 5\text{ }\mu\text{m}$, $h = 20\text{ }\mu\text{m}$ (see Figure 6.6) and a plate of $l_2 = 100\text{ }\mu\text{m}$, $w_1 = 150\text{ }\mu\text{m}$. The size of the piezoresistors was $50 \times 3 \times 20\text{ }\mu\text{m}^3$. The sensor exhibited a first mode resonance at 19 kHz. The excitation loop was driven by a voltage of 1 V amplitude at this frequency. With the 20 k Ω series resistors (Figure 6.2) the excitation current amplitude was $\hat{i}_e = 50\text{ }\mu\text{A}$. The readout circuit was driven at a different frequency, $f_r = 99.4\text{ kHz}$, and a voltage amplitude of $\hat{u}_r = 1\text{ V}$. As described in Section 6.6.3, the differential output voltage has spectral components at f_r due to the inequalities of the piezoresistors and at f_e due to the 3 Ω resistance of the common ground connection. According to equation (6.62), a deflection of the plate at the excitation frequency f_e leads to spectral components at $f_r - f_e$ and $f_r + f_e$. A lock-in amplifier was locked to the differential frequency $f_r - f_e = 80.4\text{ kHz}$ and measured the output voltage u_d . No magnetic field was provided in the experiment, therefore lock-in amplifier output voltage should vanish. Nevertheless, a voltage of approximately 6 μV was detected by the lock-in amplifier. When the permanent magnet was added to the setup, $B_z \approx 300\text{ mT}$, this voltage changed to approximately 6.5 μV . An analysis of the measurement setup leads to the strong suspicion, that the nonlinear characteristics of the piezoresistors causes an additional mixing effect besides equation (6.62).

The characteristics of the sensor's piezoresistors were determined using a Keithley 236 Source Measure Unit. The measurement was carried out in pulsed sweep mode to prevent thermal drift of the resistance which was observed in continuous measurements. In this pulsed mode, a dc voltage was applied for one second and the resulting current was measured. Between the measurements, the voltage was turned off for 20 seconds. Figure 6.12 depicts the setup and the results. To examine the consequences of the nonlinear contact resistances, the curves of Figure 6.12 were approximated by polynomials

³Device label AC24 L-09 on the wafer mask.

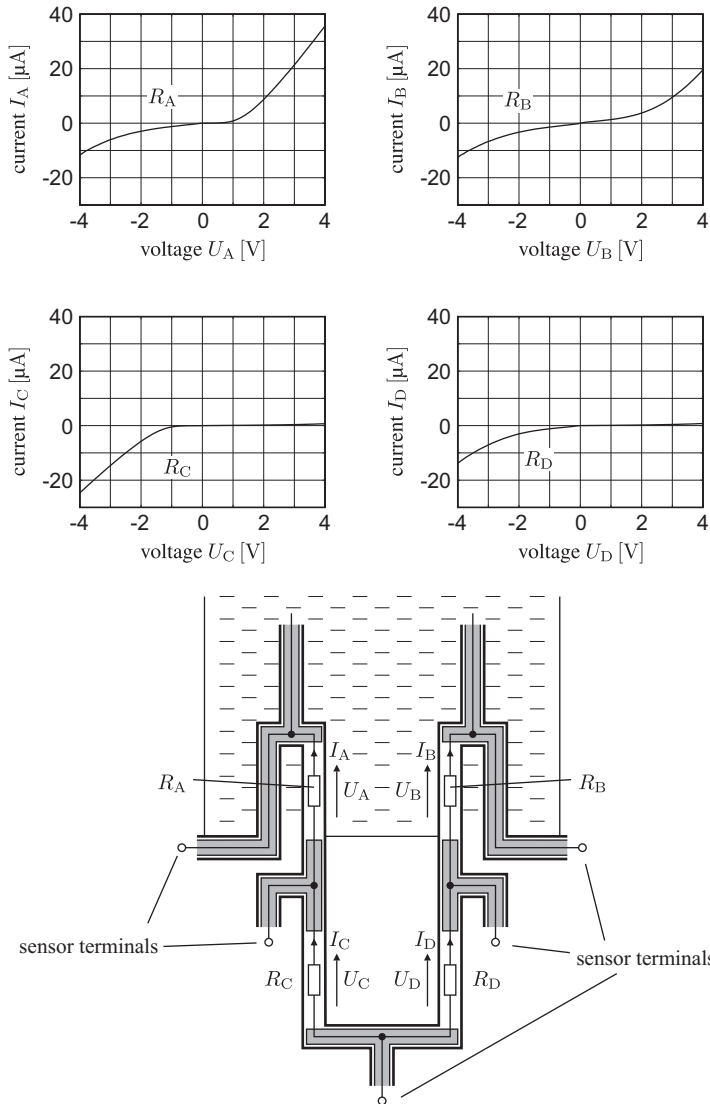


Figure 6.12: Measurement setup and the voltage–current characteristics of the four piezoresistors of a sensor with Cr–Au–Cr metallization.

```
.SUBCKT CONTACTA 10 20
G1 10 20 POLY(1) (10,20) 0 2.6726e-07 -3.41695e-07
      8.80432e-07 4.92806e-07 -6.60986e-08 -7.06326e-08
      2.81316e-09 4.29259e-09 -4.62808e-11 -9.59853e-11
Ro 20 10 100e9
.ENDS CONTACTA
```

Figure 6.13: PSPICE model [73] of the contact resistance R_A . This model implements equation (6.65) using a voltage controlled current source. An additional resistor was placed in parallel to the current source to avoid floating node errors.

using Matlab's `polyfit` function. As an example, the fit result for R_A is

$$\begin{aligned}
 I_{A,\text{fit}} = & 2.6726 \cdot 10^{-7} U_A - 3.4170 \cdot 10^{-7} U_A^2 + 8.8043 \cdot 10^{-7} U_A^3 \\
 & + 4.9281 \cdot 10^{-7} U_A^4 - 6.6099 \cdot 10^{-8} U_A^5 - 7.0633 \cdot 10^{-8} U_A^6 \\
 & + 2.8132 \cdot 10^{-9} U_A^7 + 4.2926 \cdot 10^{-9} U_A^8 - 4.6281 \cdot 10^{-11} U_A^9 \\
 & - 9.5985 \cdot 10^{-11} U_A^{10}.
 \end{aligned} \tag{6.65}$$

Equation (6.65) was then modelled in Orcad PSPICE using the voltage controlled current source `G1` [73]. A parallel output resistance `Ro` (100 GΩ) had to be added to avoid floating nodes. The model code is listed in Figure 6.13. Similar models were created for R_B , R_C , and R_D and were inserted in the schematic diagram presented in Figure 6.14. The resistance of the metal layer was determined by measurements and calculations from the device geometries. A transient simulation of the circuit and a subsequent FFT yielded the frequency spectrum of the output voltage u_d . The result is depicted in Figure 6.15. The diagram shows a dominating spectral component at 99.4 kHz (the readout frequency f_r), caused by the inequalities of the piezoresistors. As described above, the common ground connection resistance leads to a crosstalk from the excitation loop to the readout circuit. This is confirmed by the small peak at 19 kHz. Furthermore, the spectrum exhibits significant spectral components at 80.4 kHz and 118.4 kHz, although no deflection of the device is considered in the model. When the nonlinear models are replaced by resistors with an R_{calc} from equation (6.63), those components vanish. This behaviour indicates, that the nonlinear metal–semiconductors lead to a crosstalk from the excitation current to the lock-in amplifier output voltage. In order to improve the signal-to-noise ratio, ohmic contact must be established between the metal connections and the p-Si device layer. These simulation

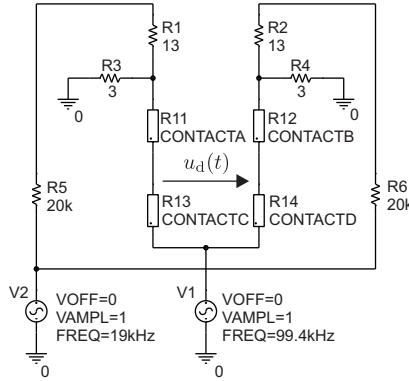


Figure 6.14: Circuit implemented in PSPICE to examine the influence of the non-linear contact resistances on the output voltage $u_d(t)$. The resistors R1, R2, R3, and R4 represent the resistance of the metal conductors, whereas R11, R12, R13 and R14 model the nonlinear behaviour of the piezoresistors (see their characteristics in Figure 6.12). The PSPICE model code of R11 is listed in Figure 6.13. R5 and R6 are external resistors limiting the excitation current, and V1 and V2 generate the readout current and the excitation voltage, respectively.

results were later confirmed by measurements using devices with aluminium metallization and ohmic contacts.

Figure 6.16 emphasizes the importance of ohmic contact resistances. The diagram depicts the resonance characteristic of a sensor with non-ohmic contact resistances⁴ immersed in ethanol. The excitation current amplitude was $250\text{ }\mu\text{A}$ and the readout was fed by a voltage of $\hat{u}_r = 1\text{ V}$ and $f_r = 99.4\text{ kHz}$. A detailed description of the measurement setup can be found below in Section 6.8. Two measurement results were recorded by a lock-in amplifier locked to the frequency $|f_e - f_r|$. In the first measurement, labelled *Ethanol* in the figure, a magnetic field density of approximately 320 mT was applied. Although highly disturbed the measurement indicates the presence of a resonance at approximately 8 kHz . Then, the magnet was removed and a second curve labelled $\vec{B} = 0$ was recorded. Since no mechanical deflections of the sensor are excited now (neglecting thermal vibration), the output voltage should be zero, see (6.62). However, due to the presence of the non-

⁴Device label AC23 M-09 on the wafer mask. The geometries were $l_1 = 450\text{ }\mu\text{m}$, $w_2 = 5\text{ }\mu\text{m}$, $h = 20\text{ }\mu\text{m}$, $l_2 = 100\text{ }\mu\text{m}$, $w_1 = 150\text{ }\mu\text{m}$ (see Figure 6.6).

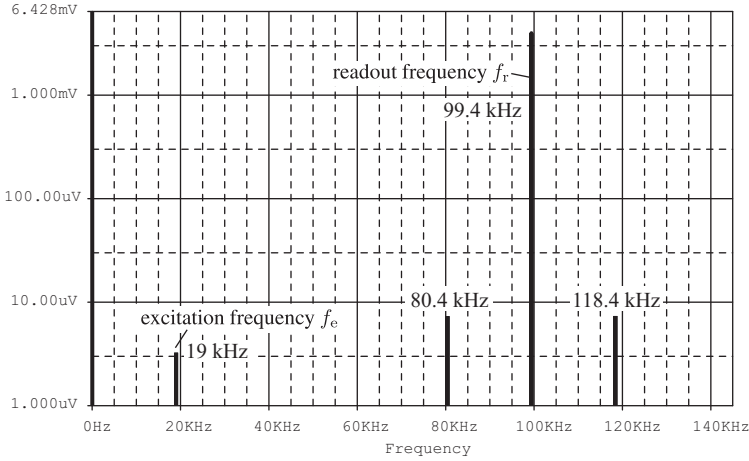


Figure 6.15: Simulation result of the circuit of Figure 6.14. The diagram depicts the spectrum of the Wheatstone bridge's output voltage $u_d(t)$ obtained from a transient analysis and subsequent FFT. Although no deflection is considered in the simulation, u_d has spectral components at 80.4 kHz and 118.4 kHz, indicating that the nonlinear contact resistances of the device act similar to diode mixers.

linear contacts a significant output voltage is detected by the lock-in amplifier, diminishing the sensitivity of the piezoresistive readout circuit.

6.6.5 Alternative Readout Configurations

Below, two layouts for a piezoresistive readout will be discussed, which could be used alternatively to the setup of Figure 6.2.

3 ω -Method

The 3 ω -method is a widely used approach for the measurement of thermal conductivity [74]. A resistor serving as heating and sensing element is driven by a well-known current. Due to the Joule heating in the resistor, its temperature increases and its resistance is changed. This change of resistance can then be detected by measuring the voltage drop at the sensor. From the measurement results, the thermal conductivity of the material surrounding the resistor is obtained. Using the 3 ω method, the resistor is fed with an ac cur-

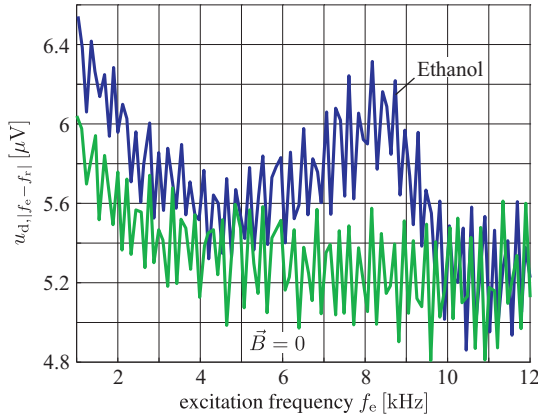


Figure 6.16: Output voltage of the Wheatstone bridge detected by a lock-in amplifier at the frequency $|f_e - f_r|$ versus excitation frequency f_e of a device with nonlinear contact resistances immersed in ethanol. The curve labelled $\vec{B} = 0$ was recorded without a magnetic field applied and indicates the high influence of the nonlinear contact resistances.

rent at a frequency ω instead of a dc current, $i(t) = I \cos(\omega t)$. Since the Joule heating power is the resistance times the squared heater current, it alternates at the frequency 2ω ,

$$P(t) = Ri^2(t) = \frac{RI^2}{2} [1 + \cos(2\omega t)]. \quad (6.66)$$

Consequently, both the increase of temperature $\Delta T(t)$ of the resistor and the change of resistance ΔR are time-dependent and alternate at 2ω ,

$$\Delta T(t) = T_0 + T_1 \cos(2\omega t + \phi), \quad (6.67)$$

where ϕ is the phase shift between the heating power and the resulting increase of temperature. T_0 and T_1 are the dc component and the amplitude of the ac component of ΔT , respectively. The resistance of the heating element reads

$$\begin{aligned} R(t) &= R_0 [1 + \alpha \Delta T(t)] = \\ &= R_0(1 + \alpha T_0) + \alpha T_1 R_0 \cos(2\omega t + \phi), \end{aligned} \quad (6.68)$$

where α is the temperature coefficient of the resistor, and has a spectral component at the frequency 2ω . Consequently, the voltage drop across the heater terminals is

$$\begin{aligned} u(t) &= R(t)i(t) = \\ &= R_0 I (1 + \alpha T_0) \cos(\omega t) + \frac{R_0 I \alpha T_1}{2} \cos(\omega t + \phi) \\ &\quad + \frac{R_0 I \alpha T_1}{2} \cos(3\omega t + \phi). \end{aligned} \quad (6.69)$$

Since $i(t)$ alternates at the frequency ω and $R(t)$ has a spectral component at 2ω , the amplitude modulation due to equation (6.69) leads to a component of the output voltage at 3ω . Therefore, this principle allows reading the temperature change of the heating element at the frequency 3ω with a lock-in amplifier, eliminating the baseline $R_0 I (1 + \alpha T_0)$ and interferences that occur during dc measurements [74].

Applied to the piezoresistive readout of the viscosity sensor, such a principle would allow to read the plate deflection at three times the excitation frequency. Thus, the device would require only a single signal source and crosstalk from the excitation current would be eliminated. To achieve a frequency triplication similar to the example explained above, a suitable configuration of piezoresistors must be found. As in former designs, the spring bar is split in two parts at its fixed end (Fig. 6.17). Both of these arms act as piezoresistive elements with an ohmic resistance R_0 . A driving current of $i(t) = I \cos(\omega_e t)$ and, therefore, a plate deflection of $\psi = \hat{\psi} \cos(\omega_e t + \phi)$ are assumed. The mean mechanical stress in the piezoresistors is $\sigma = \pm \hat{\sigma}_1 \cos(\omega_e t + \phi)$. Consequently, the electrical resistance of the piezoresistors is given by

$$R_{\text{left}}(t) = R_0 + R_1 \cos(\omega_e t + \phi) \quad \text{and} \quad (6.70)$$

$$R_{\text{right}}(t) = R_0 - R_1 \cos(\omega_e t + \phi), \quad (6.71)$$

where R_1 is the amplitude of resistance change obtained from equation (6.58),

$$R_1 = \hat{\sigma}_1 \pi_1 R_0. \quad (6.72)$$

Both resistances are electrically connected in parallel. Their total resistance is

$$R_{\text{AB}}(t) = \frac{R_{\text{left}} R_{\text{right}}}{R_{\text{left}} + R_{\text{right}}} = \frac{R_0^2 - R_1^2 \cos^2(\omega_e t + \phi)}{2R_0}. \quad (6.73)$$

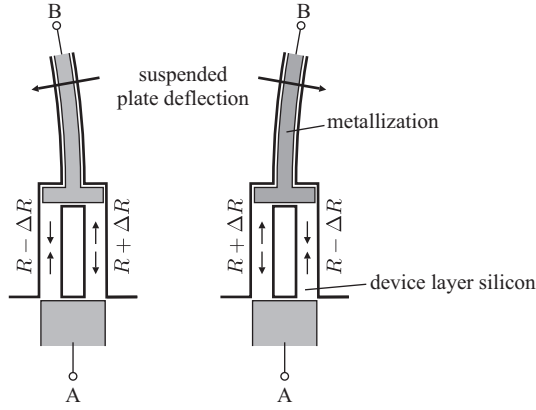


Figure 6.17: Alternative configuration of the piezoresistors to allow for utilizing the 3ω -method.

Thereby the contribution of the metallic conductive paths is neglected. Measuring the voltage between A and B yields

$$\begin{aligned}
 u_{AB}(t) = R_{AB}(t)i(t) = & \left(\frac{R_0 I}{2} - \frac{R_1^2 I}{4R_0} \right) \cos(\omega_e t) - \\
 & - \frac{R_1^2 I}{8R_0} \cos(\omega_e t + 2\phi) + \\
 & + \frac{R_1^2 I}{4R_0} \cos(3\omega_e t + 2\phi). \quad (6.74)
 \end{aligned}$$

Using a lock-in amplifier locked to the 3rd harmonic of the excitation frequency ω_e , the amplitude

$$U_{3\omega_e} = \frac{R_1^2 I}{8R_0} = \frac{1}{8} \pi_1^2 R_0 I \hat{\sigma}_1^2 \quad (6.75)$$

along with the phase shift angle 2ϕ is obtained.

However, the resulting voltage amplitudes are very small. Assuming similar values as in Section 6.6.3, i.e., $I = 1 \text{ mA}$, $R_0 = 917 \dots 1750 \, \Omega$ as calculated in equation (6.63), $\pi_1 = 7.18 \cdot 10^{-10} \text{ Pa}^{-1}$ from equation (6.57), and $\hat{\sigma}_1 = 74 \text{ kPa}$, results in output voltages in the range from 0.3 to 0.6 nV. Therefore, no devices using the described readout principles have been fabricated in the course of this thesis.

Separate Excitation and Readout Circuits

The sensor readout depicted in Figure 6.2 is based on spring beams which are forked at one side. These forks form a piezoresistive element and a ground connection each. Due to the layout of the electrical connections on the sensor, the excitation current flows over all of the four beam springs, maximizing the Lorentz force acting on the vibrating part of the sensor. However, the ground connections formed by two of the prongs are shared by the excitation loop and the readout circuit. Due to the finite resistance of this ground connection, the design is prone to crosstalk from the excitation current to the sensor output voltage u_d , see the discussions in Sections 6.6.3 and 6.6.4. Measures must be taken to eliminate this crosstalk by driving the Wheatstone bridge circuit and the excitation loop at different frequencies, using lock-in amplifiers, and ensuring ohmic contact resistances. Figure 6.18 depicts an alternative readout configuration. Here, the network of piezoresistive elements is concentrated at the fixed end of one of the beam springs. Hence, the excitation and readout circuits do not share the ground connection any more. However, since only two of the beam springs carry an excitation current, only half the Lorentz force (compared to the design in Figure 6.2) acts on the vibrating plate, decreasing the deflection amplitude. It should be noted that an electrical connection between the readout and the excitation circuit exists at one single point: On the rectangular plate, the both conductors are electrically connected by the p-doped device layer, which must be considered in the design of the readout electronics. Alternatively, the fabrication process (Section 6.2) could be modified to add an insulation between the excitation loop and the device layer.

6.7 Device Operation in Air

First, the operation of the sensor in air was investigated. As the structure vibrates in an in-plane mode, the stroboscopic planar motion mode of a Polytec MSA-400 microsystem analyser was used. The result of a frequency sweep measurement is depicted in Figure 6.19. The sensor was excited with a current amplitude of $10\text{ }\mu\text{A}$ supplied by the MSA-400 and the magnetic flux density of 320 mT of a permanent magnet. The resulting vibration amplitude is rather low and the optical measurement is highly influenced by thermal drift and noise, but higher mechanical deflections lead to nonlinear spring effects like the Duffing behaviour (Figure 6.20). Nevertheless, the results give an idea of the sensor's resonance frequency $f_0 = \omega_0/(2\pi)$, quality factor Q , and

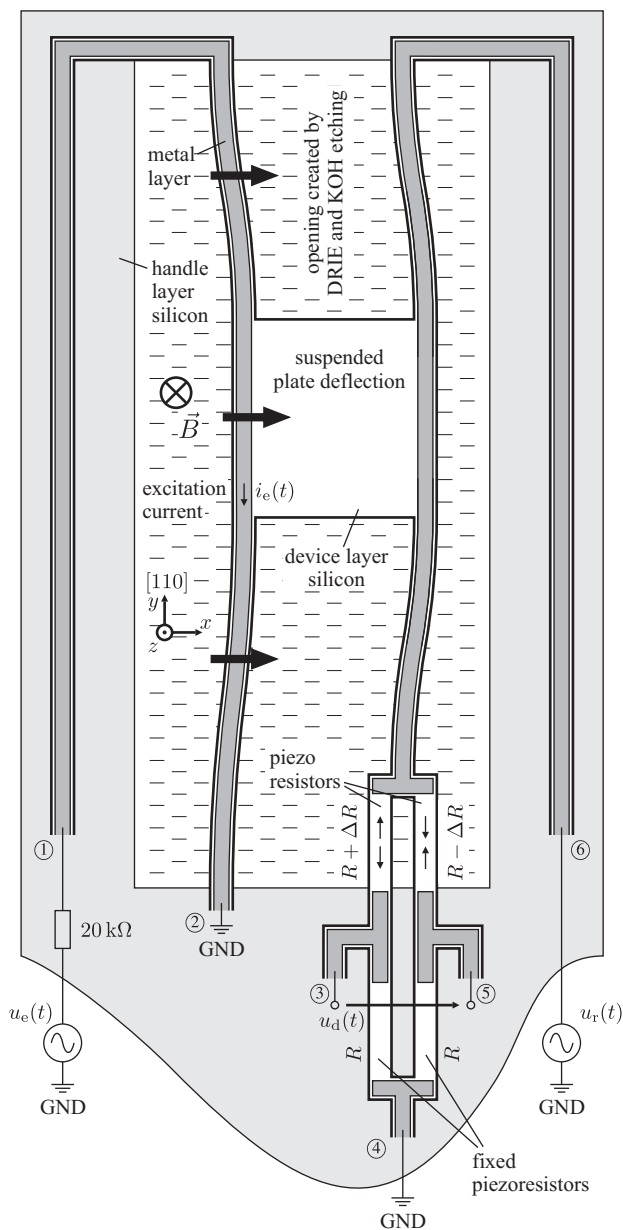


Figure 6.18: Alternative sensor layout featuring separate connections for the readout and the excitation circuit. The left hand beam springs carry the excitation current $i_e(t)$, whereas the right hand springs are part of the Wheatstone bridge used for detecting the plate deflection.

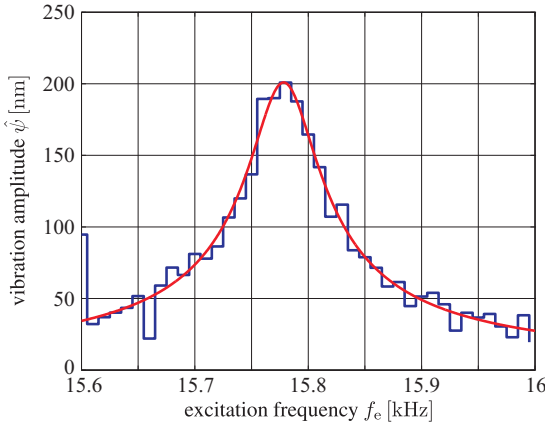


Figure 6.19: Frequency sweep measurement of the vibration amplitude of the sensor vibrating in air (excitation current amplitude $\hat{i}_e = 10 \mu\text{A}$, flux density $B_z = 320 \text{ mT}$), obtained from the planar motion analyser of a Polytec MSA-400 microsystem analyser. The stair plot depicts the measurement results, whereas the solid line represents a second order system fit. The resonance frequency of the sensor vibrating in air is 15.77 kHz , the quality factor Q is 256.

maximum vibration amplitude $\hat{\psi}(\omega_0) = A_0 Q$. These parameters have been extracted by fitting the amplitude of a second order system,

$$\hat{\psi}(\omega) = \left| \frac{A_0}{1 + j\omega \frac{1}{Q\omega_0} - \frac{\omega^2}{\omega_0^2}} \right|, \quad (6.76)$$

to the results. The fit is represented by the solid line in Figure 6.19, and results in $f_0 = 15.77 \text{ kHz}$, $Q = 256$, and $\hat{\psi}(\omega_0) = 200 \text{ nm}$. The static beam deflection for excitation with $10 \mu\text{A}$ dc is obtained from $A_0 = 0.78 \text{ nm}$. In addition, the plate movements in the out-of-plane direction were investigated using the scanning laser vibrometer of the microsystem analyser. No vibrations could be detected in the vicinity of the sensor's first-mode resonance frequency. As the vertical resolution of the vibrometer is in the range of several tens of femtometres, this measurement confirms the assumption of pure in-plane vibrations.

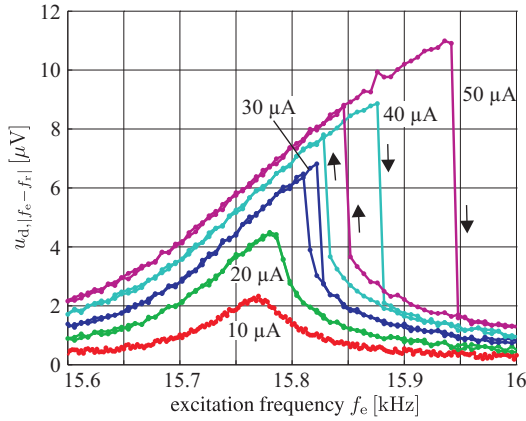


Figure 6.20: Frequency sweep experiments demonstrating the nonlinear Duffing behaviour of the resonator operating in air. Depending on the sweep direction indicated by the arrows the sensor exhibits a different frequency characteristic [75] at higher excitation currents. $u_{d, |f_e - f_r|}$ is the output voltage of the sensor's piezoresistive readout. The sensor was driven by a magnetic flux density of 320 mT and the excitation current amplitude stated in the figure. The results recorded for 10 μ A correspond to the optical measurements of Figure 6.19.

6.8 Measurements in Liquids

To characterize the piezoresistive elements electrically, their voltage–current characteristics were measured. In the range from -4 to 4 V, the resistances behave fairly ohmic. However, the measured resistance of 8.1 k Ω did not fit the calculated values, see (6.63). This result indicates the presence of high contact resistances between the aluminium metallization and the semiconductor. Such sheet resistances R_S increase the total resistance in equation (6.60): R_0 must be replaced by $R_0 + 2R_S$, whereas ΔR is still given by equation (6.58), $\Delta R = \sigma_1 \pi_1 R_0$. Thus, the additional sheet resistances decrease the sensitivity of the piezoresistive readout.

Figure 6.21 describes the measurement setup for the integrated piezoresistive readout. Two Agilent 33220A function generators provided the excitation voltage u_e and the readout voltage u_r . In the excitation loops, 20 k Ω series resistors were placed (see also Figure 6.2) to limit the excitation current. The maximum output voltage of the function generators was 10 V and, therefore, the excitation current was limited to 500 μ A. A Stanford Research

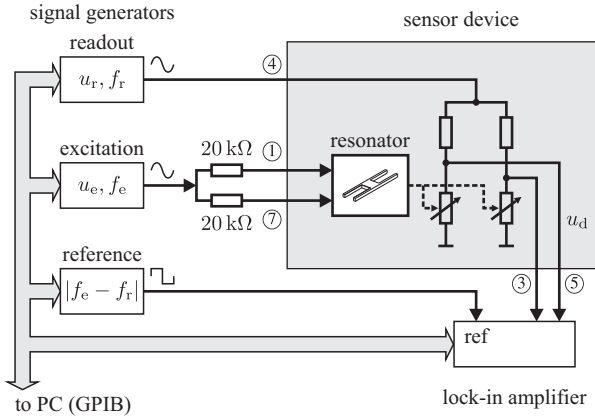


Figure 6.21: Schematic diagram of the measurement setup utilizing piezoresistive sensor readout. The circled numbers correspond to the sensor terminals in Figure 6.2.

SR830 lock-in amplifier measured the differential voltage output of the sensor's bridge circuit. Due to the amplitude modulation in the piezoresistors (see Section 6.6.3), the lock-in amplifier must be locked to either $|f_e - f_r|$ or $|f_e + f_r|$. A suitable reference signal for the lock-in amplifier can be generated by multiplying u_e with u_r . This multiplication results in a superposition of two sinusoidal signals at the frequencies $f_r - f_e$ and $f_r + f_e$. Subsequent filtering is required to suppress one of these signals. Due to the low excitation frequency f_e (down to 1 kHz) both the signals are very close in the frequency spectrum. A high order filter is required, which could potentially introduce an additional phase shift. To keep the setup simple in these first measurements a third function generator was used instead and set to $f_r - f_e$. The difference frequency was selected due to the limited frequency range of the lock-in amplifier. However, using a third signal source inhibits the measurement of the phase shift between excitation current and mechanical vibration, as the mutual phase shifts of the function generators are arbitrary.

In the first experiments, the excitation currents were kept low (amplitude of $500\text{ }\mu\text{A}$), making sure not to destroy the device. To still achieve reasonable vibration amplitudes and output voltages, an electromagnet applied a strong magnetic field with a flux density of 1.9 T . In later experiments, we determined a maximum excitation current of several mA, which allows reducing the magnetic flux density to a level that can easily be generated by a per-

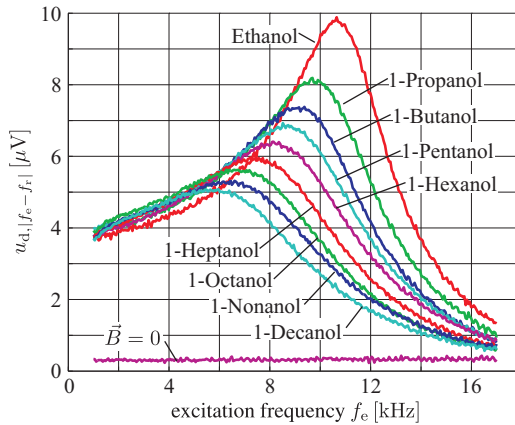


Figure 6.22: Resonance peaks of the suspended plate sensor immersed in a variety of nine sample liquids. The tenth result labeled $\vec{B} = 0$ was recorded with the electromagnet turned off, and indicates the noise floor.

manent magnet. In comparison to the microsystem analyser measurement, the excitation current was increased by a factor of 50, and the magnetic flux density by 5.9. An estimation of the static plate deflection in the following experiment is therefore $A_0 = 0.78 \text{ nm} \cdot 50 \cdot 5.9 = 230 \text{ nm}$.

The readout circuit was fed with a voltage of 1 V (amplitude) and a frequency of 99.4 kHz. The function generator that fed the excitation loop was set at an amplitude of 10 V, resulting in an excitation current amplitude of 500 μA . The lock-in amplifier was configured for differential input, a sensitivity of 10 μV , and a time constant of 300 ms. A computer performed a frequency sweep measurement in the range from 2 to 17 kHz by changing the function generator frequencies f_e , and $|f_e - f_r|$, accordingly. The sensor was mounted in the centre of a $1 \times 1 \times 1 \text{ cm}^3$ liquid container. This container was placed between the poles of the electromagnet and filled with 1 ml of the sample liquid. The temperature of the liquids was 25 $^\circ\text{C}$. A cover lid prevented evaporation of the liquids. As sample liquids, a variety of alcohols was chosen, which exhibit viscosities in the range from 1.06 to 10.97 mPa·s and densities from 789 to 830 kg/m^3 (Table 6.2). The sensor device was cleaned with ethanol between the measurements.

Figure 6.22 depicts the measurement results obtained with the nine sample liquids. To give an idea of the noise floor, a tenth curve was recorded

Table 6.5: Results of curve-fitting a second order system, equation (6.76), to the measurement data of Figure 6.22: f_0 is the resonance frequency, Q is the quality factor, and $D = 1/(2Q)$ is the damping factor.

Liquid	f_0 [kHz]	D	Q
ethanol	10.71	0.18	2.75
1-propanol	9.92	0.23	2.17
1-butanol	9.38	0.26	1.92
1-pentanol	9.11	0.29	1.75
1-hexanol	8.82	0.30	1.64
1-heptanol	8.24	0.33	1.50
1-octanol	7.76	0.36	1.36
1-nonanol	7.64	0.39	1.27
1-decanol	7.18	0.42	1.20

with the sensor immersed in 1-decanol and with the electromagnet turned off. Again, the second order system, (6.76), was curve-fitted to the results of Figure 6.22. Table 6.5 lists the resonance frequencies and the quality factors of the suspended plate sensor immersed in the liquids. The deviation of the modelling results (Section 6.4) from the measurements are 15 % for the quality factor, and up to 30 % for the resonance frequency.

For the dynamic viscosity of the liquids and the damping factor $D = 1/(2Q)$, a simple relation was found, similar to that for the clamped-clamped beam, equation (5.10). Figure 6.23 depicts both parameters in logarithmic scale.

6.9 Simultaneous Measurement of Viscosity and Density

The theoretical model devised in Section 6.3 was found to be a suitable tool for the design of the suspended plate device. However, the deviations between the modelling results and the device behaviour revealed in the experiments are still quite large. Although the relation between the viscosity η and the damping factor D can be approximated by

$$D = 1.998\eta^{0.346} \quad (\text{see Figure 6.23}) \quad (6.77)$$

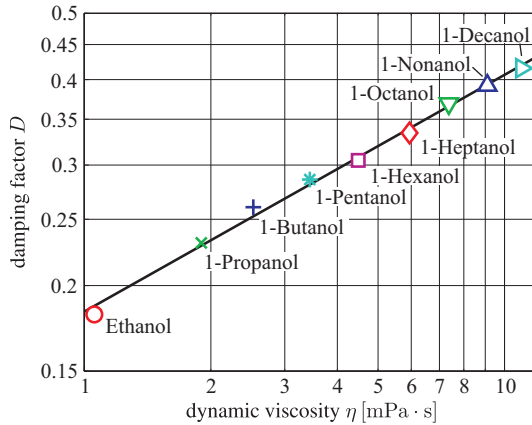


Figure 6.23: Damping factor D versus viscosity η . The markers indicate the measurement results, whereas the solid line depicts the fit result, $D = 1.998\eta^{0.346}$.

a suitable model is required for the simultaneous measurement of the liquid's mass density.

In Chapter 3 a generalized model based on the forces acting on a sphere that is oscillating in a liquid is devised. Below its feasibility for the evaluation of the measurement results of the suspended plate device shall be examined. In general, micromachined silicon devices feature high quality factors in vacuum, the intrinsic damping, b_i in (3.10) on page 26, is therefore considered to be zero. Analogously to (3.14) and (3.15) we have

$$\omega_0^2 = \frac{K}{m_e + m_i} \quad \text{and} \quad T = \frac{b_i}{K}, \quad (6.78)$$

where ω_0 is the angular resonance frequency, T is the damping time constant, m_e is the effective mass of the resonator and m_i and b_i are the induced mass and damping due to the liquid surrounding the moving plate and the beam. In air, the resonance frequency and the damping time constant are

$$\omega_{0,\text{air}}^2 = \frac{K}{m_e} \quad \text{and} \quad T_{\text{air}} = \frac{b_{i,\text{air}}}{K}. \quad (6.79)$$

With m_i and b_i from (3.11) and (3.12) and by introducing four generalized

model coefficients c_1 , c_2 , c_3 , and c_4 we obtain the model equations

$$\omega_0^2 = \omega_{0,\text{air}}^2 \frac{1}{1 + c_1\rho + c_2 \frac{1}{\sqrt{\omega_0}} \sqrt{\eta\rho}}, \quad \text{and} \quad (6.80)$$

$$T = T_{\text{air}} (c_3\eta + c_4\sqrt{\omega_0}\sqrt{\eta\rho}). \quad (6.81)$$

The measurements were carried out using a sensor featuring a beam length l_1 of 450 μm and a rectangular plate of $200 \times 200 \mu\text{m}^2$ (Figure 6.6). In contrast to the measurement results presented above, a permanent magnet was used instead of the bulky electromagnet. The magnetic field density was measured, $B_z = 320 \text{ mT}$. Excitation current amplitudes of 1.6 mA and 5 μA were applied for the measurements in liquid and air, respectively. The measurement setup depicted in Figure 6.21 was used to record frequency sweep measurements in the vicinity of the resonance frequency. From the measured values, $f_0 = \omega_0/(2\pi)$ and the damping time constant $T = 2D/\omega_0$ were obtained by curve fitting, see equation (6.76). The results are depicted in Figure 6.24. Similar to the approach of Chapter 3, the model equations (6.80) and (6.81) were fitted to these results to obtain the parameters $\omega_{0,\text{air}} = 2 \cdot \pi \cdot 10.72 \text{ kHz}$, $T_{\text{air}} = 28.9 \text{ ns}$, $c_1 = 7.493 \cdot 10^{-5} \text{ m}^3/\text{kg}$, $c_2 = 98.361 \text{ m}^2/\text{kg}$, $c_3 = 21.476 \cdot 10^3 \text{ m}\cdot\text{s}/\text{kg}$, and $c_4 = 0.5935 \text{ m}^2/\text{kg}$. In Figure 6.24 the fit results are indicated by crosses and show good agreement of the model and the measurement results.

As an example for the measurement of density and viscosity of an “unknown” liquid, the resonance frequency f_0 and the damping time constant T of the sensor immersed in 1-hexanol was measured. This liquid was not included in the fit process described above. The measurement results were $f_0 = 7.795 \text{ kHz}$ and $T = 9.75 \mu\text{s}$. Using the model parameters c_1 , c_2 , c_3 , c_4 , $f_{0,\text{air}}$, and T_{air} obtained above and solving equations (6.80) and (6.81) for η and ρ yielded the viscosity $\eta = 4.28 \text{ mPa}\cdot\text{s}$ and the density $\rho = 816 \text{ kg}/\text{m}^3$ of the fluid. The results agree well with the reference values listed in Table 6.2, 4.49 mPa·s and 814 kg/m³.

6.10 Conclusion

The feasibility of the novel viscosity sensor was demonstrated by experiments. The integrated piezoresistive readout is suitable to measure the frequency characteristics of the plate vibrations in liquids. The sensor output voltages were in the range of up to 10 μV with a noise floor at 350 nV.

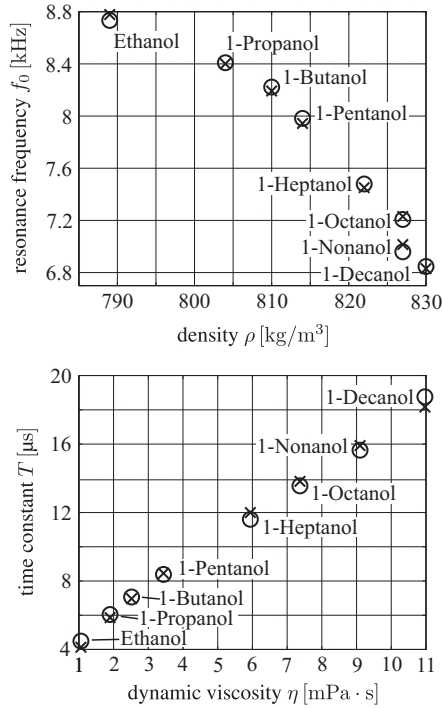


Figure 6.24: Fitting the model equation based on the sphere model to the measurement results. The circles indicate the measured resonance frequency f_0 and damping time constant T of the resonator versus the density and viscosity of the liquid, respectively. The crosses indicate the fit results of the equations (6.80) and (6.81) to the results.

At the top and bottom faces of the rectangular plate, mainly shear waves are excited. Their contribution to the damping of the device is low. Therefore, increasing the size of the plate only results in a small increase of the damping, whereas the higher resonator mass enhances the quality factor. These increased Q-factors lead to higher deflection amplitudes, and better signal-to-noise ratios of the output voltage. We illustrated this relations with a model, resulting in Q-factors of up to 10.9 at a dynamic viscosity of 1.09 mPa·s. Such a Q-factor was already achieved by a clamped–clamped beam device presented in [15]. However, this resonating beam is characterised by a large

thickness, increasing not only the moving mass, but also the stiffness of the device. Thus, the beam deflections were low, and required an extremely sensitive readout, in particular a laser vibrometer.

Varying the size of the rectangular plate also allows the design of devices that exhibit different resonance frequencies. An array of such devices on a single chip would probe the viscosity of liquids at several frequencies simultaneously, which could be a promising approach to measure the behaviour of viscoelastic fluids.

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Contributions to Conferences

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